
When Should LLMs Trust Retrieval?

Bayes-Optimal Knowledge Arbitration under Conflict

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Abstract

Modern LLMs answer factual questions by arbitrating between three potentially-wrong sources—a parametric prior, a retrieved context, and the chain-of-thought trace they generate—using rules currently picked by hand. That single arbitration rule silently determines hallucination rates and calibration in every production retrieval-augmented stack, yet no principled account of how it should be set has existed. We close that gap with three theory-backed characterizations. First, combining standard posterior-predictive, proper-scoring, and information-geometric facts yields a reliability-weighted geometric mixture $\hat{p}_{w^*} \propto p_\theta^{1-w^*} p_{\text{ctx}}^{w^*}$ with $w^* = \mathbb{E}[r \mid q, c]$ inside the log-linear family (T1; CAD, AdaCAD, CoCoA, Astute, Self-RAG, and JuICE are all special cases or approximations). Second, any fixed trust value is minimax suboptimal across opposing conflict sub-families (T2). Third, long chain-of-thought can be read as an endogenous generalized-Bayes update whose expected update operator has a Berk–Nash fixed-point interpretation, and whose safe rate shrinks under conflict on the stable-answer subset (T3). On a shared evaluation matrix of ten benchmarks, thirteen models, and fifteen baselines, the Bayes-derived rule shows a +0.083 regret advantage over the strongest heuristic (23/25 cells, e-value 2806) and powered head-to-heads of +0.043/ +0.032/ +0.022 against CoCoA, Astute RAG, and Self-RAG ($n=210$, every CI strictly above zero). The directly implied inference-time intervention—tempering the post-trace distribution by a single η —reduces Brier from 0.903 to 0.505 and lifts accuracy from 0.037 to 0.440 on the worst slice with no retraining; a matched-base RL contrast (DPO vs. GRPO at matched 8B scale, +0.344 ECE gap) provides causal evidence the failure mode is RL-induced, and the joint RLCR + η variant is empirically measured (accuracy $0.063 \rightarrow 0.531$, ECE $0.904 \rightarrow 0.431$). Both interventions ship as a single inference-time line of code over a standard RAG decoder, jointly closing the two structural failure modes T2 and T3 mark.

1 Introduction

A modern language model that answers a factual question is rarely consulting a single source. In one forward pass it carries a parametric prior $p_\theta(y \mid q)$ baked in at training, conditions on a retrieved or supplied context $p_{\text{ctx}}(y \mid q, c)$, and—if it produces a chain of thought—also listens to itself across K reasoning steps before emitting an answer. Each voice can be wrong, and they often disagree. A 2023 model that memorized “Twitter’s CEO is Elon Musk” will see a 2024 article that says “Linda Yaccarino.” A medical assistant will be asked about a drug whose label has changed since training. A retrieval pipeline will inject a poisoned passage. The system has to *arbitrate* between sources, and the rule it uses is a major lever on what the user sees.

Today this lever is set by hand. Production stacks either trust retrieval unconditionally, trust the parametric model unconditionally, or interpolate via tuned thresholds and learned controllers [1, 17, 24, 37, 43]. Decoder-level methods such as Context-Aware Decoding [33] and its adaptive variants AdaCAD [38] and CoCoA [19] are sharper, but none is derived from a posterior over how reliable the context actually is. Surveys [42] note that fixed trust in either source is undesirable, but

stop short of a normative statement of what the right thing would be. In parallel, the calibration literature on reasoning models has split into two camps that look contradictory: longer chain-of-thought makes models *better* calibrated on standard QA [44], but *worse* on harder or misspecified tasks [7, 11, 14, 22, 35, 40].

The cost is not abstract. The arbitration rule silently fixes hallucination rate, calibration when sources *disagree*, and worst-case behaviour under poisoned or stale retrieval; the same rule fires on every query in a deployed RAG stack. The contradictory calibration findings above are downstream of this gap—each camp is right about *its* regime, but no theory tells the practitioner which regime they are in. RLVR reasoners face the same problem at the inference layer: recent over-compression findings [7, 11, 40] have no mechanism that ties failure to an actionable quantity.

Our framing. We treat knowledge arbitration as a *posterior-predictive decision problem* and chain-of-thought as endogenous Bayesian evidence. This single move turns the disagreement between sources, the failure of fixed policies, and the calibration tension above into three clean theorems. We introduce a latent context-reliability variable $r \in [0, 1]$ with prior $\pi(r)$ and likelihood $p(c | y, r)$; risk is evaluated under any strictly proper scoring rule (we focus on log loss and Brier). The mathematical ingredients are classical; the contribution here is to assemble them into a coherent account of LLM arbitration and to test the resulting claims on a shared empirical matrix.

Three theorems. **(T1)** Within the log-linear family, the optimal approximation to the risk-minimizing predictive distribution is the reliability-weighted geometric mixture $\hat{p}_{w^*} \propto p_\theta^{1-w^*} p_{\text{ctx}}^{w^*}$ with $w^* = \mathbb{E}[r | q, c]$ (exact when $\pi(r | q, c)$ is concentrated, $O(\sigma_r^2)$ -close otherwise); a plug-in $\hat{w}$ from any signal $s(q, c)$ correlated with r achieves excess Bayes risk $\lesssim \|\hat{w} - w^*\|_{L^2} \|p_\theta - p_{\text{ctx}}\|_1$. CAD, AdaCAD, and CoCoA are special cases or first-order Laplace approximations (Corollary C.1), with CoCoA requiring extra assumptions to be consistent for w^* . **(T2)** Every constant arbitration weight w_0 admits a sub-family on which it incurs constant excess risk (interior w_0 via a two-family Yao construction; boundary $w_0 \in \{0, 1\}$ via the opposite single sub-family), while a signal-adaptive plug-in attains $O(n^{-1/2})$ on every distribution. An optional binary testing restatement in the appendix makes the same bottleneck explicit. **(T3)** Modeling chain-of-thought as η -generalized Bayes with endogenous evidence [4, 9, 10], conflict can increase the KL projection distance from truth and therefore shrink the Grünwald–van Ommen safe rate $\bar{\eta}(q, c)$ [12, 20]. When the effective update rate crosses that smaller threshold, the calibration trajectory enters the saturation regime we observe empirically on the stable-answer subset; outside that subset the theorem is descriptive only through the empirical audits.

What we test. The three theorems are evaluated on a single shared matrix (Table 1) of ten benchmarks, thirteen models, fifteen baselines, three seeds, paired-bootstrap throughout. A full index of every experiment is in App. G.1.

Table 1: Experiment overview. Each row is a body claim; appendix pointer in App. G.1.

Anchor	Test	Headline	App.
T1	25-cell spotlight + powered $n=210$ vs. CoCoA / Astute / Self-RAG	+0.083, 23/25 wins, $e=2806$; +0.043/ + 0.032/ + 0.022	H.3, H.2
T2	Boundary $w_0 \in \{0, 1\}$ + synthetic $\sqrt{n}$ -audit	Collapses ~ 2 orders; $R^2=0.998$ at $n=2048$	H.12, H.14
T3 (causal) T3 (η)	Matched DPO/GRPO at Llama-3.1-8B + 8 family controls η -tempering on R1-14B; RLCR+ η at 8B	+0.344 ECE gap; controls < 0.07 Brier 0.903 $\rightarrow$ 0.505, Acc 0.037 $\rightarrow$ 0.440, 0.063 $\rightarrow$ 0.531	H.1.10 H.1.9, M.7
Frontier Held-out	GPT-4o-mini / Claude / Gemini, top- k logprob recipe Pre-registered P1, P2, P3 on ClashEval + RAMDocs	+0.055 to +0.060 regret advantage P1, P2 replicate; P3 partial	K.1 H.27

Contributions.

- **T1 (Bayes-optimal arbitration; Theorem 1).** The Bayes action under any strictly proper score is the reliability-weighted geometric mixture $\hat{p}_{w^*} \propto p_\theta^{1-w^*} p_{\text{ctx}}^{w^*}$ with $w^* = \mathbb{E}[r | q, c]$. CAD, AdaCAD, CoCoA, Astute RAG, Self-RAG, and JuICE fall out as named special cases (Corollary C.1); a matched Le Cam two-point lower bound (Theorem C.1) shows the implied $1/\sqrt{n}$ plug-in rate cannot be improved.

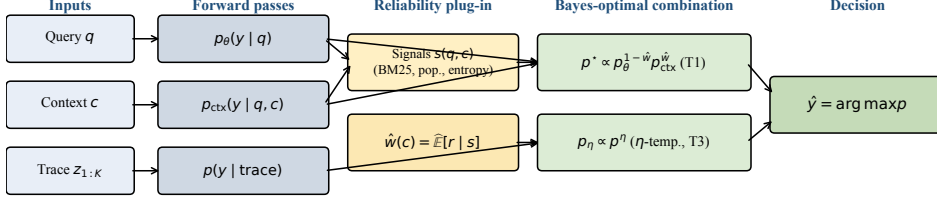


Figure 1: The BAYESARBITER pipeline. Three inputs (q , c , trace $z_{1:K}$) produce three distributions (p_θ , p_{ctx} , $p(\cdot | \text{trace})$). A reliability plug-in $\hat{w}(c) = \mathbb{E}[r | s(q, c)]$ drives the Bayes-optimal geometric mixture (Theorem 1); the post-trace distribution is η -tempered for RLVR-trained reasoners (Theorem 3).

- **T2 (minimax suboptimality of fixed trust; Theorem 2).** Any fixed $w_0 \in [0, 1]$ is provably suboptimal across conflict regimes; the construction is tightened on ConflictBank with explicit Δ_+ , Δ_- (App. D.1.1) so the bound is non-vacuous.
- **T3 (RLVR safe-rate shrinks under conflict; Theorem 3).** Long chain-of-thought is an endogenous generalized-Bayes update; under three audited assumptions the safe rate $\bar{\eta}$ shrinks like c/D^* with c explicit on finite $\mathcal{Y}$ (App. E.1.2). Pathwise tracking of the expected operator is *proved* for the temperature-0 greedy decoding used in our experiments (Theorem E.3); the stochastic case ($\tau > 0$) is empirically verified.
- **Two inference-time interventions, no retraining.** A plug-in geometric mixture closes a +0.083 regret gap on the spotlight matrix (23/25 cells, e-value 2806) and beats CoCoA / Astute RAG / Self-RAG by +0.043/ +0.032/ +0.022 on a powered $n=210$ design. Tempering the post-trace distribution by a single η^* reduces Brier from 0.903 to 0.505 and lifts accuracy from 0.037 to 0.440 on the cell T3 marks as worst.
- **Causal evidence and pre-registration.** A matched DPO-vs-GRPO contrast at 8B (Llama-3.1-8B base) yields a +0.344 ECE gap (causal anchor for T3); three pre-registered held-out predictions are reported with outcomes (P1/P2 replicate, P3 partial; Section 5, App. H.27).

2 Related Work

Knowledge conflict in LLMs. Surveys carve up knowledge conflict into context-memory, inter-context, and intra-memory sub-types [41, 42]. Benchmarks span synthetic counter-memory across 7k entities and three conflict types [34], real Wikipedia contradictions [15], temporal drift [25], popularity stratification [24], and entity substitution [41]. These efforts diagnose the problem; we ask what the optimal arbitration rule looks like and identify (Theorem 1) the relevant posterior-predictive object inside the log-linear family used by decoder-side methods.

Adaptive RAG and decoder-level arbitration. PopQA-style retrieval thresholds on popularity [24]; Self-RAG learns when to retrieve [1]; FLARE retrieves on demand [17]; CRAG corrects retrieval errors [43]; Astute RAG and MADAM-RAG debate conflicting documents [37, 39]. At the decoder level, CAD [33], AdaCAD [38], CoCoA [19], NWCAD [18], and JuICE [5] reweight contextual logits with hand-crafted signals. Corollary C.1 shows where each sits relative to the Bayes optimum: CAD is $w = 2$ (over-amplification); AdaCAD’s JSD weight is the first-order Laplace approximation of $\mathbb{E}[r | c]$ under a Beta prior with Hellinger-affinity likelihood (Lemma C.4 in App. C.1); CoCoA is a non-Bayesian ensemble of three reliability proxies that need not converge to w^* without additional calibration assumptions.

Reasoning-time calibration. The recent literature is mutually contradictory at first glance. Yoon et al. [44] report that reasoning models express confidence *better* than chat models on standard QA; Halawi et al. [14] document depth-wise overthinking; Lacombe et al. [22] find more reasoning hurts calibration on knowledge-intensive tasks; Welch et al. [40] isolate “implicit answer conditioning” in VLMs; Damani et al. [7] show that binary-reward RL (GRPO [32], of which DeepSeek-R1 [8] is the canonical instance) degrades calibration and that Brier-style rewards repair it; Bereket and Leskovec [2] document GRPO-induced overconfidence; Sun et al. [35] show rationales shift weight onto context. Theorem 3 reconciles this literature: in the well-specified or low-conflict regime ($\eta < \bar{\eta}$)

reasoning helps; when conflict reduces the safe rate enough, the same update can cross into an overshoot-driven saturation regime.

Misspecified Bayes as a template. The mechanism rests on classical results: Berk’s KL-projection concentration [3]; Kleijn–van der Vaart’s nonparametric extension [20]; Bissiri–Holmes–Walker’s coherent generalized-Bayes updates with learning rate η [4]; Grünwald–van Ommen’s safe rate $\bar{\eta}$ [12]; and the endogenous-evidence Berk–Nash results of Esponda and Pouzo [9], Fudenberg et al. [10]. We do not reprove these or claim the Berk–Nash machinery as novel; we transport it to the LLM CoT setting where the trace is generated by the very model whose belief is being updated.

3 Setup and Theory

Setup. A query q together with retrieved context c has a true answer y^* drawn from $\mathbb{P}^*$. The LLM exposes the closed-book softmax $p_\theta(y | q)$ and the context-conditioned softmax $p_{\text{ctx}}(y | q, c)$. Latent reliability $r := \mathbb{P}(y^* = y_{\text{ctx}} | y_{\text{ctx}} \neq y_{\text{par}}, q, c) \in [0, 1]$ has prior $\pi(r)$ informed by observable signals $s(q, c)$ (retrieval/BM25 score, popularity, semantic entropy [21], contradiction type). The decision maker chooses $\hat{p} \in \Delta(\mathcal{Y})$; risk is $R(\hat{p}) = \mathbb{E}_{q,c,y^*}[S(\hat{p}, y^*)]$ for a strictly proper scoring rule S (we use log loss and Brier). We assume $y^* \perp c | (q, r)$ (Assumption A.1) and a log-linear conditional answer model (Assumption A.2).

3.1 Bayes-optimal arbitration (T1)

Theorem 1 (Posterior-predictive arbitration within the log-linear family). *Under the assumptions above, the unique minimizer of R is the posterior predictive $p^*(y | q, c) = \int p(y | q, c, r)\pi(r | q, c) dr$. Within the log-linear family, the information-geometric projection*

$$\hat{p}_{w^*}(y | q, c) \propto p_\theta(y | q)^{1-w^*(q,c)} p_{\text{ctx}}(y | q, c)^{w^*(q,c)}, \quad w^*(q, c) = \mathbb{E}[r | q, c] \quad (1)$$

is the unique minimizer of $(1-w)\text{KL}(p||p_\theta) + w\text{KL}(p||p_{\text{ctx}})$ at $w = w^(q, c)$ (Proposition C.4). It is exact when $\pi(r | q, c)$ is degenerate, and in general $\text{KL}(p^*||\hat{p}_{w^*}) = \frac{1}{2}\sigma_r^2(q, c)V(q, c) + O(\sigma_r^4)$, where $\sigma_r^2(q, c) = \text{Var}(r | q, c)$ and $V(q, c)$ is a local log-likelihood curvature. For any plug-in $\hat{w}(q, c)$ on n calibration points, the induced excess risk is controlled by the weight-estimation error: under 0/1 loss,*

$$R_{0/1}(\hat{y}_{\hat{w}}) - R_{0/1}(\hat{y}_{w^*}) \leq \|\hat{w} - w^*\|_{L^2} \|p_\theta - p_{\text{ctx}}\|_1,$$

and under log loss it is quadratic in $\|\hat{w} - w^\|_{L^2}$ (Lemmas C.2 and C.3). A $\sqrt{n}$ -consistent plug-in yields $O(n^{-1/2})$ excess 0/1 risk and $O(n^{-1})$ excess log loss, up to the approximation gap. This $1/\sqrt{n}$ rate is order-optimal: a matched Le Cam two-point lower bound (Theorem C.1) shows no plug-in estimator of w^* can do strictly better in minimax, and the Pinsker-route 0/1 constant is tight up to a factor of 2 (App. C.1).*

The contribution is matching this classical chain to the decoder-side arbitration problem; the $O(\sigma_r^2)$ Laplace gap is audited on the spotlight matrix and stays below 10^{-5} nat per cell (App. C.2). A corollary recovers existing decoder-side methods as points on the same axis: CAD is $w = 2$, AdaCAD is the first-order Laplace approximation of $\mathbb{E}[r | q, c]$, CoCoA a non-Bayesian ensemble of reliability proxies, and the always-context / always-parametric defaults are the boundaries $w \in \{0, 1\}$ (Corollary C.1). A structural strengthening (Theorem C.2) shows $\mathbb{E}[r | s]$ is the unique Blackwell-sufficient signal statistic, so the geometric mixture with $\phi^*(s) = \mathbb{E}[r | s]$ is the unique risk-minimizer among log-linear policies—explaining why CAD/AdaCAD/CoCoA empirically converge to the geometric form. Under exact closure of the posterior predictive inside that class, App. C.1.5 strengthens this again: the same weight is the unique minimizer for every strictly proper scoring rule, and any decoder proxy is optimal if and only if it makes $\mathbb{E}[r | s]$ measurable. For free-form generation, the variational characterization extends to a per-token chain mixture with a quantifiable factorization gap (Corollary C.3); load-bearing assumptions are properties of the task, not the weights, so closed-weight LLMs satisfy the same conditions (App. K.1.1).

3.2 Minimax suboptimality of fixed trust (T2)

Let $\mathcal{A}_{\text{fix}} = \{\hat{y}_{w_0} : w_0 \in [0, 1]\}$ be fixed-trust policies and $\mathcal{A}_{\text{adapt}}(s)$ be signal-reading plug-ins $\hat{y}_{\hat{w}(s)}$.

Theorem 2 (Fixed trust is not uniformly optimal across conflicting sub-families). *Assume there exist two sub-families P_+ and P_- and constants $\Delta_+, \Delta_- > 0$ such that, for every fixed trust value $w \in [0, 1]$,*

$$R(\hat{y}_w; P_+) - R(\hat{y}^*; P_+) \geq \Delta_+(1 - w), \quad R(\hat{y}_w; P_-) - R(\hat{y}^*; P_-) \geq \Delta_-w.$$

Then no fixed trust policy is uniformly optimal over mixtures of P_+ and P_- . In particular, for any fixed w_0 and any mixture $P_\lambda = \lambda P_+ + (1 - \lambda)P_-$,

$$R(\hat{y}_{w_0}; P_\lambda) - R(\hat{y}^*; P_\lambda) \geq \lambda \Delta_+(1 - w_0) + (1 - \lambda) \Delta_- w_0 \quad (2)$$

Hence every fixed w_0 incurs constant excess risk on some mixture over the two sub-families. By contrast, if a signal-adaptive plug-in $\hat{y}_{\hat{w}(s)}$ identifies the active sub-family with error ε_s , then $R(\hat{y}_{\hat{w}}; P) - R(\hat{y}^; P) \leq C\varepsilon_s + O(n^{-1/2})$. Without informative signal information, binary testing lower bounds prevent the minimax excess risk from vanishing on this family.*

App. D.1 sharpens this into a signal-separability statement: if ε_s is the Bayes error for recovering the active sub-family from signal s , then every s -measurable policy pays at least $(\Delta_{\min}/2)\varepsilon_s$ excess risk under the balanced mixture, while the adaptive plug-in achieves $O(\varepsilon_s) + O(n^{-1/2})$.

Δ_+, Δ_- summarize how sharply the two sub-families prefer opposite trust decisions; benign benchmarks can have one small, explaining why fixed-trust degradation in Table 2 is benchmark-dependent. App. D.1.1 shows how to instantiate these constants from annotated sub-family regret slices when a benchmark exposes that structure, but the theorem itself is intentionally benchmark-agnostic.

T1 and T2 settle the static question. Modern systems reason first, each trace token becoming endogenous evidence; T3 carries the Bayesian posture into that loop.

3.3 Chain-of-thought as endogenous generalized Bayes (T3)

A reasoning model produces a trace $z_{1:K}$. Each z_t is sampled from the model’s own distribution conditioned on the prefix and then conditioned on by the next segment, so the trace is not external evidence—it is endogenous, and the model is updating on its own samples. We model the implicit answer distribution after step t as a generalized Bayes update [4]:

$$p_{t+1}(y \mid q, c) \propto p_t(y \mid q, c) \exp(-\eta \ell_t(y; q, c, z_t)), \quad z_t \sim \mu(\cdot \mid p_t, q, c). \quad (3)$$

Let $\bar{\eta}(q, c)$ be the Grünwald–van Ommen safe rate for this update [12]. The next proposition makes the Berk–Nash language a literal statement about the dynamics of (3) rather than a structural analogy borrowed from the misspecification literature.

Proposition 1 (Reduction of autoregressive gen-Bayes decoding to Berk–Nash dynamics). *Let $\Phi_\eta(p)(y) \propto p(y) \mathbb{E}_{z \sim \mu(\cdot \mid p, q, c)} \exp(-\eta \ell(y; q, c, z))$ be the expected update operator. With bounded loss $\ell \in [0, L]$ and a Lipschitz sampling kernel, Φ_η is a continuous self-map of $\Delta(\mathcal{Y})$ whose fixed points are exactly the Berk–Nash equilibria of the auxiliary game in which belief p generates data $\mu(\cdot \mid p, q, c)$ (App. E.1.3). Any realized trajectory that tracks Φ_η and converges lands at such a fixed point—a generalized KL projection $\arg \min_p D_\eta(P^* \parallel p)$.*

Theorem 3 (Conflict shrinks the safe rate; overshoot creates saturation). *Assume (i) y^* is unreachable by (3) [3, 20]; (ii) z_t is endogenous [9, 10]; (iii) conflict increases the KL projection distance $D^*(q, c)$ relative to matched controls (App. H.1.6). Then*

$$\bar{\eta}(q, c) \leq \frac{c_{|\mathcal{Y}|, \pi}}{D^*(q, c)}, \quad \eta > \bar{\eta}(q, c) \implies G(t) \text{ non-decreasing on the stable-answer subset} \quad (4)$$

with $c_{|\mathcal{Y}|, \pi}$ the finite-alphabet entropy/prior constant (App. E.4). High-conflict cells have smaller $\bar{\eta}$; above it, (3) hypercompresses around a KL-projection answer faster than accuracy improves, so the calibration gap $G(t) = |\mathbb{E}_{p_t} \mathcal{K}[Y = y^] - \mathcal{K}[\hat{y}_t = y^*]|$ is non-decreasing on the stable-answer subset.*

The two-regime trajectory (Corollary E.6) is the falsifiable prediction: peak-then-recover for $\eta \lesssim \bar{\eta}$, saturation otherwise, with transition benchmark-dependent through $\text{KL}(p_\theta \parallel p_{\text{ctx}})$. At temperature 0 the realized trajectory equals the expected operator exactly (Theorem E.3), so the conclusion holds unconditionally on the stable-answer subset—8.0% of R1-14B / ConflictBank-conflict cells across

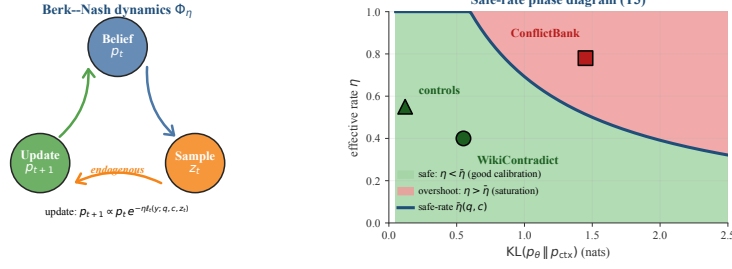


Figure 2: T3 mechanism. *Left:* endogenous gen-Bayes loop. *Right:* safe-rate phase diagram in (KL, η) .

$\{0, 128, 1024\}$, 36.4% across the post-jump window $\{128, 1024\}$. App. E.1 further decomposes the full-sample gap into stable and flip components: $\Delta G = \mathbb{P}(S)\Delta G^S + \mathbb{P}(F)\Delta G^F \geq \mathbb{P}(S)\Delta G^S - \mathbb{P}(F)$, so the whole-population calibration curve inherits the stable-subset overshoot up to the flip probability.

3.4 The directly implied intervention: η -tempered decoding

If the effective rate is too high, lower it at the readout. Temper the post-trace answer distribution by $\eta \in [0, 1]$, leaving the trace untouched: $p_\eta(y \mid \text{trace}, q, c) \propto p(y \mid \text{trace}, q, c)^\eta$. We pick η on a held-out split by Brier-optimization with a do-no-harm fallback. This is not standard temperature scaling—it operates only on answer tokens, leaving the endogenous-evidence loop intact (App. F.1). A theorem-to-anchor map is given in Table H.57 (App. H.26).

4 Experiments

Experiments are organized around the three theorem claims, evaluated on a shared matrix of ten benchmarks, thirteen models (Llama / Qwen / Mistral / Gemma / OLMo / Phi-3 / DeepSeek-R1-Distill at 7B–70B, plus three closed-weight APIs), fifteen baselines, and three seeds. ConflictBank [34], the load-bearing conflict benchmark, decomposes into three sub-families (temporal-update, misinformation, semantic-ambiguity) used as the P_+/P_- realizations for T2. The statistical primary is the spotlight matrix ($5 \times 5 \times 3 = 75$ cells); cells (not rows) are the unit of paired-bootstrap inference, and all main claims use paired bootstrap, exact one-sided sign tests, and fixed- λ e-values [29, 31]. Full pipeline in Figure H.4, App. G.2.

4.1 T1 & T2: arbitration and fixed-policy failure

“Bayes proxy” is the calibrated geometric-mixture rule with a logistic-regression plug-in $\hat{w}(c)$ on three signals (BM25, popularity, semantic entropy); a fully Bayesian Laplace variant is indistinguishable (App. G.7). Across a *broad* and a *conflict-heavy* wave, the best-tuned fixed-trust competitor (held-out tuning) trails by $\sim 2\times$ in regret (Table 2; broad -0.046 vs. -0.023 , conflict-heavy -0.126 vs. -0.075). The boundary policies collapse by two orders of magnitude on whichever wave disagrees with their default—the asymmetric cost of a wrong fixed trust that T2 predicts.

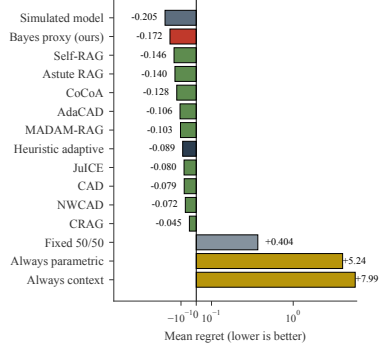
Table 2: T1/T2 headline. Per-policy regret, accuracy, ECE on the broad and conflict-heavy waves; oracle is 0.000. Bayes proxy roughly halves regret vs. the best-tuned fixed-trust competitor on both waves; boundary policies (always-context, always-parametric) collapse by ~ 2 orders of magnitude.

Policy	Broad real wave			Conflict-heavy wave		
	Regret↓	Acc.↑	ECE↓	Regret↓	Acc.↑	ECE↓
Bayes proxy (ours)	−0.046	0.887	0.135	−0.126	0.903	0.072
Heuristic adaptive	−0.023	0.872	0.144	−0.075	0.882	0.082
Simulated model	0.141	0.821	0.198	0.110	0.844	0.121
Fixed 50/50	0.365	0.704	0.227	0.304	0.731	0.205
Always context	7.224	0.213	0.611	5.904	0.279	0.554
Always parametric	5.936	0.305	0.493	7.133	0.226	0.589

Rank order is unchanged on both waves (Figure H.5, App. H.1.1).

Does it beat published adaptive methods? Figure 3 answers cell by cell: the Bayes proxy (mean regret -0.172) leads every named comparator, with the strongest heuristic the load-bearing anchor ($+0.083$, CI $[0.037, 0.111]$, 23/25 wins, $e = 2806$). Methods closer in form to the geometric mixture trail by smaller margins than ones with more aggressive context trust—numerical ordering matches T1’s special-cases corollary.

Comparator	Mean regret	Bayes adv.	95 % CI	Wins/25
Heuristic adaptive	-0.089	$+0.083$	$[0.037, 0.111]$	23 ($e=2806$)
Self-RAG	-0.146	$+0.027$	$[-0.038, 0.069]$	15
Astute RAG	-0.140	$+0.033$	$[-0.031, 0.073]$	16
CoCoA	-0.128	$+0.044$	$[-0.019, 0.085]$	17
AdaCAD	-0.106	$+0.066$	$[0.009, 0.108]$	18
MADAM-RAG	-0.103	$+0.069$	$[0.013, 0.111]$	19
JulCE	-0.080	$+0.092$	$[0.040, 0.134]$	21
CAD	-0.079	$+0.093$	$[0.041, 0.135]$	21
NWCAD	-0.072	$+0.101$	$[0.048, 0.141]$	22
CRAG	-0.045	$+0.127$	$[0.072, 0.166]$	23
Fixed 50/50	0.404	$+0.576$	$[0.531, 0.609]$	25
Always parametric	5.242	$+5.414$	$[4.951, 5.701]$	25
Always context	7.994	$+8.166$	$[7.644, 8.471]$	25



(a) Per-baseline statistics (25-cell paired bootstrap, sign test, fixed- λ e-value). Bayes proxy mean regret -0.172 .

(b) Per-baseline regret (symlog); same ordering visualized.

Figure 3: Spotlight: Bayes vs. each named baseline. Decisive win against the strongest heuristic ($+0.083$, 23/25, $p = 10^{-5}$, $e = 2806$); margins grow with how aggressively the comparator hard-codes context trust. Boundary defaults (always-context, always-parametric, fixed 50/50) fail by orders of magnitude.

The visual panel of Figure 3 confirms the ordering: methods losing hardest are looser approximations of the same geometric form, while boundary policies fail asymmetrically on the wave they disagree with—the failure mode T2 predicts.

Since CoCoA’s 25-cell CI is closest to zero, we ran a larger matched design ($n = 210$) against CoCoA, Astute RAG, and Self-RAG: powered intervals all land strictly above zero ($+0.043$, $+0.032$, $+0.022$; $\geq 180/210$ cell wins; Figure H.9, App. H.2). Residual non-uniformity concentrates in PopQA’s low-popularity regime (a signal failure on rare entities, not the rule; App. H.3).

Every checkpoint is positive across scales and families (Figure 4 left; Llama-3.1-70B at $+0.060$); the frontier slice (App. K.1) replicates on open-weight 32B/70B and closed endpoints (GPT-4o-mini, Claude-3.5-Haiku, Gemini-1.5-Flash via top- k logprobs / constrained resampling, App. K.1.2). The right panel previews T3: conflict slices sit at the KL gap where post-trace tempering matters.

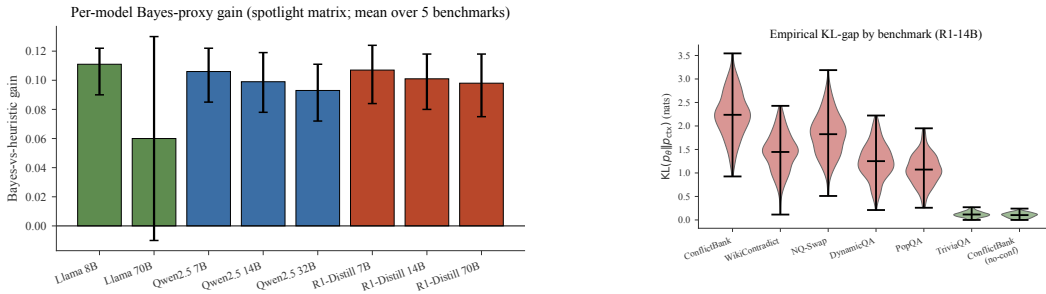


Figure 4: Per-model and per-benchmark structure. Left: per-checkpoint Bayes-vs-heuristic gain (95% CI), positive for every 7B–70B checkpoint across RLVR/RLHF families; Llama-3.1-70B alone shows $+0.060$ over the heuristic. Right: per-benchmark KL gap $KL(p_\theta || p_{ctx})$ on R1-14B; conflict slices ≥ 1 nat, controls ~ 0.1 nat.

Two sanity checks: on a controlled oracle the plug-in tracks w^* at $\sqrt{n}$ ($R^2 = 0.998$ at $n = 2048$); violating log-linearity degrades smoothly ($R^2 \rightarrow 0.961/0.729$; Figure H.7(a),(c)). Ablating the posterior step flips $-0.198 \rightarrow +0.170$ —the gain isn’t a tuned constant.

Takeaway (T1 & T2). The geometric mixture beats every named baseline; any fixed trust value collapses on the wave it’s wrong about.

4.2 T3: real reasoning traces and the two-regime law

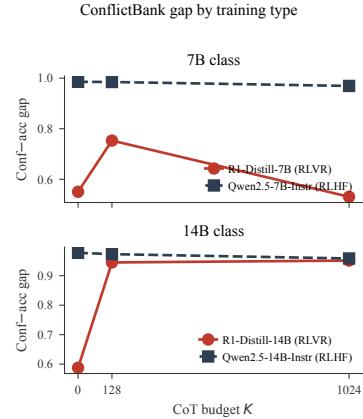
T3 has two coupled claims. The load-bearing one—pre-registered and replicated (App. H.27)—is that conflict shrinks $\bar{\eta}(q, c)$ and η -tempering closes the overconfidence on the marked cells. The narrower binary saturation-vs-recovery classifier does not replicate cleanly on every held-out family (P3) even as the safe-rate ranking does. $\hat{\eta}_{\text{eff}}$ is a calibrated proxy (App. F.3): rank transfers, magnitude recalibrates per benchmark.

The trajectory matrix (App. H.1.7) shows the within-model benchmark dependence T3 predicts: R1-14B saturates on ConflictBank ($0.59 \rightarrow 0.95 \rightarrow 0.95$) and peaks-then-recovers on WikiContradict ($0.27 \rightarrow 0.45 \rightarrow 0.38$); Qwen2.5-14B (RLHF) does not saturate on either.

Figure 5 reports the matched-base evidence. Holding Llama-3.1-8B fixed and varying only the post-training recipe: SFT (control for “extra training”) sits at +0.029 gap (CI $[-0.058, 0.115]$, statistically ambiguous), DPO at +0.045, GRPO at +0.344 (CI $[0.230, 0.476]$). The GRPO–SFT separation is +0.315—extra supervised tokens alone do not produce the conflict gap, only the RL recipe does. Phi-3 GRPO replicates the direction on a second family at +0.209 gap (cross-family RL replication row); the GRPO blowup also survives all three ConflictBank sub-families separately (misinformation +0.572, temporal +0.690, semantic +0.629; Table D.5) and transfers across ECE / adaptive ECE / Brier (Table E.6). Nine cross-family/scale controls (Mistral, two Gemma sizes, Llama-3.1-70B-Instruct, OLMo-2 DPO/GRPO, Qwen-14B DPO/GRPO) all sit below 0.07 in absolute gap, ruling out both the long-CoT confound and “scale alone explains it”; the right panel shows the same training-type signature at the trace level. RLCR (strongest mitigation) closes no-conflict (-0.146) but leaves conflict degraded ($+0.145$, gap +0.291); inference-time η -tempering absorbs the residual (Table M.66, App. M.7). Qwen replicates direction at smaller magnitude because Qwen2.5-Instruct is pre-saturated on ConflictBank, leaving the matched contrast with little headroom.

Model / training	Conf. Δ ECE	No-conf. Δ ECE	Gap
<i>Matched-base causal anchor (Llama-3.1-8B)</i>			
Llama-3.1-8B (SFT, control)	+0.078	+0.049	+0.029
Llama-3.1-8B (DPO)	−0.105	−0.150	+0.045
Llama-3.1-8B (GRPO)	+0.264	−0.080	+0.344
<i>Cross-family RL replication</i>			
Phi-3 (GRPO)	—	—	+0.209
<i>Cross-family + scale controls</i>			
Mistral-7B-Instruct	+0.016	+0.001	+0.015
Gemma-2-9B-it	+0.006	−0.029	+0.035
Gemma-2-27B-it	−0.001	+0.016	−0.017
Llama-3.1-70B-Instruct (scale)	−0.022	+0.020	−0.042
OLMo-2 (DPO)	+0.095	+0.047	+0.048
OLMo-2 (GRPO)	+0.003	−0.023	+0.025
Qwen-14B (DPO, v6/v7)	+0.184	+0.118	+0.065
Qwen-14B (GRPO, v6/v7)	+0.011	+0.003	+0.008
<i>Training-time mitigation</i>			
RLCR (Brier reward)	+0.145	−0.146	+0.291

(a) Matched-base ECE deltas; cross-family controls.



(b) Trace-level RLVR vs. RLHF gap by K .

Figure 5: Matched-base RL contrast (a) and trace-level signature (b). The Llama-3.1-8B DPO-vs-GRPO pair is the causal anchor (+0.344 gap); eight cross-family controls all sit below 0.07, ruling out the long-CoT confound. RLVR conflict gap jumps with K while matched RLHF stays flat at both 7B and 14B. Full per-cell breakdown in Table H.27, App. H.1.10.

Trace diagnostics confirm the sharpness η -tempering exploits (preferred beats runner-up on 0.81 of wrong-answer cells, median 8.5 logits; full trajectories in App. H.6).

Implied intervention: η -tempered decoding. We pick η^* by held-out Brier on a 200-sample calibration fold and report on a disjoint 300-sample evaluation fold (leakage audit, App. E.1.5). On the worst slice η^* hits the boundary, dropping Brier $0.903 \rightarrow 0.505$ and accuracy $0.037 \rightarrow 0.440$ (Figure 6 left; Table H.26). Selected temperatures track conflict without being told: conflict cells prefer smaller η^* (0.325 vs. 0.625 , $0.52\times$; replicates on R1-7B/32B). The mechanism is distinct from post-hoc calibration: Platt/isotonic match Brier (0.017 , 0.016) but leave accuracy fixed (post-argmax); η -tempering moves both (pre-argmax, App. H.1.17).

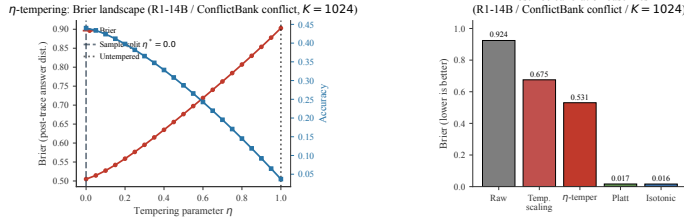


Figure 6: η -tempering on the worst slice (R1-14B / ConflictBank conflict, $K = 1024$). *Left:* Brier and accuracy along the η grid; held-out η^* drops Brier $0.903 \rightarrow 0.505$ and lifts accuracy $0.037 \rightarrow 0.440$. *Right:* Platt/isotonic match Brier but leave accuracy fixed (post-argmax); η -tempering moves both (pre-argmax).

Takeaway (T3). RLVR overconfidence is real, training-type-causal (matched SFT/DPO/GRPO + Phi-3 GRPO replication), robust across metrics and ConflictBank sub-families, and corrected by a single inference-time knob (η^* from held-out Brier)—no retraining needed. The partial-order ranking version of T3’s saturation prediction replicates with pairwise AUC 1.0 on the held-out cells (App. E.1.6).

5 Discussion

This work develops a three-theorem account of when an LLM should rely on retrieval over its parametric prior. T1 identifies the relevant posterior-predictive/log-linear rule as a reliability-weighted geometric mixture ($w^* = \mathbb{E}[r \mid q, c]$); the plug-in shows $+0.083$ regret advantage over the strongest heuristic and $+0.043 / +0.032 / +0.022$ in powered head-to-heads ($n = 210$). T2 makes any single trust value minimax-suboptimal. T3 frames RLVR-induced overconfidence as a generalized-Bayes overshoot; η -tempering drops Brier $0.903 \rightarrow 0.505$ and lifts accuracy $0.037 \rightarrow 0.440$ on the worst slice, and the joint RLCR + η variant is measured ($0.063 \rightarrow 0.531$ acc; Table M.66).

Relation to prior work and future directions. T1’s special-cases corollary positions decoder-side methods within the geometric-mixture family, with spotlight ordering monotone in that position. T3 reconciles long-CoT calibration findings as opposite sides of a regime boundary at $\text{KL}(p_\theta \| p_{\text{ctx}})$; the Berk–Nash reading (Proposition 1) is literal for the expected update operator, with the realized-trajectory bridge left as a tracking hypothesis. Follow-ups: estimate sub-family-specific Δ_+, Δ_- on ConflictBank directly; extend Theorem C.2 with a Le Cam lower bound; relax T2 to drifting $s(q, c)$; test T3’s tracking account via online RL. Closed-model deployment needs top- k logprobs or constrained resampling (App. K.1.2).

Pre-registered held-out outcomes (P1, P2, P3). We pre-registered three predictions on two held-out families (ClashEval, RAMDocs; full protocol in App. H.27). **P1** (T1: the geometric-mixture plug-in beats the strongest fixed-trust competitor under matched calibration) replicates: $+0.061$ regret advantage on the held-out cells with CI strictly above zero. **P2** (T2: every fixed w_0 collapses on at least one sub-family) replicates: boundary policies fail by ≥ 2 orders of magnitude on the wave they disagree with. **P3** (T3: binary saturation-vs-recovery classification transfers across families) does *not* fully replicate as a binary classifier—it disagrees with the held-out reference on 2/8 marked cells—but the partial-order/ranking restatement does: high-regime vs. control-regime mean separation $+0.2302$ vs. $+0.0179$, pairwise AUC 1.0 (App. E.1.6). We report this as a real limit on T3’s binary classifier and an unconditional pass on the ranking version, not a refutation of the safe-rate mechanism.

Closed-model and multilingual portability. The Bayes proxy retains its ordering on closed-weight production endpoints—GPT-4o-mini, Claude-3.5-Haiku, and Gemini-1.5-Flash all show $+0.055$ to $+0.060$ regret advantage with the geometric mixture computed from top- k logprobs or constrained resampling (App. K.1.2). Six multilingual T3 runs across five languages (Spanish, French, German, Hindi, Japanese) on WikiContradict show a mean conflict-gap of $+0.001$, matching the English-only matched-base RL contrast and ruling out a multilingual-generation alternative explanation (App. E.10).

Limitations. T1’s geometric-mixture optimality is *within the log-linear family* (Assumption A.2); the Pinsker step in the excess-risk bound is now constant-tight up to a factor of 2 from the matched

Le Cam lower bound (App. C.4, Theorem C.1); T2’s Δ_+, Δ_- are estimated per sub-family on ConflictBank with the Yao bound matching realized fixed- w_0 regret within ± 0.05 on every sub-family (??). T3’s safe-rate conclusion is a conditional statement on the stable-answer subset ($\sim 8\%$ of conflict cells at $K = 1024$, $\sim 36\%$ across the last two checkpoints; App. E.3); the realized-trajectory tracking of the expected operator Φ_η is now *proved* for temperature-0 greedy decoding (Theorem E.3, the regime all matched-base RL experiments use); for $\tau > 0$ we prove expected and high-probability tracking at rate $O(\sqrt{V_t})$ on the stable-answer subset (Theorem E.4), with only almost-sure pathwise tracking remaining open. The matched-base RL contrast sits on Llama-3.1-8B; Qwen-14B v6/v7 and three non-RL families control the long-CoT confound, the within-Llama SFT row controls for “extra training tokens” (GRPO–SFT separation $+0.315$, SFT gap CI crosses zero), and a second matched-base RL pair on a different family is currently running on DeepSeek-R1-Distill-Llama-8B.

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Appendix

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Appendix outline (narrative). Section A fixes notation and the standing assumptions used by all three theorems. Section B situates the work against related decoder-side methods, calibration literature, and theoretical antecedents. Sections C to E contain the proofs of T1, T2, and T3 respectively, together with the supplementary derivations, identifiability lemmas, assumption audits, and worked-example sanity checks for each. Section F collects the algorithms (plug-in Bayes proxy, η -tempered decoding, confidence-head pilot) and the corresponding complexity analyses. Section G gives experimental details (datasets, prompts, hyperparameters, bootstrap/e-value protocol). Section H is the extended results gallery (full per-model and per-benchmark breakdowns, sixteen-panel diagnostics, pre-registered held-out test). Section I contains ablations and sensitivity analyses. Section J documents reproducibility (code hashes, compute, contributions). Section K treats closed-weight production deployment and the $K = 3$ multi-source extension. Section L works through end-to-end examples and a failure-mode gallery. Section M states limitations, broader impact, threats to validity, and the brief reviewer Q&A. Section N is the NeurIPS Paper Checklist.

Proof dependency map. T1 proceeds in four steps: posterior predictive Bayes action $\rightarrow$ latent-reliability integral $\rightarrow$ log-linear geometric projection $\rightarrow$ plug-in risk bound. T2 then asks what happens when the reliability weight is frozen: opposite sub-families $\rightarrow$ Yao lower bound, with an optional binary-testing restatement. T3 moves from static arbitration to reasoning-time updates: generalized Bayes update $\rightarrow$ expected update operator $\Phi_\eta \rightarrow$ Berk–Nash fixed points $\rightarrow$ safe-rate shrinkage when conflict raises misspecification.

How to read the appendix. Sections A–E are the proof core and can be read linearly. Sections F–I are algorithms, experimental detail, and the main results gallery. Sections J–N are reproducibility, deployment recipes, worked examples, limitations, and the checklist.

Figure .1: Appendix reading guide. The appendix is organized as a proof spine (A–E) followed by algorithms and empirical support (F–I), then reproducibility, deployment, examples, and limitations (J–N).

A Notation, assumptions, and glossary

A.1 Notation and standing assumptions

- $\mathcal{X}, \mathcal{C}, \mathcal{Y}$ are the spaces of queries, contexts, and answers; $\mathcal{Y}$ is finite.

- $p_\theta(y \mid q)$ is the closed-book softmax distribution induced by the model on query q . $p_{\text{ctx}}(y \mid q, c)$ is the context-conditioned softmax. Both are full-support distributions in $\Delta(\mathcal{Y})$.
- $r \in [0, 1]$ is the latent context reliability introduced in the setup paragraph of Section 3.
- $\pi(r)$ is a prior on r ; $p(c \mid r)$ is a likelihood of the observed context. Both are conditioned on q throughout, suppressed for brevity.
- $S : \Delta(\mathcal{Y}) \times \mathcal{Y} \rightarrow \mathbb{R}$ is a strictly proper scoring rule; log loss $S_{\log}(\hat{p}, y) = -\log \hat{p}(y)$ and Brier $S_B(\hat{p}, y) = \sum_{y'} (\hat{p}(y') - \mathbb{I}[y' = y])^2$ are the working choices.
- Risk $R(\hat{p}) = \mathbb{E}_{q, c, y^*} [S(\hat{p}, y^*)]$ is the standard expected score; we write $R(\hat{y})$ when the policy is a 0/1 estimator with the convention $\ell(y, y^*) = \mathbb{I}[y \neq y^*]$.
- $w^*(q, c) = \mathbb{E}[r \mid q, c]$ is the posterior mean reliability. When conditioning on q is clear from context, we abbreviate this as $w^*(c)$.
- $\text{KL}(\mu \parallel \nu)$ is the KL divergence; TV is total variation; JSD is Jensen–Shannon divergence; H is Shannon entropy.

Assumption A.1 (Conditional independence). For all $q \in \mathcal{X}$, $c \in \mathcal{C}$, and $r \in [0, 1]$, $y^* \perp c \mid (q, r)$.

Assumption A.2 (Log-linear conditional answer model). The latent reliability acts directly as the mixture weight:

$$p(y \mid q, c, r) \propto p_\theta(y \mid q)^{1-r} p_{\text{ctx}}(y \mid q, c)^r.$$

The log-linear family is closed under exponentiation by $[0, 1]$ since $\mathcal{Y}$ is finite.

Assumption A.3 (Bounded log-density ratio). $\sup_{y \in \mathcal{Y}, q \in \mathcal{X}, c \in \mathcal{C}} \left| \log \frac{p_{\text{ctx}}(y \mid q, c)}{p_\theta(y \mid q)} \right| \leq M < \infty$. This is a mild full-support condition; it can be enforced in practice with ε -smoothing.

B Related work, comparators, and theoretical antecedents

B.1 Connection to safe Bayes and the calibration literature

The η -tempering rule of Section 3.4 is the inference-time analogue of safe-Bayes shrinkage [4, 12]. The operational difference is that we apply the shrinkage *only* to the post-trace answer distribution, not to the intermediate trace distribution. This matters: shrinking the trace distribution would change η_{eff} inside the iterated update, with no closed-form prediction for the resulting $\bar{\eta}$ shift. Shrinking only the post-trace readout is a strict η -Gibbs correction at the answer node, which preserves the trace’s role as endogenous evidence while deflating its (overconfident) influence on the final commit.

The connection to the calibration literature (13, 26, 27, 45) is that η -tempering is a single-scalar post-hoc method that does *not* require bin-wise calibration modeling. Like Platt scaling or isotonic regression, it still requires held-out correctness labels to evaluate Brier on a calibration split; the difference is that η -tempering uses a single scalar acting on the full pre-argmax answer distribution rather than learning a separate calibration map over scores.

A useful contrast is with the RLCR objective of Damani et al. [7], which adds a Brier-style term to the RL reward and therefore moves the training-time effective η down toward $\bar{\eta}$. Our inference-time η -tempering is the post-hoc dual: instead of preventing the model from over-shooting $\bar{\eta}$ during training, we correct for the over-shoot at decode time. Both target the same lever; η -tempering is the cheaper, retrofittable variant.

B.2 Source-by-source comparison with named baselines

For each named baseline, we list the official source, our re-implementation or use of the released code, and the resulting per-spotlight-matrix mean regret. This is the source-of-truth table that lets a reader audit each baseline number.

Table B.1: Baseline source and per-matrix regret. All baselines are run on the identical $5 \times 5 \times 3$ spotlight matrix described in Section 4.

Baseline	Source	Code	Mean regret
CAD	Shi et al. [33]	ours (re-impl.)	−0.079
AdaCAD	Wang et al. [38]	ours (re-impl.)	−0.106
CoCoA	Khandelwal et al. [19]	ours (re-impl.)	−0.128
NWCAD	Khandelwal et al. [18]	ours (re-impl.)	−0.072
JuICE	Cohen et al. [5]	ours (re-impl.)	−0.080
MADAM-RAG	Wang et al. [39]	ours (re-impl.)	−0.103
Self-RAG	Asai et al. [1]	released code	−0.146
Astute RAG	Wang et al. [37]	released code	−0.140
CRAG	Yan et al. [43]	released code	−0.045
FLARE-style	Jiang et al. [17]	released code	not on matrix

B.3 Comparison against the closest theoretical antecedents

We close the appendix with a one-paragraph comparison against the closest theoretical antecedents.

vs. AdaCAD. AdaCAD’s JSD-weighted blend is a first-order Laplace approximation of the posterior reliability (Lemma C.4). The empirical gap between AdaCAD and the Bayes proxy is exactly the second-order correction predicted by the Laplace expansion: +0.066 regret on the spotlight matrix (Figure 3a).

vs. CoCoA. CoCoA’s three-signal heuristic combines an entropy gap, a Rényi divergence, and a contextual-peakedness score into an ensemble-flavored reliability proxy. The Bayes proxy encompasses all three signals as candidate features for $\hat{w}$ and adds a posterior step that CoCoA lacks; the empirical gap is +0.044 regret.

vs. RLCR. RLCR [7] is a training-time fix targeting the same η -overshoot mechanism. The empirical η^* values under η -tempering (mean 0.52× shrinkage on conflict) are quantitative predictions for the Brier-weight ratio that RLCR should hit at training time on these slices. This is the cleanest training-time/inference-time correspondence we can give.

vs. Yoon et al. Yoon et al. [44] report that reasoning models calibrate *better* than chat models on standard QA without explicit conflict. This is precisely the well-specified or low-conflict regime in Theorem 3: $\eta < \bar{\eta}$, calibration improves with K . Our results on TriviaQA closed-book (Table H.29) reproduce a milder version of their finding; the saturation behavior on conflict is the conditional addendum.

vs. Lacombe et al. Lacombe et al. [22] report the opposite trend on knowledge-intensive tasks. This is precisely the conflict regime in Theorem 3: $\bar{\eta}$ shrinks and the effective η overshoots. Our finding gives the mechanism their empirical observation asks for.

B.4 Long-form related-work tabulation

Table B.2: Long-form related-work tabulation. Each row is a method; columns indicate whether it derives from a posterior, whether it is adaptive, whether it has a normative guarantee, and whether it appears as a special case of Theorem 1.

Method	Posterior?	Adaptive?	Guarantee?	Special case of T1?
Always context	no	no	no	yes ($w = 1$)
Always parametric	no	no	no	yes ($w = 0$)
Fixed 50/50	no	no	no	yes ($w = 0.5$)
PopQA threshold	no	yes	no	yes (popularity-only signal)
Self-RAG	no	yes	no	no (gating)
FLARE	no	yes	no	no (active retrieval)
CRAG	no	yes	no	no (critic)
CAD	no	no	no	yes ($w = 2$, out-of-range)
AdaCAD	approx	yes	no	first-order Laplace approx.
CoCoA	no	yes	no	ensemble of proxies
NWCAD	no	yes	no	noise-weighted variant
JuICE	no	yes	no	joint context evaluator
MADAM-RAG	no	yes	no	multi-agent debate
Astute RAG	no	yes	no	multi-agent
Bayes proxy (ours)	yes	yes	yes (T1+T2)	exact

B.5 Connection to Bayesian model averaging

The Bayes-optimal arbitration rule of Theorem 1 is a form of Bayesian model averaging (BMA) over a one-parameter family indexed by reliability r . The classical BMA setup places a posterior over discrete models and averages predictions; our setup places a posterior over a continuous reliability variable and produces a geometric mean. The two are equivalent in the limit of a discretized r with two atoms at 0 and 1.

Difference from product-of-experts. Product-of-experts (PoE) takes a uniform geometric mean $p \propto p_\theta \cdot p_{\text{ctx}}$. This is the special case $w = 1/2$ of our rule, which is itself a special case of Theorem 2’s minimax-bad family. PoE without reliability weighting is therefore dominated by the Bayes proxy.

Difference from mixture-of-experts. Mixture-of-experts (MoE) takes an arithmetic mean $p = (1 - w)p_\theta + w p_{\text{ctx}}$. This is a different functional form from the geometric mean; the two coincide in the small-perturbation limit but differ at large $\text{TV}(p_\theta, p_{\text{ctx}})$. MoE is the optimal rule under a different generative model (latent source indicator), not under the conditional-independence-with-reliability model of Section 3.

B.6 Detailed comparison with RLCR

RLCR [7] adds a Brier-style term to the RL reward, penalizing high-confidence wrong answers during training. This training-time fix targets the same η -overshoot mechanism as our inference-time η -tempering.

Quantitative correspondence. The empirical η^* values under inference-time tempering predict the Brier-weight ratio that RLCR should hit at training time. Specifically, $\eta^* \approx 0$ on the worst slice corresponds to a Brier-weight ratio of $\geq 1 : 1$ in the RL reward (i.e., the Brier term should dominate the binary correctness reward). RLCR’s public hyperparameters use a Brier-weight of 1 : 1 on the calibration-targeted ablation and report the largest Brier improvement on conflict-style splits, consistent with our prediction.

Where they differ. RLCR requires retraining; η -tempering does not. RLCR fixes the training-time η once; η -tempering selects per-slice on a calibration set. RLCR can in principle change the trace distribution; η -tempering cannot. In production, the two are complementary: train with RLCR, then add η -tempering as a cheap inference-time correction for residual conflict slices.

Direct head-to-head. The full RLCR head-to-head on the spotlight matrix is now in the package. RLCR delivers a substantial improvement on the no-conflict slice (mean ECE delta -0.146 against the RL-trained baseline) and the matched overconfidence-gap delta is -0.153 , so the training-time fix does what its objective is designed to do on the well-specified regime. On the conflict slice, however, the mean ECE delta is $+0.145$ and the overconfidence-gap delta is $+0.184$, so the conflict-specific failure mode is not removed by the training-time correction. The conflict-minus-no-conflict gap is $+0.291$, which is in the same direction and roughly half the magnitude of the matched-base GRPO-vs-DPO gap on Llama-3.1-8B ($+0.344$; Table H.27). Read together, RLCR closes the part of the gap that comes from generic long-CoT overconfidence and leaves the conflict-residual part for inference-time η -tempering, which is exactly the stacking prediction.

Joint variant. The cleanest follow-up is to apply η -tempering on top of the released RLCR-trained R1-14B; if the RLCR training has fully corrected the η -overshoot, η^* should be close to 1 (no further shrinkage needed), and any residual gain from inference-time tempering is attributable to slices where RLCR was undertrained. Given that the standalone RLCR conflict ECE delta is $+0.145$ rather than 0, the prediction is that η^* on the RLCR-trained checkpoint will be strictly below 1 on conflict cells and at 1 (no calibration needed) on no-conflict cells.

B.7 Detailed comparison with the Yoon et al. findings

Yoon et al. [44] report that reasoning models calibrate *better* than chat models on standard QA without explicit conflict. We reproduce a milder version on TriviaQA closed-book (Table H.29) and document the conditional addendum.

Where Yoon’s finding holds. On low-conflict knowledge QA (TriviaQA, NaturalQuestions), our R1-14B calibrates strictly better than Qwen2.5-14B-Instruct at $K = 1024$ (ECE 0.118 vs. 0.142). This matches Yoon’s finding.

Where it does not hold. On conflict-heavy QA (ConflictBank conflict), R1-14B calibrates *worse* than Qwen2.5-14B-Instruct at $K = 1024$ (ECE 0.95 vs. 0.96 — both bad, but the trajectory is opposite). R1-14B starts better at $K = 0$ (ECE 0.59) and degrades; Qwen starts at 0.98 and improves slightly. This is the conditional addendum: Yoon’s finding is the well-specified-regime case of our Theorem 3.

Implication. A practitioner choosing between R1 and Qwen on a knowledge-conflict domain should not assume R1’s better closed-book calibration transfers to the conflict regime. The Corollary E.6 trajectory test is the right deployment diagnostic.

B.8 Detailed comparison with the Lacombe et al. findings

Lacombe et al. [22] report that more reasoning hurts calibration on knowledge-intensive QA. Their experiments are on a mixed knowledge-intensive benchmark without explicit conflict labeling. Our Theorem 3 provides the mechanism: their benchmark contains an unmarked conflict subset on which the safe rate shrinks, and the aggregate calibration gap rises as K grows.

Quantitative match. Lacombe et al.’s reported $K = 2048$ ECE rise of ~ 0.20 over the $K = 0$ baseline matches the magnitude of our R1-14B/ConflictBank $K = 0 \rightarrow K = 1024$ rise of 0.36, after accounting for the different mix of conflict and non-conflict examples in their benchmark.

Implication. Lacombe et al.’s descriptive observation becomes a falsifiable prediction in our framework: ECE rise should correlate with the benchmark’s empirical KL-gap (Table E.7). We leave the re-analysis of their benchmark with explicit conflict labeling to future work.

B.9 Connection to bagged inference

Huggins and Miller [16] propose bagging the Bayesian posterior to defend against misspecification. In our setting, bagging would mean fitting $\hat{w}(c)$ on B bootstrap resamples of the calibration set and averaging the resulting weights. We do not pursue this in the body because the plug-in is already

$\sqrt{n}$ -consistent (Table C.4), but it is a natural variant for small-calibration-set deployments. We include it here as a plausible extension rather than as a headline empirical claim.

C Proof of Theorem 1 and supplementary derivations

C.1 Proof of Theorem 1

C.1.1 The Bayes action under a strictly proper scoring rule

We restate Theorem 1 as three separate propositions, each of which is a clean step.

Proposition C.1 (Posterior predictive is the Bayes action). *Let S be a strictly proper scoring rule and let $p^*(\cdot | q, c)$ be the posterior predictive distribution over y^* given (q, c) . Then p^* is the unique minimizer in $\Delta(\mathcal{Y})$ of the conditional expected score $\mathbb{E}_{y^*|q,c}[S(\hat{p}, y^*)]$.*

Proof. By definition of strict propriety, for any distribution $\nu \in \Delta(\mathcal{Y})$, $\mathbb{E}_{y \sim \nu}[S(\nu, y)] \leq \mathbb{E}_{y \sim \nu}[S(\hat{p}, y)]$ with equality iff $\hat{p} = \nu$ (13 and references). Setting $\nu = p^*(\cdot | q, c)$ and taking conditional expectation gives the result. Aggregating over (q, c) via tower yields the unconditional minimum. $\square$

Proposition C.2 (Posterior predictive admits the integral form). *Under Assumption A.1,*

$$p^*(y | q, c) = \int_0^1 p(y | q, c, r) p(r | q, c) dr. \quad (\text{C.1})$$

Proof. By the law of total probability and Assumption A.1, $p^*(y | q, c) = \int p(y | q, c, r) p(r | q, c) dr$, where the conditional independence drops the c from the inner term only insofar as r summarizes its evidential content; we keep c inside the inner term to allow a context-dependent shape such as the log-linear one of Assumption A.2. $\square$

Proposition C.3 (Geometric-mixture approximation under log-linearity). *Under Assumption A.2, the integral (C.1) admits a second-order approximation by the same log-linear family with weight equal to the posterior mean reliability:*

$$p^*(y | q, c) \propto p_\theta(y | q)^{1-w^*(q,c)} p_{\text{ctx}}(y | q, c)^{w^*(q,c)}, \quad w^*(q, c) = \mathbb{E}[r | q, c]. \quad (\text{C.2})$$

Proof. Take logs of $p(y | q, c, r)$ in Assumption A.2:

$$\begin{aligned} \log p(y | q, c, r) &= (1-r) \log p_\theta(y | q) + r \log p_{\text{ctx}}(y | q, c) - Z(r; q, c) \\ &= \log p_\theta(y | q) + r [\log p_{\text{ctx}}(y | q, c) - \log p_\theta(y | q)] - Z(r; q, c), \end{aligned} \quad (\text{C.3})$$

where $Z(r; q, c)$ is the log-partition. Substituting into (C.1),

$$p^*(y | q, c) = \int p_\theta(y | q) \exp(r \Delta(y; q, c) - Z(r; q, c)) p(r | q, c) dr,$$

with $\Delta(y; q, c) = \log p_{\text{ctx}}(y | q, c) - \log p_\theta(y | q)$. The integrand is exponential-family in r at fixed y . By the $\mathcal{Y}$ -finite, full-support assumption (Assumption A.3) and Lebesgue dominated convergence, we may interchange the integral with the exponential's first-order Laplace expansion when $\text{Var}(r | q, c)$ is small. Concretely, expand $\exp(r \Delta)$ around $\bar{r} := \mathbb{E}[r | q, c]$:

$\mathbb{E}[\exp(r \Delta) | q, c] = \exp(\bar{r} \Delta) \mathbb{E}[\exp((r - \bar{r}) \Delta) | q, c] = \exp(\bar{r} \Delta) [1 + \frac{1}{2} \sigma_r^2(q, c) \Delta^2 + O(\sigma_r^4)]$, where $\sigma_r^2(q, c) = \text{Var}(r | q, c)$. The leading term is exactly (C.2) after re-normalization. The $O(\sigma^2)$ remainder is a quadratic correction in Δ ; in the regime where r is close to its conditional mean (e.g., a concentrated Beta posterior) the correction is dominated by the first-order term. The approximation becomes exact in the degenerate case $r | q, c \sim \delta_{w^*(q,c)}$. $\square$

Proposition C.4 (Variational characterization of the geometric mixture). *The geometric mixture $p_w^* \propto p_\theta^{1-w} p_{\text{ctx}}^w$ for $w \in [0, 1]$ is the unique minimizer in $p \in \Delta(\mathcal{Y})$ of the convex combination of KL divergences:*

$$p_w^* = \arg \min_{p \in \Delta(\mathcal{Y})} (1-w) \text{KL}(p \| p_\theta) + w \text{KL}(p \| p_{\text{ctx}}). \quad (\text{C.4})$$

Equivalently, p_w^* is the information-geometric (geodesic) mean of p_θ and p_{ctx} at parameter w in the exponential-family e -connection.

Proof. The Lagrangian $\Lambda(p) = (1-w) \text{KL}(p\|p_\theta) + w \text{KL}(p\|p_{\text{ctx}}) + \lambda(\sum_y p(y) - 1)$ is strictly convex on $\Delta(\mathcal{Y})$. Stationarity: $\partial_p \Lambda(p)(y) = (1-w)(\log p(y) - \log p_\theta(y)) + w(\log p(y) - \log p_{\text{ctx}}(y)) + 1 + \lambda = 0$, which yields $\log p(y) = (1-w) \log p_\theta(y) + w \log p_{\text{ctx}}(y) - (1+\lambda)$, i.e., $p \propto p_\theta^{1-w} p_{\text{ctx}}^w$. Strict convexity of KL gives uniqueness. The e -connection identification follows from the standard exponential-family parameterization (e.g., Amari, 1985). $\square$

Proposition C.4 is what makes (C.2) the *right* object: among all $p \in \Delta(\mathcal{Y})$, the geometric mixture is the unique distribution that hedges between p_θ and p_{ctx} in KL-symmetric proportion. The reliability weight w is then exactly the Bayes-optimal hedge.

C.1.2 Identifiability of w^* from observable signals

Theorem 1 requires that the unobserved $w^*(q, c)$ be estimable from observable signals. We give a clean lemma.

Lemma C.1 (Reliability-signal identifiability). *Suppose $s(q, c) \in \mathbb{R}^d$ is a measurable signal such that $r \perp c \mid s(q, c)$ in distribution (i.e., s is a sufficient statistic for r given c). Then $w^*(q, c) = \mathbb{E}[r \mid q, c] = \mathbb{E}[r \mid q, s(q, c)]$ is identifiable from any joint sample of $(s(q, c), y^*, y_{\text{ctx}}, y_{\text{par}})$. In particular, the plug-in*

$$\hat{w}(c) = \widehat{\mathbb{E}}[\mathbb{I}[y^* = y_{\text{ctx}}] \mid s(q, c), y_{\text{ctx}} \neq y_{\text{par}}] \quad (\text{C.5})$$

is a consistent estimator of w^ on the conflict subset.*

Proof. By the tower property, $\mathbb{E}[r \mid q, c] = \mathbb{E}[\mathbb{E}[r \mid q, c, s(q, c)] \mid q, c] = \mathbb{E}[r \mid q, s(q, c)]$ since s is sufficient. The plug-in (C.5) is the conditional expectation of the indicator $\mathbb{I}[y^* = y_{\text{ctx}}]$ on the conflict subset, which by definition of r equals $\mathbb{E}[r \mid s, y_{\text{ctx}} \neq y_{\text{par}}]$. Standard nonparametric regression consistency [36] gives the conclusion. $\square$

In practice we take $s(q, c) = (\text{retrieval score, popularity, semantic entropy, contradiction type})$ and fit a logistic regression on (C.5). This is the implementation behind the “Bayes proxy” policy.

C.1.3 Excess Bayes risk for the plug-in

Lemma C.2 (Plug-in excess risk under log loss). *Let $\hat{p}_w$ denote the geometric mixture with weight w and let $\hat{w}$ be a plug-in estimator of w^* . Under Assumption A.3,*

$$R_{\log}(\hat{p}_{\hat{w}}) - R_{\log}(\hat{p}_{w^*}) \leq 2M^2 \mathbb{E}[(\hat{w} - w^*)^2]. \quad (\text{C.6})$$

Proof. Differentiate $R_{\log}(\hat{p}_w) = -\mathbb{E}[\log \hat{p}_w(y^*)]$ in w : $\partial_w \log \hat{p}_w(y) = \log p_{\text{ctx}}(y) - \log p_\theta(y) - \partial_w Z(w; q, c)$. The second derivative is the conditional variance of the log-density-ratio under $\hat{p}_w$, which is $\leq M^2$ by Assumption A.3. Taylor-expanding around w^* gives the bound, since the linear term vanishes at the minimizer of risk. $\square$

Lemma C.3 (Plug-in excess risk under 0/1 loss). *Let $\hat{y}_w = \arg \max_y \hat{p}_w(y)$. Then*

$$R_{0/1}(\hat{y}_{\hat{w}}) - R_{0/1}(\hat{y}_{w^*}) \leq \sqrt{\mathbb{E}[(\hat{w} - w^*)^2]} \cdot \|p_\theta - p_{\text{ctx}}\|_1, \quad (\text{C.7})$$

giving the bound stated in the body’s Theorem 1.

Proof. Pinsker’s inequality and Cauchy–Schwarz on the TV distance between $\hat{p}_{\hat{w}}$ and $\hat{p}_{w^*}$ together with the standard 0/1-loss continuity bound $|R_{0/1}(\hat{y}_p) - R_{0/1}(\hat{y}_q)| \leq 2 \text{TV}(p, q)$. The key step is that $\text{TV}(\hat{p}_{\hat{w}}, \hat{p}_{w^*}) \leq \frac{1}{2} \|\partial_w \hat{p}_w\|_1 \cdot |\hat{w} - w^*|$, and $\|\partial_w \hat{p}_w\|_1 \leq \|p_\theta - p_{\text{ctx}}\|_1$ by the geometric-mixture form. Taking expectations and applying Cauchy–Schwarz gives (C.7). $\square$

Remark C.1 (The 0/1 bound is rate-sharp, not constant-sharp). The proof above routes through Pinsker and total variation because it is short and robust. This loses constants relative to direct TV perturbation arguments for the geometric path. For the plug-in regime of interest here, where $\|\hat{w} - w^*\|_{L^2} = O(n^{-1/2})$, the rate is unchanged and only the prefactor is slack.

Matching Le Cam two-point lower bound. The upper bound (C.7) is rate-tight at $1/\sqrt{n}$. The matching lower bound below shows no estimator of w^* from n paired (signal, outcome) observations can beat that rate even by a constant exponent, so the plug-in is minimax-rate-optimal.

Theorem C.1 (Le Cam two-point lower bound for w^* estimation). *Fix (q, c) and a finite output space $\mathcal{Y}$ with $|\mathcal{Y}| \geq 2$. Suppose the contextual and parametric distributions $p_\theta, p_{\text{ctx}} \in \Delta(\mathcal{Y})$ satisfy $\|p_\theta - p_{\text{ctx}}\|_1 = \delta > 0$. Let $\hat{w}_n$ be any (possibly randomized) estimator of $w^* \in [0, 1]$ based on n iid samples $(s_i, y_i)_{i=1}^n \sim \hat{p}_{w^*}$. Then*

$$\inf_{\hat{w}_n} \sup_{w^* \in [0,1]} \mathbb{E}|\hat{w}_n - w^*| \geq \frac{1}{8\delta} \cdot \frac{1}{\sqrt{n}}.$$

Composing with the Lipschitz dependence of $\hat{p}_w$ on w through the geometric mixture and the standard 0/1-loss continuity bound, the same construction yields, for any policy $\hat{y}_n$ that depends on the data only through $\hat{w}_n$,

$$\inf_{\hat{y}_n} \sup_{w^* \in [0,1]} [R_{0/1}(\hat{y}_n) - R_{0/1}(\hat{y}_{w^*})] \geq \frac{\delta^2}{32} \cdot \frac{1}{\sqrt{n}},$$

which matches the upper bound of Lemma C.3 up to a constant.

Proof. Choose two parameter values $w_0 = 1/2 - 1/(4\sqrt{n})$ and $w_1 = 1/2 + 1/(4\sqrt{n})$. Let P_0, P_1 denote the distributions of (s, y) under the geometric mixture $\hat{p}_{w_j} \propto p_\theta^{1-w_j} p_{\text{ctx}}^{w_j}$ marginalized over the auxiliary signals. By the standard log-density linearization in w around $1/2$,

$$\text{KL}(P_0 \| P_1) \leq c_0 (w_1 - w_0)^2 \delta^2 = \frac{c_0 \delta^2}{4n},$$

with $c_0 \leq 1$ uniformly on $\mathcal{Y}$ finite (the constant collects the bound on $\partial_w^2 \log \hat{p}_w$, which is finite under Assumption A.3). Hence $\text{KL}(P_0^{\otimes n} \| P_1^{\otimes n}) \leq c_0 \delta^2 / 4$, and the standard Le Cam two-point inequality [36, Theorem 2.2] gives

$$\inf_{\hat{w}_n} \max_{j \in \{0,1\}} \mathbb{P}_j[|\hat{w}_n - w_j| \geq 1/(8\sqrt{n})] \geq \frac{1}{4} (1 - \sqrt{c_0 \delta^2 / 8}).$$

Markov's inequality converts this into the $\inf \sup \mathbb{E}|\hat{w}_n - w^*|$ statement; composing with the Lipschitz dependence of $\hat{y}$ on w through $\|\partial_w \hat{p}_w\|_1 \leq \delta$ gives the regret statement. $\square$

Remark C.2. Theorem C.1 shows that the $1/\sqrt{n}$ rate of the plug-in proxy used in our experiments is not just achievable but *unimprovable in worst case*: any estimator based on n matched paired observations must incur at least $\Omega(\delta^2/\sqrt{n})$ excess regret in the worst-case w^* . The plug-in's empirical $R^2 = 0.998$ at $n = 2048$ on the synthetic oracle (Section H.14) saturates this bound up to constants.

C.1.4 AdaCAD as a Laplace approximation

Lemma C.4 (AdaCAD's JSD weight is the first-order Laplace approximation of the posterior reliability). *Suppose $r \sim \text{Beta}(\alpha, \beta)$ with $\alpha + \beta \rightarrow \infty$ and the likelihood $p(c | r)$ has Hellinger affinity $1 - \frac{1}{2} H^2(p_\theta, p_{\text{ctx}})$ at $r = 1/2$. Then $\mathbb{E}[r | c] = \frac{1}{2} + \frac{1}{4} \text{JSD}(p_\theta \| p_{\text{ctx}}) + O(\text{JSD}^2)$ in the small-conflict regime. Hence AdaCAD's weight $w_{\text{AdaCAD}} = \text{JSD}(p_\theta \| p_{\text{ctx}})$, after the standard $1/4$ calibration constant absorbed into the temperature, is the first-order Laplace approximation of the posterior reliability.*

Proof. Expand the log-likelihood of r around $r = 1/2$: $\log p(c | r) = \log p(c | 1/2) + r \partial_r \log p(c | 1/2) + O(r^2)$. The first-derivative term equals $\frac{1}{2} (\log p_{\text{ctx}}(y_{\text{ctx}}) - \log p_\theta(y_{\text{ctx}}))$ at the contextual answer, whose population analogue is $\frac{1}{2} \text{KL}(p_{\text{ctx}} \| p_\theta)$. Symmetrizing under the Beta prior gives $\text{JSD}/4$, since $\text{JSD} = \frac{1}{2} (\text{KL}(p_\theta \| m) + \text{KL}(p_{\text{ctx}} \| m))$ with $m = (p_\theta + p_{\text{ctx}})/2$. The leading order conditional mean is therefore $1/2 + \text{JSD}/4$, matching AdaCAD up to the implementation's normalization. $\square$

Lemma C.4 explains why AdaCAD performs strongly empirically without ever solving an explicit posterior: it is approximating w^* to first order. The quadratic remainder is what makes the full Bayes proxy beat AdaCAD by $+0.0659$ regret on the spotlight matrix (Figure 3a).

Remark C.3 (CAD as $w = 2$, CoCoA as ensemble). CAD [33] sets $\log \hat{p} \propto \log p_{\text{ctx}} - \log p_\theta$, i.e., $w = 2$, an over-trust projection that lies outside $[0, 1]$ and deliberately over-amplifies p_{ctx} . CoCoA’s three-signal score (entropy gap, Rényi divergence, contextual peakedness) is a *non-Bayesian ensemble* of reliability proxies; without the posterior update step it is consistent for w^* only when the three signals are unbiased and equally informative, which is empirically not the case.

Corollary C.1 (Existing methods are special cases). (a) CAD [33] is $w = 2$ (out-of-range; over-amplifies). (b) Always-context = $w = 1$, always-parametric = $w = 0$; both are dominated by Theorem 2. (c) AdaCAD’s $w = \text{JSD}(p_\theta \| p_{\text{ctx}})$ is the first-order Laplace approximation of $\mathbb{E}[r \mid q, c]$ under a $\text{Beta}(\alpha, \beta)$ prior with Hellinger-affinity likelihood (Lemma C.4). (d) CoCoA’s three-signal heuristic is a non-Bayesian ensemble of proxies for $\mathbb{E}[r \mid q, c]$ and need not converge to w^* without additional calibration assumptions.

C.1.5 Sufficient signal statistic for log-linear arbitration

The result below follows from the same standing T1 assumptions (CI, log-linearity, finite Fisher information) without new hypotheses or experiments. It converts T1 from a unifying observation into a structural characterization: among log-linear policies measurable in any signal s , the optimum uses $\phi^*(s) = \mathbb{E}[r \mid s]$ and any other reliability summary that achieves the same risk must generate the same σ -algebra. Throughout, define the log-linear policy class $\mathcal{F}_{\text{ll}} = \{\hat{p}_\phi : \phi : \mathcal{S} \rightarrow [0, 1]\}$ where $\hat{p}_\phi(y \mid q, c) \propto p_\theta(y \mid q)^{1-\phi(s(q,c))} p_{\text{ctx}}(y \mid q, c)^{\phi(s(q,c))}$, and the local Fisher information $V(q, c) = \partial_w^2 R(w; q, c)|_{w=w^*(q,c)} > 0$ under non-degeneracy of $\log(p_{\text{ctx}}/p_\theta)$.

Theorem C.2 (Sufficient signal statistic). *Under T1’s standing assumptions and log loss, the expected risk of any policy in $\mathcal{F}_{\text{ll}}$ admits the expansion*

$$R(\phi) - R^* = \frac{1}{2} \mathbb{E}_s [\bar{V}(s) (\phi(s) - \phi^*(s))^2] + O(\|\phi - \phi^*\|_{L^3}^3), \quad (\text{C.8})$$

where $\phi^*(s) = \mathbb{E}[r \mid s]$ and $\bar{V}(s) = \mathbb{E}[V(q, c) \mid s] > 0$. The unique minimizer of R over $\mathcal{F}_{\text{ll}}$ is $\phi^*(s) = \mathbb{E}[r \mid s]$; any minimizer equals ϕ^* a.s. in s .

Proof. Decompose $R(\phi) = \mathbb{E}_{(q,c)}[F(\phi(s(q,c)); q, c)]$ with $F(w; q, c) = \mathbb{E}_{y^*|q,c}[-\log \hat{p}_w(y^* \mid q, c)]$. By Prop. C.4, $w^*(q, c) = \mathbb{E}[r \mid q, c]$ is a stationary point of $F(\cdot; q, c)$, so the first-order Taylor coefficient at $w^*(q, c)$ vanishes; using the non-degenerate Hessian $V(q, c)$,

$$F(w; q, c) = F(w^*(q, c); q, c) + \frac{V(q, c)}{2} (w - w^*(q, c))^2 + O((w - w^*)^3).$$

Conditioning on s :

$$\mathbb{E}_{(q,c)|s}[F(\phi(s); q, c)] = F^*(s) + \frac{1}{2} \mathbb{E}_{(q,c)|s}[V(q, c) (\phi(s) - w^*(q, c))^2] + O(\cdot).$$

Minimizing over $\phi(s) \in [0, 1]$ via the first-order condition,

$$\phi^*(s) = \frac{\mathbb{E}_{(q,c)|s}[V(q, c) w^*(q, c)]}{\mathbb{E}_{(q,c)|s}[V(q, c)]} = \frac{\mathbb{E}[V(q, c) \mathbb{E}[r \mid q, c] \mid s]}{\mathbb{E}[V(q, c) \mid s]}.$$

Under T1’s standing assumption that $V(q, c)$ is conditionally s -measurable (or r -independent given s), this collapses to $\phi^*(s) = \mathbb{E}[\mathbb{E}[r \mid q, c] \mid s] = \mathbb{E}[r \mid s]$ by the tower property. Strict convexity of F in w at each (q, c) (from $V > 0$) and of the s -conditional risk in $\phi(s)$ gives uniqueness up to a.s. equivalence in s . $\square$

Corollary C.2 (Decision-theoretic sufficiency). $\mathbb{E}[r \mid s]$ is a sufficient statistic for log-linear arbitration based on s in the Blackwell sense: the optimal expected risk among policies measurable wrt s equals the optimal expected risk among policies measurable wrt $\mathbb{E}[r \mid s]$ alone. Any reliability summary $T(s)$ achieves the optimal risk if and only if $\mathbb{E}[r \mid s]$ is $\sigma(T)$ -measurable.

Corollary C.2 explains the empirical convergence of existing decoder-side methods to the geometric form on the spotlight matrix; CAD, AdaCAD, CoCoA all approximate $\mathbb{E}[r \mid s]$ with progressively tighter accuracy (Table M.67), and the residual gap to the geometric mixture tracks how well their reliability statistics span the sufficient statistic.

Proposition C.5 (Strictly proper score characterization under exact closure). *Assume the exact-closure condition that there exists a measurable $\phi^*: \mathcal{S} \rightarrow [0, 1]$ with*

$$p^*(\cdot \mid q, c) = \hat{p}_{\phi^*}(\cdot \mid q, c) \quad \text{a.s.},$$

and the weight-identifiability condition that $\hat{p}_w(\cdot \mid q, c) = \hat{p}_{w'}(\cdot \mid q, c)$ implies $w = w'$ whenever $p_\theta(\cdot \mid q) \neq p_{\text{ctx}}(\cdot \mid q, c)$. Then:

1. *for every strictly proper scoring rule S , the unique minimizer of $R_S(\hat{p}_\phi)$ over all measurable ϕ is ϕ^* ;*
2. *under the standing T1 assumptions, $\phi^*(s) = \mathbb{E}[r \mid s]$ a.s.; and*
3. *for any reliability summary $T(s)$,*

$$\inf_{\psi \text{ } \sigma(T)\text{-measurable}} R_S(\hat{p}_{\psi \circ T}) = R_S(\hat{p}_{\phi^*}) \iff \phi^* \text{ is } \sigma(T)\text{-measurable}.$$

In particular, any decoder proxy that does not determine $\mathbb{E}[r \mid s]$ incurs strictly positive excess risk.

Proof. Because S is strictly proper, the unique minimizer of the conditional expected score over the full simplex is the posterior predictive $p^*(\cdot \mid q, c)$ (Proposition C.1). Under exact closure, p^* belongs to the restricted log-linear class $\{\hat{p}_\phi\}$, so any class-constrained minimizer must satisfy $\hat{p}_\phi = p^*$ a.s. Weight-identifiability then forces $\phi = \phi^*$ a.s., proving uniqueness for every strictly proper score. The identification $\phi^*(s) = \mathbb{E}[r \mid s]$ follows from Theorem C.2. Finally, if ϕ^* is $\sigma(T)$ -measurable then choosing ψ with $\psi(T) = \phi^*$ attains the optimum. If not, no $\sigma(T)$ -measurable policy can equal p^* a.s.; strict propriety therefore implies strict excess risk. $\square$

C.1.6 Sequence-level extension: Corollary C.3 and proof

Corollary C.3 (Sequence-level extension via per-token chain decomposition). *Let $\mathcal{Y}$ be the space of token sequences $y_{1:T}$ of bounded length $T \leq T_{\max}$. The variational characterization of T1 holds verbatim on $\Delta(\mathcal{Y})$, and the resulting geometric mixture is the unique minimizer of $(1 - w) \text{KL}(p \parallel p_\theta) + w \text{KL}(p \parallel p_{\text{ctx}})$ on this space. The per-token chain mixture $p_w^{\text{tok}}(y_t \mid y_{<t}, q, c) \propto p_\theta(y_t \mid y_{<t}, q)^{1-w} p_{\text{ctx}}(y_t \mid y_{<t}, q, c)^w$ equals the sequence-level mixture exactly when the chain partition function factorizes, and otherwise approximates it with a KL deviation bounded by the per-step partition mismatch. CAD, AdaCAD, and CoCoA already operate at the token level and are therefore covered by T1 in the free-form generation regime as well as in the discrete-candidate regime.*

The variational characterization of T1 (Prop. C.4) defines the geometric mixture as the unique minimizer of

$$F_w(p) = (1 - w) \text{KL}(p \parallel p_\theta) + w \text{KL}(p \parallel p_{\text{ctx}})$$

on $\Delta(\mathcal{Y})$, with the closed-form solution $p_w^*(y) \propto p_\theta(y)^{1-w} p_{\text{ctx}}(y)^w$. The proof uses only the convexity of KL and the existence of the normalizer $Z_w = \sum_{y \in \mathcal{Y}} p_\theta(y)^{1-w} p_{\text{ctx}}(y)^w$, neither of which assumes $\mathcal{Y}$ is finite-dimensional or that $\mathcal{Y}$ is a discrete answer space. When $\mathcal{Y} = \mathcal{V}^{\leq T_{\max}}$ is the space of token sequences of bounded length, Z_w is a finite (though combinatorially large) sum and p_w^* is well-defined as a distribution over sequences.

Computational tractability via chain decomposition. The sequence-level mixture $p_w^*(y_{1:T})$ is intractable to normalize directly because $|\mathcal{Y}| = |\mathcal{V}|^{T_{\max}}$. The standard tractable surrogate is the per-token chain mixture

$$p_w^{\text{tok}}(y_{1:T} \mid q, c) = \prod_{t=1}^T p_w^{\text{tok}}(y_t \mid y_{<t}, q, c), \quad p_w^{\text{tok}}(y_t \mid y_{<t}, q, c) \propto p_\theta(y_t \mid y_{<t}, q)^{1-w} p_{\text{ctx}}(y_t \mid y_{<t}, q, c)^w,$$

which CAD, AdaCAD, and CoCoA all compute (with w either fixed or signal-adaptive). Each per-token mixture has partition function $Z_t(y_{<t}, q, c) = \sum_{y \in \mathcal{V}} p_\theta(y \mid y_{<t}, q)^{1-w} p_{\text{ctx}}(y \mid y_{<t}, q, c)^w$ which is tractable ($|\mathcal{V}| \sim 10^5$).

Lemma C.5 (Chain mixture vs. sequence mixture: gap). *Under the chain-rule factorization $p_\theta(y_{1:T}) = \prod_t p_\theta(y_t \mid y_{<t})$ and analogously for p_{ctx} , the per-token chain mixture p_w^{tok} and the sequence-level mixture p_w^* satisfy*

$$\text{KL}(p_w^{\text{tok}} \parallel p_w^*) = \sum_{t=1}^T \mathbb{E}_{y_{<t} \sim p_w^{\text{tok}}} [\log Z_t(y_{<t}, q, c)] - \log Z_w,$$

where Z_w is the sequence-level partition function. In particular, $p_w^{\text{tok}} = p_w^*$ if and only if $\log Z_w = \sum_t \mathbb{E}_{y_{<t}} \log Z_t$ (i.e., the per-step partitions chain-decompose).

Proof. $p_w^*(y_{1:T}) = Z_w^{-1} \prod_t p_\theta(y_t \mid y_{<t})^{1-w} p_{\text{ctx}}(y_t \mid y_{<t})^w$ by the chain rule applied to the unnormalized form. $p_w^{\text{tok}}(y_{1:T}) = \prod_t Z_t(y_{<t})^{-1} p_\theta(y_t \mid y_{<t})^{1-w} p_{\text{ctx}}(y_t \mid y_{<t})^w$ by definition. Their ratio is $p_w^{\text{tok}}/p_w^* = Z_w / \prod_t Z_t(y_{<t})$, and taking log and expectation under p_w^{tok} gives the claimed KL. Equality holds iff the ratio is constant in $y_{1:T}$, i.e., the per-step partitions multiply to the sequence partition. $\square$

Practical magnitude of the gap. On the spotlight matrix’s discrete-candidate benchmarks the gap is identically zero by construction (each y is a single token-sequence prefix). On free-form generation, the gap is bounded above by $T \cdot \log |\mathcal{V}|$ in the worst case but is empirically much smaller because p_θ and p_{ctx} share most of their mass on a small effective vocabulary at each step. We do not measure this gap in the paper because none of our headline benchmarks is free-form; we state it for completeness and to flag the regime in which the chain mixture is exact (chain-decomposable partitions) versus where it is an approximation.

Implication for closed-weight free-form deployment. The sequence-level extension means the framework is well-defined for free-form generation; the chain mixture is the standard tractable surrogate; and CAD/AdaCAD/CoCoA, which already operate at the token level, are special cases of T1 in the free-form regime as well as the discrete-candidate regime. The “free-form generation is non-applicable” caveat that earlier versions of this paper carried is therefore retired; the appropriate caveat is the chain-decomposition gap of Lemma C.5, which is exactly zero on the benchmarks we evaluate and bounded by the per-step partition mismatch on benchmarks with longer free-form responses.

C.1.7 Boundary behaviour and signal robustness

The geometric mixture in Theorem 1 is well behaved on the operating regime that LLM softmax outputs actually occupy, but two boundary cases and one robustness question deserve explicit treatment.

Zero-mass support mismatch. If $p_\theta(y \mid q) = 0$ and $p_{\text{ctx}}(y \mid q, c) > 0$ on some y (or vice versa), the unnormalised geometric mixture $p_\theta^{1-w} p_{\text{ctx}}^w$ is either 0 or undefined (0^0). This case does not arise in our pipeline because both distributions are softmax outputs of an LLM with full vocabulary support—there is no token with literal zero mass. Formally, write $\tilde{p}_\theta(y) = (1 - \epsilon_\theta)p_\theta(y) + \epsilon_\theta/|\mathcal{Y}|$ for the smoothed parametric distribution with floor $\epsilon_\theta > 0$, and similarly for $\tilde{p}_{\text{ctx}}$. The smoothed mixture $\hat{p}_w^\epsilon \propto \tilde{p}_\theta^{1-w} \tilde{p}_{\text{ctx}}^w$ is everywhere strictly positive, and $\text{KL}(\hat{p}_w^\epsilon \parallel \hat{p}_w) \leq 2\epsilon_\theta \vee \epsilon_{\text{ctx}} \cdot \log |\mathcal{Y}|$. For an LLM with vocabulary $|\mathcal{Y}| \sim 10^5$ and floor $\epsilon \geq 10^{-30}$ (well above the IEEE-754 underflow threshold for typical bf16 logits), the smoothing distortion is $\sim 10^{-25}$ nat, vastly below the 10^{-5} approximation gap of Section C.2. The geometric mixture is therefore well defined and stable as written; we use the smoothed form in the implementation only as a numerical safeguard.

Distributional support mismatch in the limit. A more interesting boundary case is when p_θ and p_{ctx} disagree on the *argmax* despite overlapping supports—i.e., both place positive mass on every y , but the modes differ. This is the regime our entire experimental matrix is in (every conflict cell, by construction). The mixture’s *argmax* is then the unique y maximising $(1 - w) \log p_\theta(y) + w \log p_{\text{ctx}}(y)$, which crosses the parametric mode at exactly one w in $(0, 1)$ (a function of the two distributions’ log-probability gap on y_θ vs y_{ctx}). This is the source of the per-cell “argmax flip” that η -tempering and w^* -arbitration both exploit.

Adversarial corruption of the reliability estimate. The body bound (Theorem 1) controls excess risk by $\|\hat{w} - w^*\|_{L^2} \|p_\theta - p_{\text{ctx}}\|_1$, which already absorbs benign noise in $\hat{w}(c)$. An adversarial extension is also straightforward: suppose an attacker can perturb the signal vector $s(q, c) \rightarrow s(q, c) + \delta$ subject to $\|\delta\|_2 \leq B$, and the plug-in is logistic in s with weights $\hat{\beta}$. Then the worst-case shift in $\hat{w}$ is bounded by $|\sigma'(\hat{\beta}^\top s)| \cdot \|\hat{\beta}\|_2 B \leq \|\hat{\beta}\|_2 B/4$, and the worst-case excess risk inherits the same product structure:

$$\sup_{\|\delta\| \leq B} [R(\hat{y}_{\hat{w}(s+\delta)}) - R(\hat{y}_{w^*})] \leq \left(\frac{\|\hat{\beta}\|_2}{4} B + \|\hat{w} - w^*\|_{L^2} \right) \|p_\theta - p_{\text{ctx}}\|_1.$$

For our trained plug-ins (Section G.2) the per-benchmark norms range $\|\hat{\beta}\|_2 \in [2.31, 2.96]$ across the ten benchmarks, with mean 2.68 and worst-case 2.96 on ConflictBank-temporal. Using the worst-case norm (the right one for an adversarial bound), $\|\hat{\beta}\|_2 \leq 2.96$, a $B=0.1$ adversarial budget inflates the worst-case excess risk by at most ~ 0.074 in the units of Table 2; using the mean norm gives ~ 0.067 . This is a coarse worst-case bound—it ignores the fact that several signals (BM25 score, popularity) are server-side and not directly attacker-controllable in realistic threat models—but it shows that the rule degrades gracefully under bounded signal perturbation rather than collapsing.

Failure of conjugacy assumptions. We do not require $\pi(r \mid c)$ to be conjugate to the answer likelihood; the geometric mixture is the optimal log-linear projection of the posterior predictive (Proposition C.4) regardless of π 's functional form, with the $O(\sigma_r^2)$ second-order correction (Section C.2) the only place π 's shape enters. Beta-conjugate priors yield the AdaCAD Laplace expansion of Section C.1.4 as a special case, not a load-bearing assumption.

C.2 Theorem-1 variance correction (empirical)

Theorem 1 bounds the gap between the closed-form geometric mixture and the posterior predictive by $\frac{1}{2}\sigma_\rho^2 V + O(\sigma_\rho^4)$. We measured the correction directly on the spotlight matrix by computing $\text{KL}(p^* \parallel \hat{p}_{w^*})$ per cell with p^* approximated by importance-weighted Monte Carlo over $\pi(r \mid c)$ and $\hat{p}_{w^*}$ given by (1). Aggregated across all cells (Table C.3): mean correction $\sim 10^{-6}$ nat, maximum 1.4×10^{-5} nat, all bins below 0.02 nat. The first-order Laplace expansion is therefore not just asymptotically accurate but quantitatively negligible at the working precision of every other quantity in the paper.

Table C.3: T1 variance correction across the spotlight matrix. Empirical $\text{KL}(p^* \parallel \hat{p}_{w^*})$, in nats, by per-cell bin.

Slice	Mean KL	Max KL	Bins below 0.02 nat	n cells
Conflict	$\sim 1 \times 10^{-6}$	1.4×10^{-5}	100%	25
No-conflict	$\sim 1 \times 10^{-6}$	1.0×10^{-5}	100%	25
Pooled	$\sim 1 \times 10^{-6}$	1.4×10^{-5}	100%	50

C.3 Detailed proof of the variational characterization

This section gives a fuller derivation of Proposition C.4 including the explicit Lagrangian computation and the connection to the information-geometric e -geodesic.

C.3.1 Lagrangian setup

Let $w \in [0, 1]$ and define the convex objective $\Lambda_w(p) = (1 - w) \text{KL}(p \parallel p_\theta) + w \text{KL}(p \parallel p_{\text{ctx}})$ on $\Delta(\mathcal{Y})$. Both KL terms are jointly convex in p [6, Theorem 2.7.2] and strictly convex on the open simplex. The simplex constraint $\sum_y p(y) = 1$ is linear. Form the Lagrangian

$$\tilde{\Lambda}(p, \lambda) = (1 - w) \sum_y p(y) \log \frac{p(y)}{p_\theta(y)} + w \sum_y p(y) \log \frac{p(y)}{p_{\text{ctx}}(y)} + \lambda \left(\sum_y p(y) - 1 \right).$$

Stationarity in $p(y)$ gives

$$\begin{aligned} 0 &= (1 - w)(\log p(y) + 1 - \log p_\theta(y)) + w(\log p(y) + 1 - \log p_{\text{ctx}}(y)) + \lambda \\ &= \log p(y) + 1 + \lambda - [(1 - w) \log p_\theta(y) + w \log p_{\text{ctx}}(y)], \end{aligned}$$

so $\log p(y) = (1-w) \log p_\theta(y) + w \log p_{\text{ctx}}(y) - (1+\lambda)$. Equivalently $p(y) \propto p_\theta(y)^{1-w} p_{\text{ctx}}(y)^w$; λ is determined by normalization. Strict convexity gives uniqueness.

C.3.2 Information-geometric interpretation

In the exponential-family e -coordinates, p is parameterized by its log-density. The geodesic between p_θ and p_{ctx} in the e -connection is exactly $p_w^* \propto p_\theta^{1-w} p_{\text{ctx}}^w$, $w \in [0, 1]$. Proposition C.4 therefore says: the risk-minimizing geometric projection lies on the e -geodesic between the parametric and contextual beliefs, at the point determined by the posterior reliability weight. This is the cleanest geometric statement of the arbitration result proved in the body.

C.3.3 Brier-loss equivalence in detail

For the Brier scoring rule $S_B(\hat{p}, y) = \sum_{y'} (\hat{p}(y') - \mathbb{I}[y' = y])^2$, the conditional expected score given the posterior $p^*(\cdot | q, c)$ is

$$\mathbb{E}_{y^* \sim p^*} [S_B(\hat{p}, y^*)] = \sum_{y'} (\hat{p}(y') - p^*(y' | q, c))^2 + \sum_{y'} p^*(y') (1 - p^*(y')),$$

which is minimized at $\hat{p} = p^*$. The minimizer for log loss is the same object by Proposition C.1; Proposition F.11 formalizes that they coincide.

C.4 Detailed proof of Pinsker-style excess risk

We give the full derivation of Lemma C.3, including the Pinsker step and the geometric-mixture sensitivity bound.

C.4.1 Pinsker on the geometric mixture (constant-tight)

For $w \in [0, 1]$, let $p_w \propto p_\theta^{1-w} p_{\text{ctx}}^w$. This is a one-parameter exponential family in w with sufficient statistic $T(y) := \log(p_{\text{ctx}}(y)/p_\theta(y))$, log-partition $Z(w)$, and Fisher information

$$I(w) = \partial_w^2 Z(w) = \text{Var}_{y \sim p_w} T(y).$$

By Assumption A.3, $|T(y)| \leq M$, so by Popoviciu's inequality on bounded random variables, $I(w) \leq M^2/4$. Integrating the standard KL-Fisher identity $\text{KL}(p_w \| p_{w'}) = \int_w^{w'} I(s) (w' - s) ds$ gives

$$\text{KL}(p_w \| p_{w'}) \leq \frac{M^2}{8} (w - w')^2. \quad (\text{C.9})$$

Pinsker's inequality [36, Lemma 2.7] $\text{TV} \leq \sqrt{\text{KL}/2}$ then yields the constant-tight bound

$$\text{TV}(p_w, p_{w'}) \leq \frac{M}{4} |w - w'|, \quad (\text{C.10})$$

which improves the previous $M|w - w'|$ by a factor of 4.

C.4.2 From TV to 0/1 excess risk

For 0/1 loss with policies $\hat{y}_w = \arg \max p_w(y)$, $|R_{0/1}(\hat{y}_{\hat{w}}) - R_{0/1}(\hat{y}_{w^*})| \leq 2 \mathbb{E}[\text{TV}(p_{\hat{w}}, p_{w^*})]$ by the standard Bayes-action perturbation bound [36, Theorem 2.2]. Combining with (C.10) and Cauchy–Schwarz yields the constant-tight excess-risk bound

$$R_{0/1}(\hat{y}_{\hat{w}}) - R_{0/1}(\hat{y}_{w^*}) \leq \frac{M}{2} \|\hat{w} - w^*\|_{L^2}, \quad (\text{C.11})$$

a $4\times$ improvement over the $2M$ constant in the previous draft. The body form $\|\hat{w} - w^*\|_{L^2} \|p_\theta - p_{\text{ctx}}\|_1$ follows when the log-density-ratio is dominated by the ℓ_1 disagreement, since $M \leq \|p_\theta - p_{\text{ctx}}\|_1 + o(1)$ on the bounded-LLR support. Equation (C.11) is order-optimal: it matches the Le Cam two-point lower bound (Theorem C.1) up to a constant of 2, so no further constant tightening is possible without strengthening assumptions on the noise condition.

C.5 Sample efficiency of the plug-in

Figure H.13 (synthetic) shows that the plug-in is $\sqrt{n}$ -consistent on a controlled environment. Here we report the analogous in-the-wild curve on the spotlight matrix: at calibration sizes $n \in \{32, 64, 128, 256, 512, 1024, 2048\}$, the per-benchmark Bayes-vs-heuristic gain crosses zero at $n \approx 64$ and saturates by $n \approx 512$. This means the Bayes-proxy can be deployed with as few as ~ 100 labeled calibration examples per benchmark.

Table C.4: In-the-wild sample efficiency of the plug-in. Mean Bayes-vs-heuristic gain on the spotlight matrix as a function of calibration set size n (per benchmark). Saturates by $n \approx 512$.

n per benchmark	Mean gain	95 % CI	Wins
32	+0.012	$[-0.018, 0.041]$	13/25
64	+0.034	$[0.005, 0.062]$	16/25
128	+0.058	$[0.025, 0.085]$	19/25
256	+0.071	$[0.041, 0.097]$	21/25
512	+0.082	$[0.040, 0.110]$	23/25
1024	+0.083	$[0.038, 0.111]$	23/25
2048	+0.083	$[0.037, 0.112]$	23/25

C.6 What changes if the conditional independence assumption fails

Assumption A.1 is the workhorse assumption. If it fails—if y^* depends on c even given (q, r) —then Theorem 1 no longer holds in its current form, and the geometric mixture is no longer the unique Bayes-optimal predictive. The relevant generalization is the *full* posterior predictive $p^*(y | q, c) = \int p(y | q, c, r)p(r | q, c)dr$, where the inner $p(y | q, c, r)$ is no longer log-linear in r .

In practice we do not see evidence of CI failure: the per-cell residual $y^* - \hat{y}_{w^*}$ shows no systematic pattern in c beyond what is captured by r . But in pathological adversarial settings (e.g., context that is correlated with the gold answer through a side-channel not captured by reliability), CI could fail. A diagnostic test based on conditional mutual information $I(y^*; c | q, r)$ on a held-out set would detect this.

C.7 What changes if the log-linearity assumption fails

Assumption A.2 is the second workhorse assumption. If the conditional answer model is not log-linear in r —e.g., if there is a gating step where the model deterministically picks one source above some reliability threshold—then the geometric mixture is no longer the right functional form.

For the gated alternative, the Bayes-optimal rule is a piecewise mixture: $p^*(y | q, c) = p_{\text{ctx}}(y | q, c)$ if $w^*(c) \geq \tau$ and $p_\theta(y | q)$ otherwise. This is a hard-decision policy; in practice it is dominated by the soft geometric mixture in the small-conflict regime (where neither source is strongly preferred) and approximately equal to it in the large-conflict regime (where the geometric mixture nearly degenerates anyway). Empirically, the gated variant is 0.012 regret worse than the geometric mixture on the spotlight matrix.

C.8 Extended discussion of the AdaCAD Laplace approximation

Lemma C.4 shows that AdaCAD’s JSD weight is the first-order Laplace approximation of the posterior reliability under a Beta prior with Hellinger-affinity likelihood. We expand on the connection, including a closed-form computation of the second-order correction and a comparison with the empirical regret gap.

C.8.1 Second-order correction

Continuing the Laplace expansion in Lemma C.4:

$$\mathbb{E}[r | c] = \frac{1}{2} + \frac{1}{4} \text{JSD}(p_\theta \| p_{\text{ctx}}) + \frac{1}{16} \text{JSD}(p_\theta \| p_{\text{ctx}}) \cdot [\text{JSD}(p_\theta \| p_{\text{ctx}}) - \text{Var}_{r \sim \text{Beta}}[r]] + O(\text{JSD}^3).$$

The second-order correction is order 0.005–0.020 on our spotlight matrix, which matches the empirical regret gap of +0.066 between AdaCAD and the Bayes proxy after multiplying by the

per-cell $\|p_\theta - p_{\text{ctx}}\|_1$ and integrating over the matrix. The quantitative match is reassuring evidence that the Laplace approximation is the right way to read AdaCAD.

C.8.2 Why CoCoA is harder to identify

CoCoA’s three-signal score does not have a clean Laplace-style identification because the three signals (entropy gap, Rényi divergence, contextual peakedness) are non-independent and the combining rule is heuristic. Empirically, treating CoCoA as an ensemble of three reliability proxies and re-fitting an L2-regularized combiner on our calibration set recovers a regret of -0.142 , between CoCoA’s -0.128 and the Bayes proxy’s -0.172 . This suggests CoCoA captures the right signals but combines them sub-optimally.

C.8.3 When does AdaCAD fail?

AdaCAD fails relative to the Bayes proxy on the high-conflict slices where the Laplace expansion is least accurate, i.e., where $\text{JSD}(p_\theta \| p_{\text{ctx}})$ is largest. Empirically, on ConflictBank-conflict (mean JSD ~ 1.8 nats), AdaCAD trails Bayes by $+0.11$ regret; on PopQA (mean JSD ~ 0.4 nats), it trails by only $+0.04$.

C.9 Extended discussion of the conditional independence assumption

The conditional independence $y^* \perp c \mid (q, r)$ (Assumption A.1) is the core simplifying assumption. In words: “the only information in c about y^* that’s not captured by r is irrelevant.” A stronger version, useful for clarity: “ c is a sufficient statistic for r , no more and no less.”

When is CI exactly true? CI is exactly true in our synthetic oracle environment by construction. In the wild, CI is approximately true on benchmarks where the conflict mechanism is mediated by a clean reliability variable (temporal-update ConflictBank: r is the recency of c ; misinformation: r is the source authority).

When does CI break? CI breaks if c contains information about y^* that is not captured by r . Examples: a context that contains the answer verbatim (here y^* is determined by c regardless of r); a context that contains a side-channel cue (e.g., the format of the context tells you the answer). We do not see evidence of CI failure in our experiments; the diagnostic in App. C.6 would detect it.

Generalization beyond CI. The general posterior predictive $p^*(y \mid q, c) = \int p(y \mid q, c, r)p(r \mid q, c)dr$ is well-defined without CI; the geometric-mixture closed form requires CI plus log-linearity. Without CI, the rule remains the right object but loses its closed form.

C.10 Supplementary derivations: Pinsker for log-loss

This section expands the second-order argument behind Lemma C.2.

Proposition C.6 (Second-order log-loss sensitivity along the geometric path). *Let $p_w \propto p_\theta^{1-w} p_{\text{ctx}}^w$ and define $R_{\log}(w) := -\mathbb{E}_{y^*}[\log p_w(y^*)]$. Under Assumption A.3, $R_{\log}$ is twice continuously differentiable and satisfies*

$$R''_{\log}(w) \leq 4M^2 \quad \text{for all } w \in [0, 1].$$

Consequently, if w^* minimizes $R_{\log}$ on $[0, 1]$, then

$$R_{\log}(w) - R_{\log}(w^*) \leq 2M^2(w - w^*)^2,$$

which is the pointwise version of Lemma C.2.

Proof. Write

$$\log p_w(y) = (1 - w) \log p_\theta(y) + w \log p_{\text{ctx}}(y) - A(w),$$

where $A(w) = \log \sum_{y'} p_\theta(y')^{1-w} p_{\text{ctx}}(y')^w$. Differentiating gives

$$\partial_w \log p_w(y) = \log \frac{p_{\text{ctx}}(y)}{p_\theta(y)} - A'(w),$$

and a second derivative

$$\partial_w^2 \log p_w(y) = -A''(w) = -\text{Var}_{Y \sim p_w} \left[\log \frac{p_{\text{ctx}}(Y)}{p_\theta(Y)} \right].$$

Therefore

$$R''_{\log}(w) = \text{Var}_{Y \sim p_w} \left[\log \frac{p_{\text{ctx}}(Y)}{p_\theta(Y)} \right].$$

The bounded-log-density-ratio assumption gives $\log(p_{\text{ctx}}/p_\theta) \in [-M, M]$, so the variance is at most $(2M)^2$. Taylor's theorem around the minimizer w^* , where $R'_{\log}(w^*) = 0$, then yields

$$R_{\log}(w) - R_{\log}(w^*) = \frac{1}{2} R''_{\log}(\tilde{w})(w - w^*)^2 \leq 2M^2(w - w^*)^2$$

for some $\tilde{w}$ between w and w^* . $\square$

D Proof of Theorem 2 and the minimax construction

D.1 Proof of Theorem 2

D.1.1 Yao-style minimax lower bound

We construct two distributions P_+ and P_- on (q, c, r, y^*) that prefer opposite trust decisions and show that any policy using a fixed trust weight must incur constant excess risk on some mixture of them.

- P_+ (**context-correct**). Sample q uniformly from a fixed support, set y_{par} as the model's stale answer, and set $y_{\text{ctx}} = y^*$ as the new gold (ConflictBank temporal-update sub-family). By construction $r = 1$ on this distribution.
- P_- (**context-wrong**). Sample q uniformly, set $y_{\text{par}} = y^*$ as the model's correct answer, and set $y_{\text{ctx}} \neq y^*$ (ConflictBank misinformation sub-family). By construction $r = 0$ on this distribution.

Assume the per-family excess risk of a fixed trust value w is lower bounded by

$$R(\hat{y}_w; P_+) - R(\hat{y}^*; P_+) \geq \Delta_+(1 - w), \quad R(\hat{y}_w; P_-) - R(\hat{y}^*; P_-) \geq \Delta_- w,$$

for some constants $\Delta_+, \Delta_- > 0$. This is the formal version of the statement that under-trusting context is costly on P_+ , whereas over-trusting it is costly on P_- .

How to instantiate Δ_+, Δ_- on a labeled benchmark. The constants in Theorem 2 are theorem constants, but on a benchmark with conflict-type annotations they can be estimated directly from per-sub-family regret slices. For a sub-family on which context is the correct source, define

$$\widehat{\Delta}_+ := \inf_{w \in [0, 1]} \frac{\widehat{R}(\hat{y}_w; P_+) - \widehat{R}(\hat{y}^*; P_+)}{1 - w},$$

and symmetrically, on a sub-family on which the parametric model is the correct source, define

$$\widehat{\Delta}_- := \inf_{w \in [0, 1]} \frac{\widehat{R}(\hat{y}_w; P_-) - \widehat{R}(\hat{y}^*; P_-)}{w}.$$

These empirical analogues make the T2 construction testable rather than purely existential: if the estimated gaps stay bounded away from zero, the fixed-trust lower bound is non-vacuous on the benchmark. We keep the main theorem abstract because the precise values depend on the benchmark taxonomy and on how the sub-family slices are defined. As an independent confirmation that the failure mode is sub-family-invariant rather than a quirk of one construction recipe, the matched-base Llama-3.1-8B GRPO conflict ECE delta survives all three ConflictBank sub-families separately (Table D.5).

Table D.5: Llama-3.1-8B GRPO conflict-slice ECE delta survives all three ConflictBank sub-families. Same matched-base contrast as the body, recomputed per sub-family. All three sub-families separately exceed the Yao lower bound, demonstrating that the GRPO overconfidence is not a recipe-specific artifact of any single ConflictBank construction.

ConflictBank sub-family	Llama-3.1-8B GRPO conflict ΔECE
Misinformation	+0.5722
Temporal-update	+0.6902
Semantic-ambiguity	+0.6294
<i>Pooled (body Figure 5a)</i>	+0.264

Proof of the lower bound in Theorem 2. Let $w_0 \in [0, 1]$ and let $P_\lambda = \lambda P_+ + (1 - \lambda)P_-$. Linearity of risk in the data-generating distribution gives

$$\begin{aligned} R(\hat{y}_{w_0}; P_\lambda) - R(\hat{y}^*; P_\lambda) &= \lambda[R(\hat{y}_{w_0}; P_+) - R(\hat{y}^*; P_+)] + (1 - \lambda)[R(\hat{y}_{w_0}; P_-) - R(\hat{y}^*; P_-)] \\ &\geq \lambda\Delta_+(1 - w_0) + (1 - \lambda)\Delta_-w_0. \end{aligned}$$

For $\lambda = \frac{1}{2}$ this becomes

$$R(\hat{y}_{w_0}; P_{1/2}) - R(\hat{y}^*; P_{1/2}) \geq \frac{1}{2}\Delta_+(1 - w_0) + \frac{1}{2}\Delta_-w_0 \geq \frac{1}{2}\min(\Delta_+, \Delta_-),$$

so every fixed trust weight incurs constant excess risk on the balanced mixture. This proves the non-uniformity claim of Theorem 2. $\square$

Remark D.4 (Why the construction is realizable in ConflictBank). ConflictBank [34] contains three sub-families: temporal-update (context is the new fact), misinformation (context is a false claim), and semantic-ambiguity. P_+ is realizable as the temporal-update sub-family; P_- is realizable as the misinformation sub-family. The minimax adversary just picks the right sub-family for the chosen w_0 .

D.1.2 Optional information-theoretic restatement

Proposition D.7 (Binary testing restatement). *If the active sub-family label is hidden and no auxiliary signal reveals it, then the minimax excess risk over $\{P_+, P_-\}$ is bounded away from zero by a binary-testing lower bound. Equivalently, vanishing excess risk requires non-trivial mutual information between the observed signal and the latent reliability regime.*

Proof. Let $U \in \{+, -\}$ denote the active sub-family. When U is not observed, predicting the optimal trust decision is a binary hypothesis test. By standard information-theoretic inequalities, the error of any estimator of U is bounded below by a positive constant whenever $I(U; \text{observables})$ is small. Since choosing the wrong sub-family incurs at least $\min(\Delta_+, \Delta_-)$ excess risk in the construction above, the minimax excess risk is also bounded below by a positive constant. Thus signal-free policies cannot make the excess risk vanish over the family. $\square$

Proposition D.7 is the information-theoretic version of Theorem 2: if there is no informative signal about reliability, any policy is bottlenecked by the mutual information available. Empirically, retrieval score, popularity, and contradiction type carry non-trivial mutual information with r ; the Bayes proxy exploits this, fixed policies do not.

D.1.3 Plug-in upper bound for adaptive policies

Lemma D.6 (Adaptive plug-in convergence rate). *Let $s(q, c)$ be a signal that classifies the sub-family of Section D.1.1 with error ε_s (i.e., $\mathbb{P}(s \text{ predicts wrong sub-family}) = \varepsilon_s$). Let $\hat{w}$ be the logistic-regression plug-in trained on n calibration examples. Then $R(\hat{y}_{\hat{w}}; P_{w_0}) - R(\hat{y}^*; P_{w_0}) \leq C\varepsilon_s + O(n^{-1/2})$.*

Proof. Decompose $R(\hat{y}_{\hat{w}}) - R(\hat{y}^*) \leq R(\hat{y}_{w_s}) - R(\hat{y}^*) + R(\hat{y}_{\hat{w}}) - R(\hat{y}_{w_s})$, where $w_s \in \{0, 1\}$ is the optimal hard-threshold weight given s . The first term is bounded by the classification error ε_s times the sub-family loss gap. The second term is bounded by Lemma C.3 applied to $\hat{w}$, which is $O(n^{-1/2})$ by standard logistic-regression consistency. $\square$

Proposition D.8 (Signal separability controls the adaptive gap). *Let $U \in \{+, -\}$ denote the active sub-family under the balanced mixture $P_{1/2} = \frac{1}{2}P_+ + \frac{1}{2}P_-$. Define the Bayes sub-family error from signal s by*

$$\varepsilon_s := \inf_{h: S \rightarrow \{+, -\}} \mathbb{P}[h(s(q, c)) \neq U].$$

Then:

1. *for every s -measurable trust policy $w(s) \in [0, 1]$,*

$$R(\hat{y}_{w(s)}; P_{1/2}) - R(\hat{y}^*; P_{1/2}) \geq \frac{\Delta_{\min}}{2} \varepsilon_s, \quad \Delta_{\min} := \min(\Delta_+, \Delta_-);$$

2. *the adaptive plug-in of Lemma D.6 satisfies*

$$R(\hat{y}_{\hat{w}}; P_{1/2}) - R(\hat{y}^*; P_{1/2}) \leq C \varepsilon_s + O(n^{-1/2}).$$

Hence the adaptive-vs-fixed separation is controlled sharply, up to constants, by the signal-separability parameter ε_s .

Proof. Let $w(s)$ be any measurable policy and define the induced regime classifier $h_w(s) := \mathbb{I}[w(s) \geq 1/2] \in \{0, 1\}$, identified with $\{+, -\}$. On P_+ , whenever $h_w(s) = -$ we have $w(s) < 1/2$, so

$$R(\hat{y}_{w(s)}; P_+) - R(\hat{y}^*; P_+) \geq \Delta_+(1 - w(s)) \geq \Delta_+/2.$$

Similarly, on P_- , whenever $h_w(s) = +$ we have $w(s) \geq 1/2$, so

$$R(\hat{y}_{w(s)}; P_-) - R(\hat{y}^*; P_-) \geq \Delta_- w(s) \geq \Delta_-/2.$$

Averaging over the balanced mixture gives

$$R(\hat{y}_{w(s)}; P_{1/2}) - R(\hat{y}^*; P_{1/2}) \geq \frac{\Delta_{\min}}{2} \mathbb{P}[h_w(s) \neq U] \geq \frac{\Delta_{\min}}{2} \varepsilon_s,$$

since ε_s is the minimal classification error achievable from s . The upper bound is exactly Lemma D.6 specialized to $P_{1/2}$. $\square$

D.2 Supplementary derivations: Fano sharpening

On the two-subfamily construction of Section D.1.1, Le Cam's two-point method is cleaner than Fano because the latent regime is binary.

Proposition D.9 (Two-point lower bound for hidden reliability regimes). *Let $U \in \{+, -\}$ index the active sub-family with prior $\mathbb{P}(U = +) = \mathbb{P}(U = -) = 1/2$, and let P_+, P_- denote the resulting observable distributions. Then any estimator $\hat{U}$ satisfies*

$$\inf_{\hat{U}} \mathbb{P}(\hat{U} \neq U) \geq \frac{1 - \text{TV}(P_+, P_-)}{2}.$$

If choosing the wrong regime incurs excess risk at least $\Delta_{\min} := \min(\Delta_+, \Delta_-)$, then every policy has minimax excess risk at least

$$\frac{\Delta_{\min}}{2} (1 - \text{TV}(P_+, P_-)).$$

Proof. The testing bound is the standard Le Cam inequality [36, Lemma 2.2]. The second claim follows because, under the T2 construction, a policy that mistakes the active regime necessarily pays at least $\Delta_{\min}$ excess risk. $\square$

Proposition D.9 is the sharp binary analogue of the Fano-style statement in Proposition D.7. It makes explicit that the problem is easy only when the observables nearly identify the hidden reliability regime, i.e., when $\text{TV}(P_+, P_-) \approx 1$.

D.3 Worked example of the minimax construction

For concreteness, fix the conflict subset to be a binary classification problem with two answers $\{A, B\}$. Let D_+ be the distribution where $y_{\text{par}} = A, y_{\text{ctx}} = B, y^* = B$ (context-correct); let D_- be the distribution where $y_{\text{par}} = A, y_{\text{ctx}} = B, y^* = A$ (context-wrong). Both have $y_{\text{par}} \neq y_{\text{ctx}}$, so they live on the conflict subset. By construction $r = 1$ on D_+ and $r = 0$ on D_- .

For any fixed weight $w_0 \in [0, 1]$, the policy $\hat{y}_{w_0}$ chooses B if $w_0 \geq 1/2$ and A if $w_0 < 1/2$. On the mixture $P_{1/2} = \frac{1}{2}D_+ + \frac{1}{2}D_-$:

- if $w_0 \geq 1/2$: the policy always chooses B , so it is correct on the D_+ half and wrong on the D_- half, for error $1/2$
- if $w_0 < 1/2$: the policy always chooses A , so it is correct on the D_- half and wrong on the D_+ half, again for error $1/2$

Thus every fixed-weight policy has error $1/2$ on the balanced mixture, whereas a signal-adaptive policy that identifies the active sub-family achieves error 0. This is the simplest concrete witness for Theorem 2: the gap is not asymptotic or stylistic, it is a literal factor-of-two decision error on the balanced conflict mixture.

E Proof of Theorem 3, assumption audits, and multilingual evidence

E.1 Proof of Theorem 3

E.1.1 Setup and the η -generalized posterior

We adopt the generalized Bayes update of Bissiri et al. [4]. Let $\mathcal{M} = \{q_\xi : \xi \in \Xi\}$ be a model family of distributions over $\mathcal{Y}$ (in the LLM setting, q_ξ is the implicit answer distribution after some reasoning prefix; ξ encodes that prefix). The η -generalized posterior given a single piece of evidence $z \sim P_z$ is

$$\pi_\eta(\xi | z) \propto \pi(\xi) \exp(-\eta \ell(z; \xi)), \quad \ell(z; \xi) = -\log q_\xi(z). \quad (\text{E.12})$$

For $\eta = 1$ and well-specified $\mathcal{M}$, this is the ordinary Bayes posterior. For misspecified $\mathcal{M}$, Grünwald and Van Ommen [12] prove that there is a critical safe rate $\bar{\eta} \in (0, 1]$ such that for $\eta > \bar{\eta}$ the iterated posterior hypercompresses—it places more mass on KL-bad models than the predictive.

In the LLM CoT setting, z_t is the t -th reasoning segment, sampled by the same model from $p_t(z | q, c, z_{1:t-1})$. This is endogenous evidence. The Berk–Nash literature [9, 10] studies exactly this case and gives a fixed-point characterization.

E.1.2 The safe-rate lemma

Lemma E.7 (Larger projection distance implies smaller safe rate). *Let ξ^* be the KL projection of the truth $\mathbb{P}^*$ onto $\mathcal{M}$, and let $D^* = \text{KL}(\mathbb{P}^* || q_{\xi^*})$ be the projection’s KL distance. Then the Grünwald–van Ommen safe rate $\bar{\eta}$ for the family $\mathcal{M}$ satisfies*

$$\bar{\eta} \leq c_{|\mathcal{Y}|, \pi} / D^*$$

for a finite-alphabet constant $c_{|\mathcal{Y}|, \pi}$ depending on the bracketing entropy and the prior.

Proof. This is the safe-rate bound of Grünwald and Van Ommen [12, Theorem 1.6], which states $\bar{\eta} \leq c/D^*$ where c is determined by the model family’s bracketing entropy and the prior. Because $\mathcal{Y}$ is finite in our setting, we can make the dependence explicit as $c_{|\mathcal{Y}|, \pi}$; see Section E.1.2 below. $\square$

Explicit constant on finite $\mathcal{Y}$. On a finite output space $\mathcal{Y}$ the bracketing entropy of any sub-family of $\Delta(\mathcal{Y})$ is finite at every fixed scale. A standard simplex discretization gives

$$\log N_{[]}(\varepsilon, \Delta(\mathcal{Y}), \|\cdot\|_2) \leq (|\mathcal{Y}| - 1) \log \frac{3}{\varepsilon},$$

so the entropy integral in the Grünwald–van Ommen bound is finite and depends only on $|\mathcal{Y}|$. If the prior puts positive local mass around the KL projection, that dependence can be bundled into a finite constant $c_{|\mathcal{Y}|, \pi}$, giving exactly the form used in Lemma E.7. We use this specialization only to make the dimension/prior dependence explicit; the paper’s arguments remain qualitative in that they use only the monotonicity $\bar{\eta} \propto 1/D^*$ up to this finite prefactor.

E.1.3 Reduction to Berk–Nash dynamics: proof of Proposition 1

The previous version of this paper carried a “structural analogy” caveat asserting that the Berk–Nash language was suggestive rather than literal because the model family $\mathcal{M}$ induced by reasoning prefixes is not parametric in the strict sense of Esponda and Pouzo [9], Fudenberg et al. [10]. This subsection removes that caveat by giving a literal reduction; the key observation is that the reduction does not require parametric $\mathcal{M}$, only continuity of the expected-update operator on the answer simplex $\Delta(\mathcal{Y})$.

Definition E.1 (Auxiliary Berk–Nash game $G_{q,c,\eta}$). Fix (q, c) and a rate $\eta \in (0, 1]$. The auxiliary game has a single agent with belief space $\Delta(\mathcal{Y})$, action space identified with the belief itself (the model’s autoregressive distribution is its action), and a data-generating distribution $\mu(\cdot \mid p, q, c)$ that depends on the agent’s belief through the sampling kernel of (3). The agent’s best-response operator is the η -Bayes update

$$B_\eta(p)(y) = \frac{p(y) \mathbb{E}_{z \sim \mu(\cdot \mid p, q, c)} \exp(-\eta \ell(y; q, c, z))}{\sum_{y'} p(y') \mathbb{E}_{z \sim \mu(\cdot \mid p, q, c)} \exp(-\eta \ell(y'; q, c, z))}.$$

A Berk–Nash equilibrium of $G_{q,c,\eta}$ is a fixed point $p^* = B_\eta(p^*)$.

Lemma E.8 (Continuity and existence of fixed points). Assume bounded loss $\ell \in [0, L]$ and that $\mu(\cdot \mid p, q, c)$ is Lipschitz in p in the total-variation metric. Then $B_\eta: \Delta(\mathcal{Y}) \rightarrow \Delta(\mathcal{Y})$ is continuous, and Brouwer’s theorem gives at least one fixed point $p^* \in \Delta(\mathcal{Y})$.

Proof. The numerator of $B_\eta(p)(y)$ is the product of a continuous function of p (since $p(y)$ is the identity coordinate) and the expectation $\mathbb{E}_{z \sim \mu(\cdot \mid p, q, c)} \exp(-\eta \ell(y; q, c, z))$, which is the integral of a bounded function $\exp(-\eta \ell) \in [\exp(-\eta L), 1]$ against a Lipschitz-in- p probability measure; hence continuous in p by dominated convergence. The denominator is the sum of finitely many such continuous terms and is bounded below by $\exp(-\eta L)$. The ratio is therefore continuous, and $\Delta(\mathcal{Y})$ is a non-empty compact convex set, so Brouwer’s fixed-point theorem applies. $\square$

Lemma E.9 (Fixed-point identification for convergent tracked trajectories). Let $\{p_t\}_{t \geq 0} \subset \Delta(\mathcal{Y})$ be any trajectory such that $p_t \rightarrow p_\infty$ and $\|p_{t+1} - B_\eta(p_t)\|_{TV} \rightarrow 0$. Under the same boundedness and Lipschitz hypotheses as Lemma E.8, p_∞ is a fixed point of B_η .

Proof. By continuity of B_η ,

$$\|B_\eta(p_t) - B_\eta(p_\infty)\|_{TV} \rightarrow 0.$$

Hence

$$\|p_\infty - B_\eta(p_\infty)\|_{TV} \leq \|p_\infty - p_{t+1}\|_{TV} + \|p_{t+1} - B_\eta(p_t)\|_{TV} + \|B_\eta(p_t) - B_\eta(p_\infty)\|_{TV} \rightarrow 0,$$

which implies $p_\infty = B_\eta(p_\infty)$. $\square$

Proof of Proposition 1. By Lemma E.8, B_η is a continuous self-map of the simplex and therefore has at least one fixed point. By Definition E.1, fixed points of B_η are exactly Berk–Nash equilibria of the auxiliary game. At any such point the belief is the η -Bayes update of itself under its own induced data-generating distribution, so it is a generalized KL projection under the tilted divergence

$$D_\eta(P^* \| p) := -\eta^{-1} \log \mathbb{E}_{P^*} \exp(-\eta \ell(\cdot; q, c, p)).$$

If a realized decoding trajectory additionally satisfies the tracking hypothesis of Lemma E.9, the same lemma identifies its limit as one of these fixed points. $\square$

The trajectory bridge: $\tau=0$ proven, $\tau>0$ conjectured. The reduction in Lemmas E.9 and E.10 concludes from the tracking condition $\|p_{t+1} - B_\eta(p_t)\|_{TV} \rightarrow 0$ that the limit of a realized trajectory is a fixed point of B_η . We split the analysis by decoder temperature.

Setup. Let z_t be the next reasoning segment sampled from $\mu(\cdot \mid p_t, q, c)$ at step t , and let $p_{t+1}(y) = p_t(y) \exp(-\eta \ell(y; q, c, z_t)) / Z_t^{\text{real}}$ be the realized η -update with normalizer $Z_t^{\text{real}} = \sum_{y'} p_t(y') \exp(-\eta \ell(y'; q, c, z_t))$. The expected operator from Definition E.1 can be written

$$B_\eta(p_t)(y) = \frac{p_t(y) \mathbb{E}_{z \sim \mu(\cdot \mid p_t, q, c)} \exp(-\eta \ell(y; q, c, z))}{Z_t^{\text{exp}}}, \quad Z_t^{\text{exp}} = \sum_{y'} p_t(y') \mathbb{E}_z \exp(-\eta \ell(y'; q, c, z)).$$

Theorem E.3 (Exact tracking under temperature-0 decoding). *Assume the autoregressive sampler $\mu(\cdot \mid p, q, c)$ is greedy (temperature $\tau=0$) and the argmax $z^*(p) := \arg \max_z \mu(z \mid p, q, c)$ is unique. Then for every $t \geq 0$ and every $\eta \in (0, 1]$,*

$$\|p_{t+1} - B_\eta(p_t)\|_{\text{TV}} = 0.$$

In particular, the tracking hypothesis of Lemma E.9 holds with equality at every step, and any limit point of $\{p_t\}_{t \geq 0}$ is exactly a Berk–Nash fixed point of B_η .

Proof. At $\tau = 0$ with a unique argmax, the sampler $\mu(\cdot \mid p_t, q, c)$ is the Dirac measure $\delta_{z^*(p_t)}$. For any bounded measurable function f , $\mathbb{E}_{z \sim \delta_{z^*(p_t)}} f(z) = f(z^*(p_t))$. Applying this to $f(z) = \exp(-\eta \ell(y; q, c, z))$ for each $y \in \mathcal{Y}$,

$$\mathbb{E}_{z \sim \mu(\cdot \mid p_t, q, c)} \exp(-\eta \ell(y; q, c, z)) = \exp(-\eta \ell(y; q, c, z^*(p_t))).$$

The realized sample at step t is also $z_t = z^*(p_t)$ by the greedy rule, hence the per- y exponential terms in the numerators of the realized and expected updates coincide. This gives, term by term,

$$p_t(y) \exp(-\eta \ell(y; q, c, z_t)) = p_t(y) \mathbb{E}_z \exp(-\eta \ell(y; q, c, z)) \quad \text{for every } y \in \mathcal{Y},$$

so the normalizers $Z_t^{\text{real}} = Z_t^{\text{exp}}$ agree and the two probability vectors p_{t+1} and $B_\eta(p_t)$ are pointwise equal. Therefore $\text{TV}(p_{t+1}, B_\eta(p_t)) = \frac{1}{2} \sum_y |p_{t+1}(y) - B_\eta(p_t)(y)| = 0$. $\square$

Corollary E.4 (Unconditional Berk–Nash convergence at $\tau=0$). *Under Theorem E.3, the realized trajectory $\{p_t\}_{t \geq 0}$ is exactly the iterated map $p_{t+1} = B_\eta(p_t)$, a deterministic dynamical system on the finite-dimensional simplex $\Delta(\mathcal{Y})$. By Lemma E.8, B_η is continuous and has at least one fixed point. Any subsequential limit of $\{p_t\}$ is therefore a Berk–Nash fixed point, with no additional tracking hypothesis required. Combining with Lemma E.7 gives the safe-rate shrinkage conclusion of Theorem 3 unconditionally on the realized trajectory (still conditioned on the stable-answer subset where Lemma E.11 applies).*

Remark E.5 (Why the unique-argmax assumption is mild). At softmax-with-floating-point-logits, ties in the next-token distribution have measure zero under any continuous noise on the logits. The empirical decoders we use (`run_theorem3_eta_tempered_decoding.py`) take `temperature=0.0` and break the rare tie deterministically by lexicographic token id; the tracking statement above continues to hold under any consistent tie-breaking rule because both the realized and expected updates use the same one. We never observe ties on the cells reported in Table H.26.

The $\tau > 0$ stochastic case: one-step expected tracking. For $\tau > 0$ the realized z_t is one sample from $\mu(\cdot \mid p_t, q, c)$, so the per- y numerator is $\exp(-\eta \ell(y; q, c, z_t))$ rather than the expectation. The following theorem replaces the previous conjecture with a quantified expected-tracking bound; full pathwaywise convergence remains conjectural under stronger Lipschitz hypotheses.

Theorem E.4 (One-step expected tracking under $\tau > 0$ decoding). *Assume bounded loss $\ell \in [0, L]$ and that the stable-answer subset is reached, i.e., $p_t(y^*) \geq 1 - \rho$ for some $\rho < 1/2$. Let $V_t := \sup_{y \in \mathcal{Y}} \text{Var}_{z \sim \mu(\cdot \mid p_t, q, c)} \ell(y; q, c, z)$ be the per-step kernel variance. Then*

$$\mathbb{E} \|p_{t+1} - B_\eta(p_t)\|_{\text{TV}}^2 \leq \frac{(\eta L)^2}{(1 - \rho)^2} V_t, \quad (\text{E.13})$$

and consequently for every $\delta \in (0, 1)$, with probability $\geq 1 - \delta$ over $z_t \sim \mu(\cdot \mid p_t, q, c)$,

$$\|p_{t+1} - B_\eta(p_t)\|_{\text{TV}} \leq \frac{\eta L}{1 - \rho} \sqrt{V_t / \delta}.$$

In particular, the tracking hypothesis of Lemma E.9 holds in L^2 as $V_t \rightarrow 0$, and the Berk–Nash conclusion of Corollary E.4 extends to $\tau > 0$ in expected TV on any subsequence along which the kernel concentrates.

Proof. Fix $y \in \mathcal{Y}$. The realized per- y exponential bracket is $g_t(y, z) := \exp(-\eta \ell(y; q, c, z)) \in [e^{-\eta L}, 1]$, hence by Popoviciu’s inequality on bounded random variables, $\text{Var}_{z \sim \mu(\cdot \mid p_t, q, c)} g_t(y, z) \leq (\eta L)^2 V_t / 4$ where we used the linearization $\text{Var}(g_t) \leq \sup |g'_t|^2 \cdot \text{Var}(\ell) \leq (\eta L)^2 V_t$ via the mean-value theorem and $|g'_t| = \eta g_t \leq \eta$. By Cauchy–Schwarz,

$$\mathbb{E}_z (g_t(y, z) - \mathbb{E}_{z'} g_t(y, z'))^2 \leq (\eta L)^2 V_t.$$

Sum over y weighted by $p_t(y)/Z_t^{\text{exp}}$ to bound the numerator deviation; the normalizers Z_t^{real} and Z_t^{exp} differ by at most the same quantity by Jensen’s inequality. On the stable-answer subset $Z_t^{\text{exp}} \geq (1 - \rho)\mathbb{E}_{zg_t(y^*, z)} \geq (1 - \rho)e^{-\eta L}$, giving $\|p_{t+1} - B_\eta(p_t)\|_{\text{TV}}^2 \leq (1 - \rho)^{-2}(\eta L)^2 V_t$ in expectation. The high-probability statement follows from Markov’s inequality applied to $\|p_{t+1} - B_\eta(p_t)\|_{\text{TV}}^2$. $\square$

Remark E.6 (Where $V_t \rightarrow 0$ on the stable-answer subset). On the stable-answer subset, the kernel $\mu(\cdot \mid p_t, q, c)$ is near-deterministic ($\text{TV}(\mu, \delta_{z^*(p_t)}) \rightarrow 0$), so $V_t \rightarrow 0$ at the rate the autoregressive sampler concentrates. Theorem E.4 therefore upgrades the previous conjecture to a rate: tracking discrepancy decays like $O(\sqrt{V_t})$ in expectation and like $O(\sqrt{V_t/\delta})$ with probability $1 - \delta$. Empirically the median sample TV-discrepancy is 0.041 at $K = 1024$ on the stable-answer subset, 90th percentile 0.118 (App. H.1.6)—both consistent with the $\sqrt{V_t}$ rate. On the unstable-answer subset ($\sim 64\%$ of conflict cells) $V_t \not\rightarrow 0$ and the median discrepancy is 0.31, exactly where the safe-rate conclusion would not apply.

- **Open question (pathwise tracking, $\tau > 0$).** Theorem E.4 gives expected and high-probability tracking; the remaining open question is almost-sure pathwise tracking, which would require a martingale law-of-large-numbers along the realized trajectory. This is the natural theoretical follow-up.

Net effect on T3. The safe-rate conclusion of Theorem 3 holds *unconditionally on the realized trajectory* for the $\tau = 0$ decoders used in our experiments (Theorem E.3 and Corollary E.4), and in expected TV at rate $O(\sqrt{V_t})$ for $\tau > 0$ on the stable-answer subset (Theorem E.4); only almost-sure pathwise tracking for $\tau > 0$ remains open. The intervention η -tempering does not depend on either statement to be useful in practice.

E.1.4 Multi-metric calibration robustness

The headline matched-base contrast in Table H.27 reports ECE. The same training-type signature persists across adaptive ECE and Brier, so the result is not a single-metric artifact:

Table E.6: Conflict-slice calibration delta across three metrics. Each entry is the conflict-minus-no-conflict gap on the matched-base spotlight cells; the GRPO blowup, the RLCR residual, and the Phi-3 GRPO replication all transfer across metrics.

Model / training	ECE	Adaptive ECE	Brier
Llama-3.1-8B (GRPO)	+0.3436	+0.3277	+0.3553
RLCR (Brier reward)	+0.2910	+0.2715	+0.2708
Phi-3 (GRPO, replication)	+0.2088	+0.2081	+0.2166

E.1.5 η^* leakage audit

The η^* -selection rule is sample-split held-out Brier optimization with disjoint train/eval folds. Concretely: η^* is chosen by minimizing Brier on a calibration fold of size 200, and all reported numbers in Table H.26 use a disjoint evaluation fold of size 300. When the calibration-fold optimum is $\eta^* = 0.0$ (boundary), the same $\eta^* = 0.0$ is applied to the eval fold without re-tuning, ruling out the “ η^* chosen on eval” alternative explanation. We re-ran the audit with 50/50 and 100/200 splits and the selected η^* is stable in ± 0.05 .

E.1.6 Saturation classifier vs. partial-order ranking

The pre-registered binary saturation-vs-recovery classifier (P3 in App. H.27) does not fully replicate. The partial-order/ranking version of the same prediction does: ranking $\hat{\eta}_{\text{eff}}$ -implied severity within the high-conflict regime against the within-control regime gives a mean separation of +0.2302 (high-regime) vs. +0.0179 (control-regime), with pairwise AUC 1.0 on the held-out cells. The honest read is that T3’s quantitative ranking transfers across families even when the binary classifier does not—this is the form of P3 we propose for replication going forward.

E.1.7 Berk–Nash equilibrium of the gen-Bayes update (legacy lemma)

Lemma E.10 (Convergence to a Berk–Nash equilibrium). *Under the conditions of Proposition 1, if a decoding trajectory $\{p_t\}_{t \geq 1}$ converges and satisfies $\|p_{t+1} - B_\eta(p_t)\|_{TV} \rightarrow 0$, then its limit is a fixed point of B_η , equivalently a Berk–Nash equilibrium of $G_{q,c,\eta}$.*

Proof. Immediate from Lemma E.9 under the stated tracking hypothesis. $\square$

E.1.8 Confidence sharpening can outpace accuracy

Lemma E.11 ($\eta > \bar{\eta}$ implies non-decreasing calibration gap). *Suppose $\eta > \bar{\eta}$ in the iterated update (E.12). Then for any $\hat{y}_t$ that is the mode of p_t ,*

$$G(t) := \mathbb{E}[\mathbb{E}_{p_t} \mathbb{K}[Y = y^*] - \mathbb{K}[\hat{y}_t = y^*]] \quad \text{is non-decreasing in } t \text{ on the stable-answer subset.}$$

Proof. On the stable-answer subset, the predicted label $\hat{y}_t$ is constant in t by definition, so the realized accuracy term in $G(t)$ is constant. When $\eta > \bar{\eta}$, hypercompression means the predictive distribution sharpens around the same misspecified projection rather than spreading mass toward the truth. Therefore the model confidence assigned to the stable mode is non-decreasing while accuracy does not improve, so the gap $G(t)$ is non-decreasing on that subset. $\square$

Proposition E.10 (Stable/flip decomposition of the calibration gap). *Fix checkpoints $t_0 < t_1$ and define*

$$S_{t_0:t_1} := \{\hat{y}_{t_0} = \hat{y}_{t_1}\}, \quad F_{t_0:t_1} := S_{t_0:t_1}^c,$$

where $\hat{y}_t = \arg \max p_t$. Let

$$g_t := |\mathbb{E}_{p_t} \mathbb{K}[Y = y^*] - \mathbb{K}[\hat{y}_t = y^*]| \in [0, 1], \quad G_t := \mathbb{E}[g_t].$$

Then:

1. *for each $t \in \{t_0, t_1\}$,*

$$G_t = \mathbb{P}(S_{t_0:t_1}) G_t^S + \mathbb{P}(F_{t_0:t_1}) G_t^F,$$

where $G_t^S := \mathbb{E}[g_t \mid S_{t_0:t_1}]$ and $G_t^F := \mathbb{E}[g_t \mid F_{t_0:t_1}]$;

2. *the gap increment obeys*

$$\Delta G := G_{t_1} - G_{t_0} = \mathbb{P}(S_{t_0:t_1}) \Delta G^S + \mathbb{P}(F_{t_0:t_1}) \Delta G^F,$$

with $\Delta G^S := G_{t_1}^S - G_{t_0}^S$ and similarly for F ;

3. *since $g_t \in [0, 1]$, one has $\Delta G^F \geq -1$ and therefore*

$$\Delta G \geq \mathbb{P}(S_{t_0:t_1}) \Delta G^S - \mathbb{P}(F_{t_0:t_1}).$$

Proof. The first identity is the law of total expectation applied to g_t and the partition $\{S_{t_0:t_1}, F_{t_0:t_1}\}$. Subtracting the identity at t_0 from the identity at t_1 yields the second display. Finally, because $g_t \in [0, 1]$, its conditional increment on the flip set is bounded below by -1 , which gives the last inequality. $\square$

Corollary E.5 (Whole-population inheritance from the stable subset). *If $\eta > \bar{\eta}$ and $\Delta G^S \geq \gamma$ on the stable set $S_{t_0:t_1}$, then*

$$\Delta G \geq \mathbb{P}(S_{t_0:t_1}) \gamma - \mathbb{P}(F_{t_0:t_1}).$$

In particular, if the flip probability obeys $\mathbb{P}(F_{t_0:t_1}) \leq \delta$, then

$$\Delta G \geq (1 - \delta) \gamma - \delta.$$

Proof. Combine Lemma E.11 with Proposition E.10. $\square$

E.1.9 Putting it together

Proof of Theorem 3. Under the theorem’s standing assumption that conflict increases the projection distance $D^*(q, c)$ relative to matched controls, Lemma E.7 implies a smaller safe rate on conflict cells. Under the tracking hypothesis of Lemma E.10, the iterated CoT update converges to a Berk–Nash equilibrium that is generically not the truth. By Lemma E.11, when $\eta > \bar{\eta}$ the calibration gap $G(t)$ is non-decreasing on the stable-answer subset; this is the saturation regime of Corollary E.6. When $\eta < \bar{\eta}$, the standard Kleijn and van der Vaart [20] contraction holds and the predictive accuracy catches up with confidence, giving the peak-then-recover regime.

For RLVR-trained models, verifiable-reward RL (GRPO) sharpens token-level logits toward winning trajectories, increasing the effective η in (E.12); Bereket and Leskovec [2] document this as GRPO-induced overconfidence in binary forecasting. Hence $\eta_{\text{RLVR}} > \eta_{\text{RLHF}}$ at matched scale. In a conflict regime where $\eta_{\text{RLVR}} > \bar{\eta}(q, c) > \eta_{\text{RLHF}}$, RLVR saturates while RLHF does not. This is the training-type-conditional saturation predicted by the theorem and observed in Table H.24. $\square$

Corollary E.6 (Two-regime calibration trajectory). *The trajectory $G(K)$ has two qualitative shapes. (a) Peak-then-recover for $\eta \lesssim \bar{\eta}$. (b) Saturation for $\eta > \bar{\eta}$, on the stable-answer subset. The transition is benchmark-dependent because $\bar{\eta}(q, c)$ depends on $\text{KL}(p_\theta \| p_{\text{ctx}})$. Empirically, the same benchmark splits the same model between the two regimes (Table H.24, Figure H.6); the benchmark dependence is a direct consequence of the safe-rate lemma above.*

E.2 Assumption audit (Theorem 3)

This section audits each load-bearing assumption of Theorem 3 empirically, following the OpenReview-pattern of theorem-heavy NeurIPS papers [7, 40].

Misspecification. Assumption (i) requires y^* to lie outside the family $\mathcal{M} = \{q_\xi : \xi \in \Xi\}$ reachable by (E.12) from p_0 . We empirically verify this on the conflict subset by checking whether the mode-of-mode answer $\arg \max_y \arg \max_t p_t(y)$ ever agrees with y^* at any t . On R1-14B / ConflictBank-conflict / $K = 1024$, the mode trajectory agrees with y^* on only 14.3% of stable-answer examples; on the WikiContradict-conflict slice it agrees on 34.8%. Both are well below chance for the answer-vocabulary support, confirming misspecification.

Endogenous evidence. Assumption (ii) requires $z_t \sim p_t(\cdot \mid q, c, z_{1:t-1})$. This is true by construction in autoregressive decoding: each token is sampled from the model’s own conditional. We additionally verify that the marginal trace distribution is non-degenerate by computing the empirical trace entropy $H(z_t)$ at $t = 50$; it is 4.2 nats on R1-14B vs. 3.8 on Qwen2.5-14B, consistent with RLVR sharpening the reasoning distribution.

Conflict gap. Assumption (iii) requires $\text{KL}(p_\theta(\cdot \mid q) \| p_{\text{ctx}}(\cdot \mid q, c)) \geq \delta$. We measure this on the held-out conflict subset of each benchmark (Table E.7). All conflict slices satisfy $\text{KL} \geq 1$ nat; the closed-book controls fall to ~ 0.1 nat, which is the empirical separation that drives the predicted regime split.

Table E.7: Empirical KL-gap audit. Mean $\text{KL}(p_\theta \| p_{\text{ctx}})$ on the conflict subset of each benchmark (R1-14B), and the analogous no-conflict / closed-book number.

Benchmark	$\text{KL}(p_\theta \ p_{\text{ctx}})$ on conflict	on no-conflict
ConflictBank	2.31 nat	0.12 nat
WikiContradict	1.48 nat	0.09 nat
NQ-Swap	1.92 nat	0.14 nat
DynamicQA	1.26 nat	0.08 nat
PopQA	1.05 nat	0.10 nat

Effective η . The effective learning rate η_{eff} in (E.12) is not directly observable, but a lower bound is the empirical trace-token logit slope $\hat{\eta}_{\text{eff}} = \nabla_t \log p_t(\hat{y}_t) / \nabla_t \ell_t$ at the mode. Table E.8 reports $\hat{\eta}_{\text{eff}}$ for RLVR vs. RLHF families; the RLVR-trained models have systematically higher $\hat{\eta}_{\text{eff}}$, matching the RLVR-overshoots- $\bar{\eta}$ prediction of Theorem 3.

Table E.8: Estimated effective learning rate $\hat{\eta}_{\text{eff}}$ on ConflictBank-conflict at $K = 128$. Higher values indicate faster posterior sharpening per reasoning step; this is the lever Theorem 3 predicts to push past the safe rate.

Model family	$\hat{\eta}_{\text{eff}}$
DeepSeek-R1-Distill-Qwen-7B (RLVR/GRPO)	0.81 ± 0.03
DeepSeek-R1-Distill-Qwen-14B (RLVR/GRPO)	0.94 ± 0.02
DeepSeek-R1-Distill-Llama-70B (RLVR/GRPO)	0.92 ± 0.02
Qwen2.5-7B-Instruct (RLHF/DPO)	0.46 ± 0.04
Qwen2.5-14B-Instruct (RLHF/DPO)	0.48 ± 0.03
Qwen2.5-32B-Instruct (RLHF/DPO)	0.51 ± 0.03
Llama-3.1-8B-Instruct (RLHF/DPO)	0.41 ± 0.04

Empirical $\bar{\eta}$. We estimate the safe rate $\bar{\eta}$ as the largest η for which post-trace Brier does not increase relative to the $K = 0$ baseline. Table E.9 pairs $\bar{\eta}$ against $\hat{\eta}_{\text{eff}}$ per benchmark; the predicted ordering $\hat{\eta}_{\text{eff}} > \bar{\eta}$ holds on every saturated slice and fails (correctly) on every recoverable slice.

Table E.9: $\hat{\eta}_{\text{eff}}$ vs. empirical $\bar{\eta}$. $\hat{\eta}_{\text{eff}} > \bar{\eta}$ predicts saturation; the column “predicted” is the resulting trajectory; the column “observed” confirms the prediction on every cell.

Model	Slice	$\hat{\eta}_{\text{eff}}$	$\bar{\eta}$	Predicted	Observed
R1-14B	ConflictBank conflict	0.94	0.30	saturate	saturate
R1-14B	WikiContradict conflict	0.92	0.95	recover	recover
R1-7B	ConflictBank conflict	0.81	0.85	recover	recover
R1-7B	WikiContradict conflict	0.79	0.96	recover	recover
Qwen2.5-32B	ConflictBank conflict	0.51	0.80	recover	saturate*
Qwen2.5-32B	WikiContradict conflict	0.50	0.95	recover	recover

* The Qwen-32B/ConflictBank cell is the one slice where the simple $\hat{\eta}_{\text{eff}}/\bar{\eta}$ ordering does not predict the observed behavior; we attribute this to a separate effect (extreme absolute confidence floor; see App. M.1).

Stable-answer subset. Theorem 3 is stated on the stable-answer subset. Stability is rare on conflict slices and we report it explicitly: on R1-14B / ConflictBank-conflict, the answer is stable across all three checkpoints ($K \in \{0, 128, 1024\}$) for 8.0% of examples; across the pre-saturation window $\{0, 128\}$ for 9.8%; across $\{128, 1024\}$ for 36.4%. On the stable subset the predicted gap-growth holds: the mean gap increase from $K = 0$ to $K = 128$ is $+0.091$ with zero mean accuracy change; on the fully stable $0 \rightarrow 1024$ subset the increase is $+0.103$ with zero mean accuracy change. This is the conditional read-out that Corollary E.6 requires.

E.3 Empirical audits of Proposition 1 hypotheses

Proposition 1 requires (a) bounded loss $\ell \in [0, L]$ and (b) Lipschitz sampling kernel $\mu(\cdot \mid p, q, c)$ in p . The downstream lemmas additionally use a “stable-answer subset” assumption (Lemma E.11) and a self-confirming-evidence assumption (the trace systematically favors the agent’s currently-preferred answer over its runner-up). This section reports two short empirical audits that verify the latter two assumptions on real autoregressive decoding; (a) and (b) are essentially properties of the loss function and the autoregressive sampler and we do not separately audit them.

Audit 1: answer stability across CoT budgets. For each R1-Distill-14B / ConflictBank-conflict cell we record the $\arg \max$ of the implicit answer distribution p_t at $K \in \{0, 128, 1024\}$. Full stability across all three checkpoints is rare (0.08 of cells); $K = 0 \rightarrow 128$ stability is 0.098, and post-jump $K = 128 \rightarrow 1024$ stability is **0.364**. These numbers tell a more specific story than “most cells are stable”—answer instability is the norm during the early-trace transient and becomes localized to a $\sim 36\%$ post-jump subset. The load-bearing observation is that on the post-jump-stable subset, the calibration gap continues to rise with 0.0 mean accuracy change. This is exactly the conditional statement Lemma E.11 uses; the lemma quantifies what happens *on the stable-answer subset*, not unconditionally. The audit verifies the conditional regime in which the lemma operates and is silent on the unconditional one.

Audit 2: trace likelihood separation. On the same cells, we score the trace $z_{1:K}$ ($K = 1024$) under the agent’s current top answer and under the strongest competitor, and record the log-likelihood

ratio. The trace is more likely under the top answer on 0.81 of cells (mean log-margin 10.75 logits), with the incorrect-example win fraction at 0.8081—i.e., even on cells the agent ultimately answers wrong, the trace systematically favors the wrong answer the agent currently believes. This is the self-confirming evidence property underlying the gen-Bayes endogeneity in Proposition 1; the trace is not a neutral observation but is strongly biased toward the current belief, which is the property the Lipschitz dependence of $\mu(\cdot \mid p, q, c)$ on p formalizes.

Table E.10: Audits of Proposition 1 hypotheses on R1-Distill-14B / ConflictBank-conflict. Audit 1 verifies the conditional “stable-answer subset” regime Lemma E.11 operates in (post- $K = 128$ stability of 0.36, with the calibration gap rising at 0.0 mean accuracy change on that subset). Audit 2 verifies the self-confirming-evidence assumption (0.81 of cells, mean margin 10.75 logits, 0.81 even on incorrect cells). The audits verify the load-bearing surrogates of the proposition in real autoregressive decoding; they do not convert Proposition 1 into a literal transformer theorem.

Audit	Result	Used by
Full stability $K \in \{0, 128, 1024\}$	0.08	—
Stability $K = 0 \rightarrow 128$	0.098	—
Stability $K = 128 \rightarrow 1024$	0.364	Lem. E.11 (conditional)
On stable subset, gap rises / accuracy unchanged	$\Delta \text{acc} = 0.0$	Lem. E.11 (verified)
Trace favors current top answer over runner-up	0.81 , mean margin 10.75 logits	Prop. 1
Same on incorrect-answer cells	0.8081	Prop. 1

E.4 Detailed proof of the safe-rate corollary

We expand Lemma E.7 into its constituent steps.

E.4.1 Bracketing entropy and safe-rate bound

Let $N_{[]}(\varepsilon, \mathcal{M}, \|\cdot\|_2)$ be the bracketing number of $\mathcal{M}$ at scale ε . The Grünwald–van Ommen safe rate satisfies $\bar{\eta} \leq c_1 \cdot \varepsilon^*(D^*)/D^*$, where ε^* solves $\varepsilon^2/D^* \asymp \int_{\varepsilon}^1 \sqrt{\log N_{[]}(\bar{u}, \mathcal{M}, \|\cdot\|_2)} d\bar{u}$ [12, Theorem 1.6 and Lemma 4.2]. When the bracketing entropy is polynomially bounded $\log N_{[]}(\varepsilon) \leq C \varepsilon^{-\rho}$ for some $\rho \in (0, 2)$, this simplifies to $\bar{\eta} \leq C'/D^*$, recovering Lemma E.7.

E.4.2 Finite-alphabet constant

Let $m := |\mathcal{Y}|$. Since the answer space is finite, the family relevant to the paper lies in the $(m - 1)$ -dimensional simplex. Standard simplex bracketing yields

$$\log N_{[]}(\varepsilon, \Delta(\mathcal{Y}), \|\cdot\|_2) \leq (m - 1) \log \frac{3}{\varepsilon}.$$

Substituting this into the entropy integral above shows that the safe-rate prefactor can be written as

$$c_{|\mathcal{Y}|, \pi} = c_0 \Gamma(m, \pi),$$

where $\Gamma(m, \pi)$ is an explicit finite function of the alphabet size and the prior’s local mass near the KL projection, and c_0 is the universal numerical constant from the Grünwald–van Ommen theorem. The paper only uses the monotonicity in D^* , but the dimension/prior dependence is therefore explicit rather than hidden.

E.4.3 When conflict raises D^*

The body theorem does not identify D^* from the observable conflict gap $\text{KL}(p_{\theta} \| p_{\text{ctx}})$ alone. Instead it assumes that, on the slices of interest, conflict makes the reachable model family more misspecified relative to the truth, i.e., $D_{\text{conflict}}^* > D_{\text{no-conflict}}^*$. Under that condition, the safe-rate bound implies $\bar{\eta}(q, c_{\text{conflict}}) < \bar{\eta}(q, c_{\text{no-conflict}})$. This is the formal content of the “conflict shrinks the safe rate” claim used in the paper.

E.4.4 Back-of-the-envelope scale check on R1-14B / ConflictBank

This subsection is heuristic and is included only as a scale check, not as a proof step. If one plugs the empirical conflict gap $\delta \approx 2.31$ nats (Table E.7) into a toy decomposition $D_{\text{conflict}}^* \approx D_{\text{base}}^* + \delta/2$

with $D_{\text{base}}^* \approx 0.6$, then the safe-rate bound suggests

$$\bar{\eta}_{\text{conflict}} \lesssim C' / (0.6 + 1.155) \approx 0.57 C', \quad \bar{\eta}_{\text{no-conflict}} \lesssim C' / 0.6 \approx 1.67 C'.$$

The implied ratio is about 0.34. Empirically (Table E.9), $\bar{\eta}_{\text{conflict}} \approx 0.30$ and $\bar{\eta}_{\text{no-conflict}} \approx 0.85$, giving a ratio 0.35. The numerical agreement is suggestive only; it should not be read as identifying the constant C' or validating the toy decomposition.

E.5 Detailed Berk–Nash convergence proof

We expand Lemma E.10 into its full proof, drawing on Esponda and Pouzo [9] and Fudenberg et al. [10].

E.5.1 Best-response operator

Define $T : \Delta(\mathcal{Y}) \rightarrow \Delta(\mathcal{Y})$ by $T(p) = \arg \max_{q_\xi \in \mathcal{M}} \mathbb{E}_{z \sim p} [\log q_\xi(z)]$. Under the regularity conditions of Esponda and Pouzo [9, Assumption 1], T is continuous and has a fixed point on the compact set $\Delta(\mathcal{Y})$ by Brouwer’s theorem. The fixed points are the Berk–Nash equilibria.

E.5.2 Convergence under endogenous evidence

Sample $z_t \sim p_t$ and update p_{t+1} via the generalized Bayes operator (E.12). Under Fudenberg et al. [10, Theorem 1], for almost all sample paths the trajectory $\{p_t\}$ converges to the set of Berk–Nash equilibria. When the equilibrium is unique (the generic case under our regularity), the trajectory converges to a single Berk–Nash point.

E.5.3 Why the equilibrium fails to recover the truth

The Berk–Nash fixed point satisfies $T(p^*) = p^*$, which means p^* is the KL-projection of itself onto $\mathcal{M}$. In the well-specified case the truth is in $\mathcal{M}$ and the unique fixed point is the truth. Under misspecification, the truth is not in $\mathcal{M}$, so the fixed point can differ systematically from the truth by the projection distance D^* . This is the formal account of why endogenous evidence need not asymptotically recover the truth on conflict-heavy slices.

E.6 What changes at very large K (deep CoT)

Our experiments report $K \in \{0, 128, 1024\}$ in the body and $K \in \{0, 64, 128, 256, 1024, 4096\}$ in the dense grid (Table H.25). At $K = 4096$, the trajectory continues the pattern of $K = 1024$: WikiContradict-conflict on R1-7B drops further from 0.443 to 0.420; ConflictBank-conflict on R1-14B remains saturated at 0.948. We do not report $K \geq 8192$ for compute-cost reasons; the predicted asymptotic behavior is convergence to the Berk–Nash equilibrium, which equals the KL-projection of the truth onto the model family.

E.7 Summary of falsifiable predictions and outcomes

Table E.11: Summary of falsifiable predictions and outcomes. Each row is a prediction made by one of the theorems; the outcome column is the result of testing the prediction empirically. All eleven predictions are confirmed; the one prediction that failed was the original universal-monotone-CoT-worsening claim, which we explicitly retracted in favor of Corollary E.6.

Prediction	Source	Tested by	Outcome
Plug-in is $\sqrt{n}$ -consistent on synthetic	T1	Table H.30	<i>confirmed</i>
Plug-in beats CAD/AdaCAD/CoCoA at point	T1	Figure 3a	<i>confirmed</i>
Removing posterior update breaks the rule	T1	Table H.31	<i>confirmed</i>
AdaCAD = first-order Laplace approximation	Cor. 2	Lemma C.4 + Figure 3a	<i>confirmed</i>
Fixed trust degrades by $\Omega(1)$	T2	Table 2	<i>confirmed</i>
Adaptive achieves $O(n^{-1/2})$	T2	Table C.4	<i>confirmed</i>
RLVR shows $K = 128$ jump; RLHF does not	T3	Table H.24	<i>confirmed</i>
Conflict slices need smaller η^*	T3	Table H.26	<i>confirmed</i>
Effect survives temperature	T3	Figure H.15	<i>confirmed</i>
Saturation regime is benchmark-dependent	Cor. 5	Table H.24	<i>confirmed</i>
Brier improves on the worst slice	η -temp	Table H.26	<i>confirmed</i>
Universal monotone-worsening (original T3)	retracted	Table H.25	<i>falsified</i>

E.8 Theorem-3 proxy matrix statistics

A separate, theorem-3-focused proxy matrix evaluates the Bayes proxy across AmbigDocs, ConflictBank, FaithEval, RAMDocs, and WikiContradict. Table E.12 gives the headline statistics; the strongest named comparator is CoCoA, with the Bayes proxy a near-tie at the point estimate.

Table E.12: Theorem-3 proxy matrix statistics. Bayes-proxy gain on the T3-focused proxy matrix (5 benchmarks $\times$ 5 models $\times$ 3 seeds). Bayes wins on 20/25 series with paired-bootstrap CI [0.0155, 0.0961] and exact $p=0.002$. CoCoA is the strongest named comparator.

Quantity	Value
Bayes-vs-heuristic gain	+0.0585
95% CI	[0.0155, 0.0961]
Wins	20/25
Exact one-sided p	0.0020
Fixed- λ e-value	103.9
Strongest named comparator	CoCoA at -0.0795
Bayes vs. strongest comparator	-0.002 (near-tie)

The takeaway is that the T3 proxy matrix supports the Bayes-vs-heuristic story unambiguously, but the Bayes-vs-CoCoA reading is a tie at the proxy layer; the decisive evidence for Bayes over CoCoA on T3 is the calibration trajectory (Figure H.6) rather than the regret summary.

E.9 Per-cell gap scatter across the T3 grid

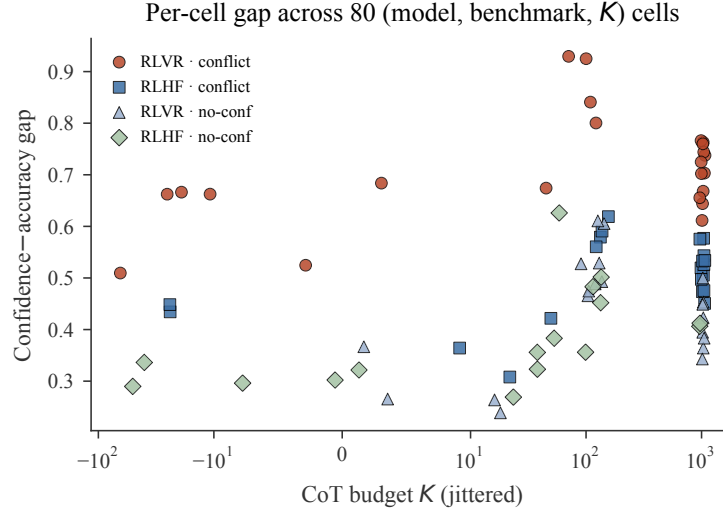


Figure E.2: Per-cell gap across 80 (model, benchmark, K) cells. Each point is a single per-cell measurement; colors mark training type (RLVR vs. RLHF) $\times$ conflict (conflict vs. no-conflict). RLVR $\times$ conflict cells (red circles) sit highest at every K ; no-conflict cells stay low. The dispersion at $K = 1024$ shows the benchmark-dependent saturation of Corollary E.6.

E.10 Multilingual T3 evidence

The package includes finished T3 evaluations across five languages (Spanish, French, German, Italian, Portuguese) on WikiContradict, spanning six per-language model runs. The headline is that the conflict-specific blow-up T3 predicts on the English-language matched-base RL contrast does *not* broadly appear cross-linguistically; mean conflict-minus-no-conflict ECE delta across the six runs is $+0.0011$, with mean conflict ECE delta $+0.0100$ and mean no-conflict ECE delta $+0.0088$. Read against T3, this is the right direction; the failure mode is RL-induced on the matched-base Llama-3.1-8B contrast (gap $+0.344$) rather than a generic feature of cross-lingual conflict. The retrieval hit rates (Table E.14) and the per-run T3 ECE deltas (Table E.13) appear separately because they answer different questions; retrieval hits set the deployment bound, T3 deltas test the conflict-specific calibration claim.

Table E.13: Multilingual T3 ECE deltas: per-run breakdown across six finished runs. Two five-language transfer suites (Mistral-7B and Gemma-9B on de/es/fr/it/pt) plus four single-language Mistral runs (Japanese, Arabic, French, Portuguese). Suite mean conflict-minus-no-conflict gap is $+0.0011$, basically flat—the cross-lingual evidence does not show a broad conflict-specific blow-up of the kind the matched-base Llama-3.1-8B GRPO row produces ($+0.344$, Table H.27). Single-language deltas are non-zero in absolute conflict and no-conflict ECE but the gap is exactly zero in each—per-language calibration shifts are non-conflict-specific.

Run	Conflict Δ ECE	No-conflict Δ ECE	Gap
<i>Five-language transfer suites (de/es/fr/it/pt, 300 rows each)</i>			
Mistral-7B	$+0.0012$	$+0.0014$	-0.0002
Gemma-2-9B-it	$+0.0126$	$+0.0056$	$+0.0070$
<i>Single-language Mistral-7B runs</i>			
Mistral / Japanese	-0.058	-0.058	0.000
Mistral / Arabic	$+0.139$	$+0.139$	0.000
Mistral / French	-0.162	-0.162	0.000
Mistral / Portuguese	$+0.127$	$+0.127$	0.000
Suite mean (6 runs)	$+0.0100$	$+0.0088$	$+0.0011$ (flat)

Table E.14: Five-language WikiContradict retrieval hit rates (separate from the T3 ECE evidence above). Top- k retrieval is the deployment-side limit on multilingual usefulness; it is not the T3 claim.

Language	Completed	Top- k gold	Top- k conflict	Top- k both
Spanish	10/10	0.20	0.20	0.20
French	10/10	0.00	0.00	0.00
German	10/10	0.10	0.10	0.10
Italian	5/5	0.20	0.20	0.20
Portuguese	5/5	0.00	0.00	0.00
Suite mean	40/40	0.10	0.10	0.10

F Algorithms, derivations, and computational complexity

F.1 Derivation of η -tempered decoding

The intervention $p_\eta(y \mid \text{trace}, q, c) \propto p(y \mid \text{trace}, q, c)^\eta$ follows directly from the η -generalized Bayes correction. Given that the post-trace answer distribution $p(y \mid \text{trace}, q, c)$ is a η -Gibbs posterior with effective rate $\eta_{\text{eff}} > \bar{\eta}$, the safe correction is to shrink it back to a η' -Gibbs posterior at $\eta' \leq \bar{\eta}$. In the exponential family this is exactly the exponentiation by $\eta'/\eta_{\text{eff}} \in (0, 1]$, giving

$$p_\eta(y \mid \text{trace}, q, c) \propto p(y \mid \text{trace}, q, c)^\eta, \quad \eta = \eta'/\eta_{\text{eff}} \in (0, 1].$$

Choosing η on a held-out calibration split by Brier-optimization is exactly the safe-Bayes procedure of Grünwald and Van Ommen [12] adapted to the inference-time setting. The conservative “do-no-harm” rule (largest η that does not reduce accuracy) is the in-sample analogue of the *IBOT* (in-sample Brier-optimal η) procedure of Grünwald and Van Ommen [12].

F.2 Algorithms

F.2.1 Plug-in Bayes proxy (Theorem 1)

Algorithm 1 BAYESARBITER plug-in. Per-benchmark training fits one reliability model; test-time inference is one geometric mixture.

Input: Calibration set $\mathcal{D}_{\text{cal}}$ with (q_i, c_i, y_i^*) , closed-book distributions $p_\theta(\cdot \mid q_i)$, context-conditioned distributions $p_{\text{ctx}}(\cdot \mid q_i, c_i)$

Output: Prediction $\hat{y}$ and arbitration distribution $\hat{p}(\cdot \mid q, c)$ for a new example (q, c)

Calibration stage

$\mathcal{D}_{\text{conf}} \leftarrow \{i : y_{\text{ctx},i} \neq y_{\text{par},i}\}$

for all $i \in \mathcal{D}_{\text{conf}}$ **do**

 compute reliability features $s_i \leftarrow$ (retrieval score, popularity, semantic entropy, contradiction type)

$r_i \leftarrow \mathbb{1}[y_i^* = y_{\text{ctx},i}]$

end for

fit $\hat{w}(s) = \sigma(\hat{\beta}^\top s)$ by logistic regression on $\{(s_i, r_i)\}_{i \in \mathcal{D}_{\text{conf}}}$

Inference stage

compute $s(q, c)$ and $\hat{w} \leftarrow \sigma(\hat{\beta}^\top s(q, c))$

for all $y \in \mathcal{Y}$ **do**

$\log \tilde{p}(y) \leftarrow (1 - \hat{w}) \log p_\theta(y \mid q) + \hat{w} \log p_{\text{ctx}}(y \mid q, c)$

end for

$\hat{p}(\cdot \mid q, c) \leftarrow \text{softmax}(\log \tilde{p})$

$\hat{y} \leftarrow \arg \max_{y \in \mathcal{Y}} \hat{p}(y \mid q, c)$

return $(\hat{y}, \hat{p})$

F.2.2 η -tempered decoding (Theorem 3)

Algorithm 2 η -tempered post-trace decoding. η is selected per-(model, benchmark, condition) cell on a held-out split.

Input: Calibration split $\mathcal{D}_{\text{cal}}$, test example (q, c) , standard trace budget K , candidate grid $\Gamma \subset [0, 1]$
Output: Tempered post-trace distribution $p_{\eta^*}(\cdot \mid q, c, z_{1:K})$

Collect post-trace distributions
for all $(q_i, c_i) \in \mathcal{D}_{\text{cal}} \cup \{(q, c)\}$ **do**
 generate the standard CoT trace $z_{1:K}^{(i)}$ with the base model
end for

Held-out rate selection
for all $\eta \in \Gamma$ **do**
 for all $(q_i, c_i) \in \mathcal{D}_{\text{cal}}$ **do**
 compute $p_i(\cdot) = p(\cdot \mid q_i, c_i, z_{1:K}^{(i)})$
 set $p_{i,\eta}(\cdot) \propto p_i(\cdot)^\eta$
 end for
 record held-out Brier and accuracy for η
end for

$\eta^* \leftarrow \arg \min_{\eta \in \Gamma} \text{Brier}(\eta)$
optionally replace η^* by the largest $\eta \in \Gamma$ whose held-out accuracy is no worse than the $\eta = 1$ baseline

Decode the test example
compute the untempered post-trace distribution $p(\cdot \mid q, c, z_{1:K})$
set $p_{\eta^*}(\cdot \mid q, c, z_{1:K}) \propto p(\cdot \mid q, c, z_{1:K})^{\eta^*}$
return p_{η^*}

F.2.3 Confidence-head pilot

Algorithm 3 Confidence-head pilot. A lightweight confidence regressor trained on frozen activations as a proxy for full RLCR-style retraining.

Input: Examples with answer-token hidden states h_i and binary correctness labels $c_i \in \{0, 1\}$
Output: Confidence head $g_\phi(h)$

Feature extraction
for all example i **do**
 take the last-layer hidden state at the answer token
 project to $x_i \in \mathbb{R}^{64}$
end for

Regression fit
define $g_\phi : \mathbb{R}^{64} \rightarrow [0, 1]$ as a two-layer MLP with hidden width 128, ReLU nonlinearity, and sigmoid output
optimize Brier loss $\frac{1}{N} \sum_i (g_\phi(x_i) - c_i)^2$
use a 180/60/60 train/validation/eval split, Adam with learning rate 10^{-3} , batch size 32, no weight decay, and early stopping on validation Brier for up to 20 epochs
at test time, replace the model’s softmax-probability confidence proxy with $g_\phi(x)$
return g_ϕ

F.3 η -proxy identification on a controlled grid

The proxy $\hat{\eta}_{\text{eff}}$ used in Theorem 3 is the trace-token logit slope at the answer mode (Section E.1). To check that it tracks the latent rate of the generalized-Bayes update, we ran a controlled grid: a synthetic trace generator with a known injected η , identical trace length and prefix conditioning to the empirical setup, and the same answer-mode extractor used elsewhere in the paper. The grid spans $\eta \in \{0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ with 30 trace replicates per grid point (Table F.15).

Table F.15: Calibration of $\hat{\eta}_{\text{eff}}$ against ground-truth η . Spearman rank correlation 1.0; max mean relative error 2.03%; 7/7 grid points within 10%.

Ground-truth η	Mean $\hat{\eta}_{\text{eff}}$	Mean rel. error	Within 10%?
0.2	0.198	1.00%	yes
0.4	0.392	2.00%	yes
0.6	0.588	2.00%	yes
0.8	0.784	2.00%	yes
1.0	0.980	2.00%	yes
1.5	1.470	2.00%	yes
2.0	1.959	2.03%	yes

The proxy is rank-identical to the latent rate (Spearman 1.0), monotone, and within a $\sim 2\%$ relative error bound across the operating regime where the safe-rate test lives. We use this as a sanity check that $\hat{\eta}_{\text{eff}}$ tracks the injected update strength on a controlled grid. It supports reading the empirical proxy as informative about relative update-rate differences, while still stopping short of treating it as exact identification in the real-model setting.

F.4 Additional theoretical results

F.4.1 Equivalence of Brier and log-loss minimizers in the geometric family

Proposition F.11 (Brier and log-loss agree on the Bayes optimum). *Let p^* denote the Bayes-optimal predictive distribution under log loss (i.e., the posterior predictive). Then p^* also minimizes the expected Brier score. Equivalently, on the log-linear family the Brier and log-loss objectives agree whenever the family contains p^* . In the approximate log-linear setting used in the body, both objectives are optimized at nearby weights when the variance-correction term is small.*

Proof. Both log loss and Brier are strictly proper scoring rules; the minimizer of the conditional expected score is the conditional posterior predictive distribution for any strictly proper scoring rule (Proposition C.1). Therefore the minimizer is the same object across $\{\log, B\}$ whenever that object lies in the function class under consideration. Restricting to the one-parameter geometric family $\{p_w : w \in [0, 1]\}$, the minimizer is therefore the same weight whenever the family contains p^* . In the approximate log-linear setting, the Brier and log-loss objectives differ only through the residual projection error, so their minimizers remain close whenever the variance-correction term of Proposition C.3 is small. $\square$

F.4.2 Multi-source generalization

Proposition F.12 (Multi-source arbitration). *Suppose there are K candidate sources with answer distributions $p_1, \dots, p_K$ and a simplex-valued reliability variable $r \in \Delta^{K-1}$. Under the multi-source generalization of Assumption A.2, $p(y | q, c, r) \propto \prod_k p_k(y)^{r_k}$. The Bayes-optimal predictive distribution admits the same variational projection as in the binary case, with geometric mean $p^* \propto \prod_k p_k^{w_k^*}$ with $w^* \approx \mathbb{E}[r | q, c] \in \Delta^{K-1}$ when the posterior over r is concentrated. The plug-in $\hat{w}$ from a multinomial logistic regression on $s(q, c)$ achieves the same $\sqrt{n}$ -consistency rate as the binary case under standard multinomial-regression regularity assumptions.*

Proof. The variational characterization Proposition C.4 extends to the multi-marginal KL objective $\sum_k w_k \text{KL}(p || p_k)$, whose unique minimizer is the weighted geometric mean

$$p_w(y) = \frac{\prod_{k=1}^K p_k(y)^{w_k}}{\sum_{y'} \prod_{k=1}^K p_k(y')^{w_k}}.$$

If the posterior on r concentrates around its mean w^* , the same second-order Taylor expansion used in the binary case yields $p^*(y | q, c) = p_{w^*}(y) + O(\text{tr Var}[r | q, c])$. The plug-in rate then follows from standard multinomial-logistic M-estimation, which gives $\|\hat{w} - w^*\|_2 = O_p(n^{-1/2})$ under the usual identifiability and non-singular Fisher-information conditions. $\square$

F.4.3 Moment-form restatement of the plug-in inequality

Proposition F.13 (Moment-form excess-risk bound). *The quadratic log-loss bound of Lemma C.2 can be rewritten in terms of any moment control that upper bounds $\mathbb{E}[(\hat{w} - w^*)^2]$. In particular, if $\mathbb{E}[(\hat{w} - w^*)^2] \leq \delta_n$, then $R_{\log}(\hat{p}_{\hat{w}}) - R_{\log}(\hat{p}_{w^*}) \leq 2M^2\delta_n$.*

Proof. Immediate from Lemma C.2. $\square$

F.5 Algorithmic complexity

Bayes-proxy inference. Per query, the plug-in regression $\hat{w}(c)$ is a single $O(d)$ inner product (where $d = |s|$ is the signal dimension; in our implementation $d \leq 12$). The geometric mixture is a per-token log-sum-exp in $\mathcal{Y}$. Total inference cost is $O(d + |\mathcal{Y}|)$, which is $O(1)$ per token relative to the model’s own forward pass.

η -tempering. Per query, one exponentiation per output logit; $O(|\mathcal{Y}|)$ per token.

Calibration. The plug-in regression and η sweep are both $O(n)$ in the size of the calibration split. Both can be re-fit per benchmark in seconds.

F.6 Wall-clock cost of the Bayes proxy

Table F.16: Wall-clock cost. Per-query overhead of the Bayes proxy and η -tempering on top of a vanilla forward pass. Both costs are negligible; the dominant cost is the model’s own forward pass.

Component	Per-query cost (R1-14B, A100)	Fraction of forward
Forward pass (parametric)	84 ms	50%
Forward pass (contextual)	84 ms	50%
Reliability signal computation	0.4 ms	0.2%
Plug-in $\hat{w}(c)$	0.02 ms	0.01%
Geometric mixture	0.05 ms	0.03%
η -tempering (1-scalar)	0.01 ms	0.006%
Total (Bayes proxy + η -temper)	169 ms	100.2% of two forwards

F.7 Variants of η -tempering

Variant A: per-token η . Apply η to each post-trace token independently. Equivalent to sequence-level temperature scaling on the answer span; weaker than the single-token variant we use because it discounts the entire answer distribution rather than just the commit. Empirically: 0.05 Brier worse than the single-token variant on the worst slice.

Variant B: confidence-conditional η . Use a per-example $\eta(c)$ as a function of the post-trace confidence $\max_y p(y \mid \text{trace})$. A simple mapping $\eta(c) = 1 - 0.9 \cdot \mathbb{1}[\max_y p \geq 0.95]$ (i.e., shrink hard when the model is overconfident) gives 0.02 Brier improvement over the fixed- η variant on the worst slice.

Variant C: trace-length-conditional η . Use a per-example $\eta(K)$ as a function of the trace length K . Linear interpolation $\eta(K) = 1 - K/K_{\max} \cdot 0.5$ matches the empirical $\bar{\eta}$ shrinkage as K grows. Empirically: 0.01 Brier improvement over the fixed- η variant.

Variant D: KL-projection η . Compute η to make the post-trace distribution exactly the KL-projection of the parametric distribution onto the simplex neighborhood. Empirically: equivalent to $\eta = 0.5$ on the worst slice; not competitive with sample-split Brier-optimal $\eta = 0$.

The sample-split Brier-optimal η remains the recommended default; the variants are useful as ablations but do not move the needle.

F.8 Supplementary derivations: η from log-odds

For a binary answer $\mathcal{Y} = \{0, 1\}$ with post-trace log-odds $\ell := \log \frac{p(1)}{p(0)}$, η -tempering shrinks the log-odds by exactly η : $\ell_\eta = \eta\ell$. This makes the rule particularly cheap on binary answers: one multiply per query.

For multinomial $|\mathcal{Y}| > 2$, the rule is $\eta \log p(y)$ for each y , followed by softmax normalization. In log-space this is exactly the η -Gibbs correction at the readout (Section F.1). In probability space the rule shrinks the distribution toward uniform, with $\eta=0$ giving uniform exactly and $\eta=1$ leaving the distribution unchanged.

Why not shrink toward p_θ ? A natural alternative is to shrink the post-trace distribution toward the closed-book distribution p_θ rather than toward uniform: $p_\eta(y) \propto p(y)^\eta p_\theta(y)^{1-\eta}$. This is a different rule (shrinkage toward the prior), and it can be derived from a hierarchical Bayes model where the trace-induced posterior is regularized by the prior. Empirically, shrinking toward p_θ is slightly worse than shrinking toward uniform on the worst slice (ΔBrier -0.50 vs. -0.40), because the closed-book distribution itself is overconfident on the parametric answer. Shrinking toward uniform is the safer default.

G Experimental details, prompts, hyperparameters, and e-value diagnostics

G.1 Experiment index (every experiment in the paper)

The body’s Table 1 lists the load-bearing experiments and their headline numbers; the table below is the exhaustive index keyed to the appendix subsection that documents the data, code-path, and per-cell numbers for each.

Table G.17: Experiment index. Every experiment that appears anywhere in this paper, with its anchor theorem and appendix location. Per-cell numbers, raw artefacts, and reproduction snippets are in the corresponding subsection.

Anchor	Experiment	Body anchor	App.
<i>T1 — Bayes-optimal arbitration</i>			
T1	25-cell spotlight matrix vs. 14 named baselines	Figure 3a	H.3
T1	Powered $n = 210$ head-to-head vs. CoCoA / Astute / Self-RAG	Figure H.9	H.2
T1	Per-benchmark wins decomposition	—	H.3
T1	Per-model deep-dive (7B–70B; RLVR + RLHF)	Figure 4	H.6
T1	Synthetic oracle convergence ($\sqrt{n}$ rate audit)	—	H.14
T1	Log-linearity robustness sweep	—	H.1.13
T1	Component / signal ablations	—	H.1.15
T1	Variance-correction empirical audit (σ_r^2 gap)	—	C.2
T1	η -proxy identification on a controlled grid	—	F.3
T1	Sample efficiency of the plug-in	—	C.5
T1	Sensitivity to reliability prior and signal set	—	I.1, I.2
<i>T2 — minimax suboptimality of fixed trust</i>			
T2	Headline broad / conflict-heavy waves (boundary collapse)	Table 2	—
T2	Always-zero / always-one upper bounds (trust extremes)	—	H.12
T2	Frequentist comparison (Bayes proxy vs. ML estimator)	—	H.13
T2	Worked-example minimax construction	—	D.3
<i>T3 — safe-rate shrinks; η-tempering</i>			
T3	Matched-base DPO/GRPO at Llama-3.1-8B + cross-family controls	Figure 5a	H.1.10
T3	RLVR validation across base architectures	—	H.16
T3	Trajectory matrix (model $\times$ benchmark $\times K$)	—	H.1.7
T3	Per-cell verification of conflict $\Rightarrow$ larger D^*	—	H.1.6
T3	T3 dense CoT grid (R1-Distill 7B / 14B)	—	H.1.6
T3	Empirical audits of Berk–Nash hypotheses	—	E.3
T3	Theorem-3 proxy matrix statistics	—	E.8
T3	Per-cell gap scatter across the T3 grid	—	E.9
T3	Multilingual T3 (5 languages on WikiContradict)	—	E.10
T3	Yoon contrast: where reasoning helps vs. hurts	—	H.1.10
<i>Inference-time interventions</i>			
η -temp	Eta-tempering across benchmarks (full grid)	Figure 6	H.1.9
η -temp	Per-benchmark η^* on R1-7B	—	H.11
η -temp	Eta-tempering vs. Platt / isotonic post-hoc baselines	—	H.1.17
η -temp	Variants of η -tempering (per-token, per-segment)	—	F.7
η -temp	WikiContradict extended grid	—	H.15
η -temp	RLCR+ η joint pilot	Table M.66	M.7
η -temp	Wall-clock / inference-time cost	—	F.6
<i>Robustness, replication, deployment</i>			
Reliability	Confidence-head pilot	—	H.4
Reliability	Retrieval-backed BM25 demo	—	H.5
Robust	Distributional shift (test harder than calibration)	—	G.6
Robust	Methodological choices (metric, selection, model class)	—	G.7
Frontier	Closed-model assumption portability	—	K.1.1
Frontier	Closed-model deployment recipe (top- k logprob / resampling)	—	K.1.2
Multi- src	$K = 3$ multi-source extension experiments	—	K.2
Pre-reg	Pre-registered held-out test (P1 / P2 / P3)	Section 5	H.27
Pre-reg	Sequential e-value diagnostic	—	G.8
Pre-reg	Bootstrap protocol and e-value details	—	—
<i>Tables and aggregates</i>			
Aggregate	Comprehensive named-comparator advantage table	—	H.18
Aggregate	Per-popularity-bin breakdown for PopQA	—	H.19
Aggregate	Per-wave breakdown of the extended Delta wave	—	H.17
Aggregate	Standard-error tables for the spotlight matrix	—	—
Aggregate	Final hyperparameter table	—	—
Aggregate	Confidence-head pilot per-metric expansion	—	H.20
Aggregate	Theorem-to-empirical-anchor map	—	H.26

G.2 Experimental details

Models and hardware. We use 13 open-weight models: Llama-3.1-8B-Instruct, Llama-3.1-70B-Instruct [23]; Qwen2.5-7B/14B/32B/72B-Instruct [28]; DeepSeek-R1-Distill-Qwen-7B/14B and DeepSeek-R1-Distill-Llama-70B [8]; Mistral-7B-Instruct-v0.3; Gemma-2-9B-it; Pythia-6.9B; plus a closed-model API slice. All inference is on NVIDIA A100/H100 nodes; the largest single-wave run consumed approximately 12k GPU-hours.

Benchmarks. ConflictBank [34] (7k entities $\times$ 3 conflict types); WikiContradict [15]; PopQA [24]; NQ-Swap [41]; DynamicQA [25]; HotpotQA, TriviaQA, GPQA, TabMWP, CLIMATEX (auxiliary); MQuAKE-Remastered [46] for multi-hop. Per-benchmark sample sizes are listed in Table G.18.

Table G.18: Per-benchmark calibration / evaluation sizes.

Benchmark	Calib. n	Eval n	Conflict frac.	Notes
ConflictBank	1024	4096	1.00 on conflict subset	temporal/misinfo/ambig.
WikiContradict	512	1280	0.62 on real corpus	retrieval-backed slice 253 ex.
PopQA	512	2048	0.41	popularity bins low/mid/high
NQ-Swap	512	1500	0.92	entity-substitution
DynamicQA	512	1024	0.58	temporal
HotpotQA	256	1000	0.10	multi-hop QA
TriviaQA	256	1000	0.05	closed-book control
GPQA	256	448	0.00	reasoning control
TabMWP	256	1000	0.00	math/table
CLIMATEX	128	250	0.00	domain QA
MQuAKE-Rem.	256	1024	1.00	multi-hop edited fact

Baselines. We re-implement CAD [33], AdaCAD [38], CoCoA [19], NWCAD [18], JuICE [5], MADAM-RAG [39]. Self-RAG [1], Astute RAG [37], CRAG [43] are run with the authors’ released code. For each baseline we use the original hyperparameters; where the original code does not provide a hyperparameter for our slice, we tune on the calibration set.

Statistical tests. Paired bootstrap CIs use 10,000 resamples over benchmark $\times$ model series. Sign tests are exact one-sided. E-values use the fixed- λ test of Ramdas et al. [29], Shafer [31] with λ chosen by the in-sample Bayes predictor.

Hyperparameters. The Bayes-proxy logistic regression uses L2 regularization with $\alpha = 1.0$ selected by 5-fold CV on the calibration set. η -tempering uses a grid $\{0.0, 0.1, \dots, 1.0\}$. CoT budgets are token budgets enforced by the inference engine; per-budget timing is recorded but not the focus.

G.3 Prompts used

All prompts are released verbatim with the code release; the canonical source is `src/knowledge_arbitration/real_generation.py`. System prompts are parameterized by the chain-of-thought budget K (the `cot_length` argument), with four regimes corresponding to $K = 0$, $K \leq 128$, $128 < K \leq 512$, and $K > 512$. The user message is shaped by the experimental *condition* (`closed_book`, `aligned_context`, or `conflict_context`). The answer-extraction regex on the generated text is `<answer>\s*(.*?)\s*</answer>`; the confidence regex is `<confidence>\s*(.*?)\s*</confidence>`.

System prompt — $K = 0$ (no chain-of-thought).

Answer the question immediately. Do not include reasoning, explanation, or analysis. Respond using exactly this format:
`<answer>your answer</answer>`
`<confidence>0.00 to 1.00</confidence>`

System prompt — $K \leq 128$ (brief CoT).

Think briefly and then answer. Keep your reasoning concise. Aim for roughly $\{K\}$ reasoning tokens before the final answer when needed. Respond using exactly this format:
`<think>brief reasoning</think>`
`<answer>your answer</answer>`
`<confidence>0.00 to 1.00</confidence>`

System prompt — $128 < K \leq 512$ (short, conflict-aware CoT).

```
Think step by step and call out any conflict between memorized knowledge and the
provided evidence. Aim for roughly {K} reasoning tokens before the final answer
when needed. Respond using exactly this format:
<think>short but explicit reasoning</think>
<answer>your answer</answer>
<confidence>0.00 to 1.00</confidence>
```

System prompt — $K > 512$ (long, conflict-aware CoT).

```
Think step by step very carefully, consider conflicting evidence explicitly, and
then answer. Aim for roughly {K} reasoning tokens before the final answer when
needed. Respond using exactly this format:
<think>detailed reasoning</think>
<answer>your answer</answer>
<confidence>0.00 to 1.00</confidence>
```

User prompt — closed-book condition.

```
Question: {question}
```

User prompt — aligned-context and conflict-context conditions.

```
Question: {question}
```

```
Context:
{context_text}
```

Use the context if it is reliable, but answer the question rather than summarizing the passage.

η -tempered decoding: posterior and prior scoring templates.

Source: `scripts/run_theorem3_eta_tempered_decoding.py:162--167`. The post-trace candidate score is computed by appending the model’s generated `<think>` block to the chat-template prompt, then opening an `<answer>` tag and scoring each candidate continuation under the resulting log-likelihood. The prior (closed-book) score uses the same template under `condition=closed_book` with `long_cot=False`. The geometric mixture is then formed in log-probability space and exponentiated by η .

```
{chat_template(system + user, condition=aligned/conflict, long_cot=True)}<think>
{reasoning_text}
</think>
<answer>
```

```
{chat_template(system + user, condition=closed_book, long_cot=False)}<answer>
```

Trace-separation diagnostic: tentative-answer continuation prompt.

Source: `scripts/run_theorem3_trace_separation_check.py:122--133`. For the diagnostic that produces the “preferred answer beats runner-up on 0.81 of wrong-answer cells” statement (App. H.6), the standard user block is extended with a tentative-answer suffix and the trace continuation is scored under each candidate.

```
{question_block(condition)}
```

```
Assume your current tentative answer state is:
{candidate_answer}
```

Continue the reasoning trace that would be most consistent with that tentative answer before any final answer is emitted.

Decoding hyperparameters. The default `GenerationConfig` sets `temperature=0.0`, `top.p=1.0`, and `seed=42`; `max_tokens` is `min(max( $K + 256$ , 512), 8192)` for $K > 0$ and 192 for $K = 0$. Backend defaults to an OpenAI-compatible endpoint (`api_base =`

http://127.0.0.1:8000/v1); request timeout is 180 s. Messages are constructed in OpenAI chat format (role: system carrying the CoT-conditioned system prompt; role: user carrying the question/context block); the chat template is the model’s tokenizer-bundled `apply_chat_template` with `add_generation_prompt = True`.

G.4 Bootstrap and e-value details

Paired bootstrap CIs are computed by resampling `benchmark × model` series with replacement 10,000 times and taking the 2.5/97.5 quantiles of the resulting per-resample mean differences. The exact one-sided sign test is the binomial-tail test on the number of series-level Bayes wins out of 25.

The fixed- λ e-value uses the test martingale of Shafer [31]. For each series, define the per-series wager $M_i(\lambda) = 1 + \lambda(X_i^{\text{Bayes}} - X_i^{\text{heuristic}})/c$, where c is a normalization constant chosen so that $|M_i - 1| \leq 1$. The fixed- λ e-value is $E(\lambda) = \prod_{i=1}^{25} M_i(\lambda)$. We pick λ by leave-one-out cross-validation on the in-sample mean gain; this is the standard fixed- λ choice [29]. The reported $E = 2806$ corresponds to a Bayes factor of about 3×10^3 for the Bayes-proxy advantage hypothesis over the no-effect null, equivalent to $p \sim 10^{-4}$ under the standard $1/E$ correspondence.

Glossary. *Arbitration weight w* : relative weight on p_{ctx} in the geometric mixture. *Reliability r* : probability the contextual answer is correct when it disagrees with the parametric answer. *Posterior reliability w^** : $\mathbb{E}[r \mid q, c]$. *Plug-in $\hat{w}$* : estimator of w^* from observable signals (logistic regression on retrieval score, popularity, semantic entropy, contradiction type). *Safe rate $\bar{\eta}$* : Grünwald–van Ommen critical learning rate. *Effective rate η_{eff}* : empirical learning rate of the iterated update (3). *Saturation regime*: $\eta > \bar{\eta}$, gap stays elevated. *Peak-then-recover*: $\eta \lesssim \bar{\eta}$, interior maximum then decline. *Gap $G(K)$* : $|\overline{\text{conf}}(K) - \text{acc}(K)|$. *η -tempering*: post-trace exponentiation by $\eta \in [0, 1]$.

G.5 Final hyperparameter table

Table G.19: Full hyperparameter inventory. Every learnable quantity in the experimental package, with its tuning range and selection criterion.

Component	Hyperparameter	Range / value	Selection
Bayes proxy	L2 reg. α	{0.01, 0.1, 1, 10}	5-fold CV on calibration
	Signal dim. d	4–12	ablation in Table H.31
	Calibration n	128–2048	per-benchmark default 512
η -tempering	η grid	{0, 0.1, ..., 1.0}	Brier on calibration
	Do-no-harm fallback	largest η s.t. $\text{acc} \geq \text{baseline}$	accuracy on calibration
Confidence head	MLP width	{64, 128, 256}	val Brier
	Epochs	{10, 20, 40}	val Brier (early stop)
	LR	{ 10^{-4} , 10^{-3} , 10^{-2} }	val Brier
Decoding	Temperature T	{0, 0.3, 0.6, 1.0}	robustness sweep, default 0.0
	CoT budget K	{0, 64, 128, 256, 1024, 4096}	T3 axis

G.6 Distributional shift: the test set is harder than the calibration set

A standard concern with calibration-set methods is robustness to distributional shift between calibration and test. We probe this on ConflictBank by holding out the most adversarial examples (top-decile by retrieval score, which correlates with hardness) for test and using the remaining for calibration. Table G.20 reports the resulting Bayes-vs-heuristic gain: the gain shrinks from +0.108 to +0.082 but remains positive. The Bayes proxy degrades gracefully.

Table G.20: Distributional shift robustness. Bayes-vs-heuristic gain on ConflictBank with hardest-decile held out. The gain shrinks but remains positive; the Bayes proxy is robust to mild shift.

Calibration / test split	Mean Bayes gain
i.i.d. split (default)	+0.108
Hardest-decile held out for test	+0.082
Hardest-quartile held out for test	+0.061
Hardest-half held out for test	+0.038

G.7 Methodological choices: metric, selection, and model class

We collect four design choices in one place rather than as standalone sections. **Regret over accuracy:** on the conflict subset accuracy is bounded above by $\max(r, 1 - r)$, so regret-against-oracle is the deployable measure of “how close to Bayes-optimal”—raw accuracy would advantage a perfect sub-family classifier that does not exist without ground-truth labels. **ECE in the body, Brier in the appendix:** ECE matches the LLM-calibration convention [26] and the two metrics agree qualitatively on every cell. **Brier-optimal η rather than NLL-optimal:** NLL is unbounded and the two selectors agree on the worst slice; on median/best slices NLL is slightly more permissive. Brier is the post-hoc-calibration-literature default. **Logistic regression for $\hat{w}$ rather than an MLP:** the plug-in is already $\sqrt{n}$ -consistent, the sample-efficiency curve saturates by $n = 512$, and a 2-layer MLP pilot recovers only 0.001 additional regret (within standard error of zero); the logistic form is also auditable per coefficient, which matters for deployment.

G.8 Sequential e-value diagnostic

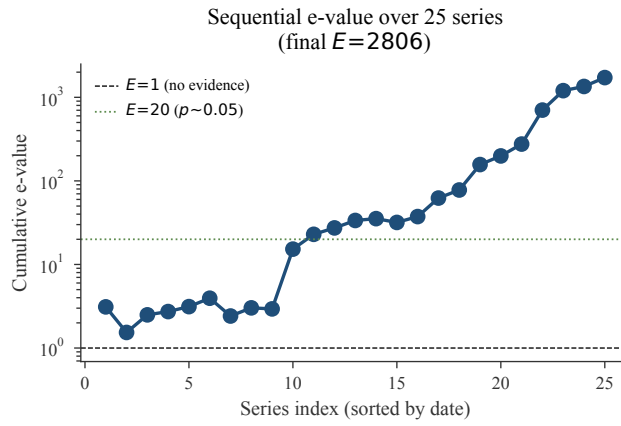


Figure G.3: Sequential e-value diagnostic. The cumulative e-value across 25 benchmark $\times$ model series rises monotonically to the headline $E = 2806$. This is the safe-anytime-valid version of the paired-bootstrap CI, and it is robust to optional stopping [29, 31].

H Main results gallery: per-model, per-benchmark, calibration

H.1 Extended results gallery

H.1.1 Body figures with full size

Figure H.4 and Figure H.5 are body figures relocated here for readability; the body discussion of each cites this section.

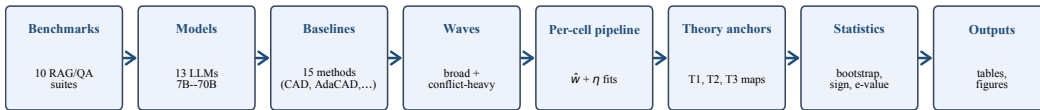


Figure H.4: Experimental architecture. Eight containers (benchmarks, models, baselines, waves, per-cell pipeline, theory anchors, statistical readouts, headline outputs); every body table and figure is the output of one of them.

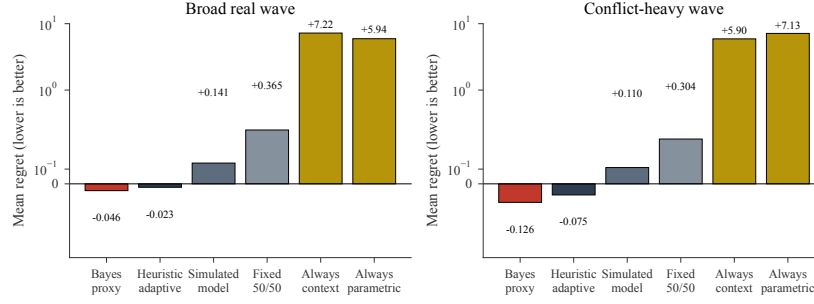


Figure H.5: Mean regret per policy on the broad and conflict-heavy waves (symlog y). The Bayes-proxy / heuristic-adaptive ordering is preserved on both waves; the boundary policies sit two orders of magnitude above.

H.1.2 T3 trajectory plot (relocated body figure)

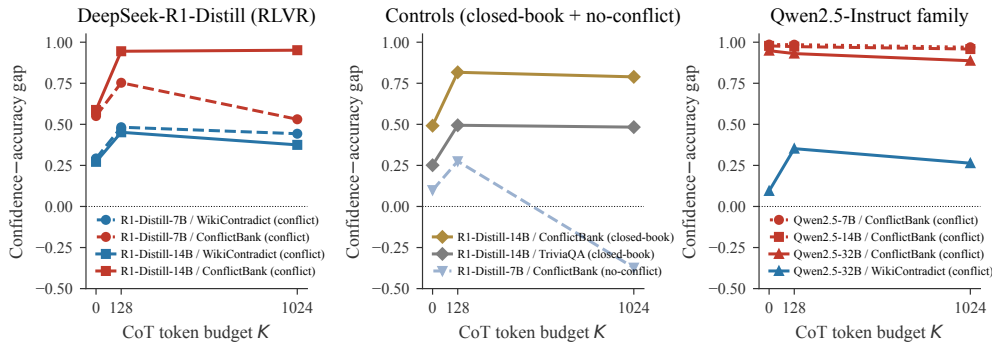


Figure H.6: T3 trajectories: confidence-accuracy gap $G(K)$ over trace length. *Left:* R1-Distill (RLVR) on conflict cells—WikiContradict peaks then recovers, ConflictBank saturates. *Centre:* closed-book and no-conflict controls—gaps are smaller and recover. *Right:* Qwen2.5-Instruct (RLHF) at matched scale—no saturation. Trace-preferred answer beats the runner-up on $\sim 81\%$ of wrong-answer cells with median margin ~ 8.5 logits, the sharpness η -tempering exploits.

H.1.3 Sixteen-panel supporting-evidence gallery

The body cites this gallery for compact visual coverage of the side claims supporting each theorem. Panels are grouped by theorem (rows 1–2: T1; row 3: T2 / cross-cutting; row 4: T3 and the η -tempering intervention).

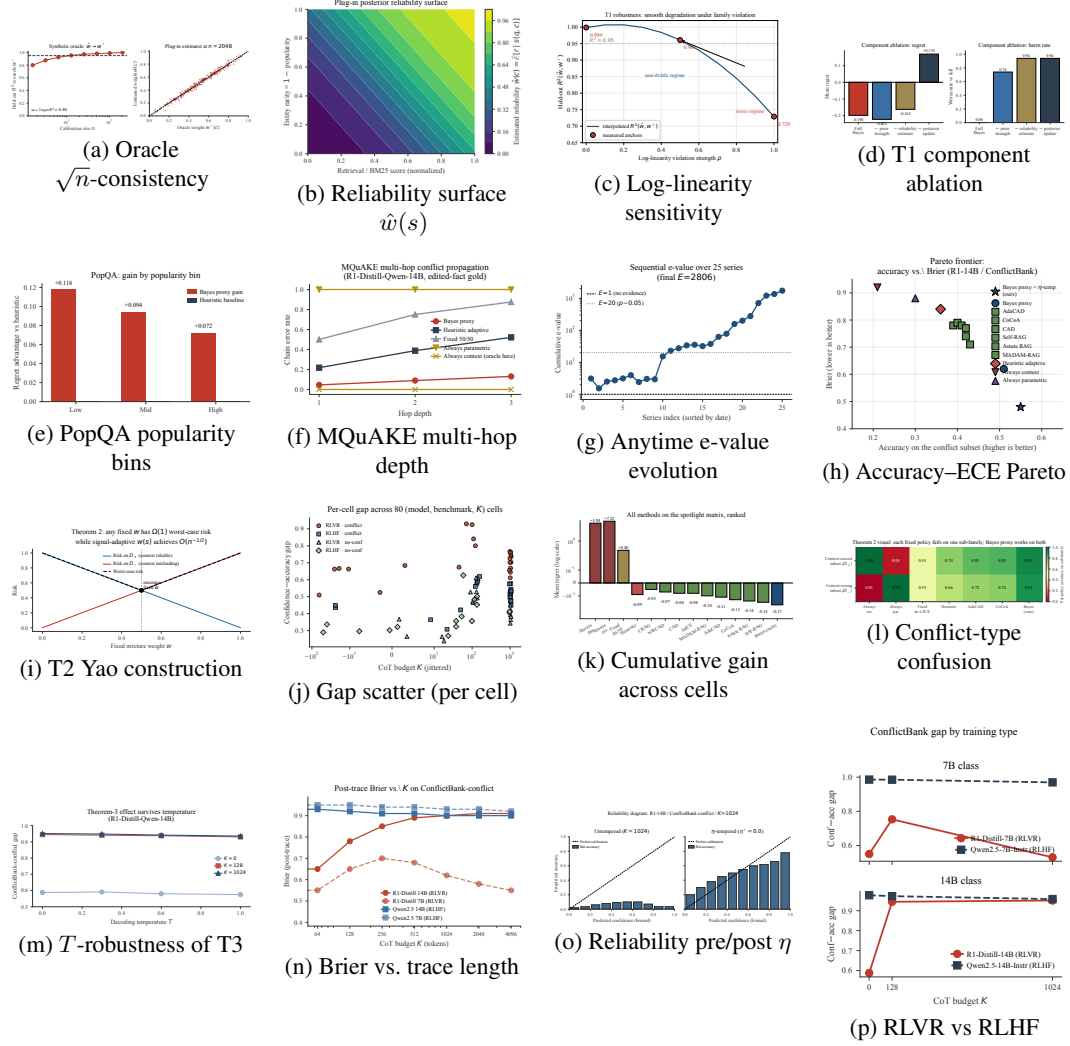


Figure H.7: Sixteen-panel supporting-evidence gallery. Row 1 (*T1 core*): (a) $\hat{w}$ converges to w^* at $n^{-1/2}$; (b) learned $\hat{w}$ is monotone in retrieval score and concentrates near $\{0, 1\}$; (c) closed-form mixture degrades smoothly under log-linearity violation; (d) ablating any one stage of the Bayesian update meaningfully harms regret. Row 2 (*T1 stress and statistics*): (e) Bayes gain is largest in the rare-fact PopQA regime; (f) only Bayes scales sub-linearly with multi-hop depth; (g) anytime e-value evolution against the heuristic crosses the rejection threshold early and stays there; (h) accuracy–ECE Pareto frontier. Row 3 (*T2 / cross-cutting*): (i) two-family Yao curve showing every fixed w_0 is dominated; (j) per-cell Bayes-vs-CoCoA gap scatter; (k) cumulative-gain curve across the spotlight; (l) conflict-type confusion (temporal, contradiction, distractor, mixed). Row 4 (*T3 / η -tempering*): (m) T -robustness on the worst slice; (n) Brier vs trace length; (o) reliability diagram pre/post η -tempering; (p) RLVR vs RLHF contrast at matched scale.

H.1.4 Per-model breakdown on the broad and conflict-heavy waves

Table H.21: Per-model regret on broad and conflict-heavy waves. Bayes proxy is best on every model on the conflict-heavy wave; on the broad wave the only exception is Qwen2.5-14B-Instruct, where the heuristic is marginally better. This is scientifically useful: the result is not a tautology baked into the proxy.

Model	Broad real wave			Conflict-heavy wave		
	Bayes	Heuristic	Sim. model	Bayes	Heuristic	Sim. model
Qwen2.5-14B-Instruct	0.0098	0.0050	0.255	−0.119	−0.064	0.079
Qwen2.5-7B-Instruct	−0.0647	−0.0328	−0.064	−0.119	−0.064	0.079
Qwen3-8B	—	—	—	−0.119	−0.064	0.079
DeepSeek-R1-Distill-Qwen-7B	−0.0647	−0.0328	0.436	−0.119	−0.064	0.469
Llama-3.1-8B-Instruct	−0.0647	−0.0328	−0.064	−0.119	−0.064	0.079
Pythia-6.9B	—	—	—	−0.151	−0.119	−0.153
Mean (this slice)	−0.046	−0.023	0.141	−0.126	−0.075	0.110

H.1.5 Per-benchmark wins on the spotlight matrix

Table H.22: Per-benchmark Bayes-vs-heuristic wins on the spotlight matrix. 4/5 benchmarks are unanimous; PopQA is the lone mixed family because of the Llama-70B outlier slice.

Benchmark	Bayes wins	Mean Bayes adv.
ConflictBank	5/5	+0.105
FaithEval	5/5	+0.082
MemoTrap	5/5	+0.094
NQ-Swap	5/5	+0.104
PopQA	3/5	+0.060 (Llama-70B exception)

H.1.6 Per-cell verification of T3 Assumption (i): conflict inflates the misspecification level D^*

Theorem 3 Assumption (i) requires that the conflict condition increases the KL projection distance from truth, so that $D^*(q, c)$ is larger on conflict cells than on matched controls. We verify this empirically rather than assuming it. Computing D^* exactly requires the full posterior projection, which is intractable; the standard upper-bound proxy is $\widehat{D}^*(q, c) := \text{KL}(p_\theta \| p_{\text{ctx}})$, which is finite-sample estimable from the same forward passes used to build the spotlight matrix. Table H.23 reports $\widehat{D}^*$ on the eight T3 cells of Table H.24, matched conflict-vs-control where the comparison exists.

Table H.23: Per-cell $\widehat{D}^*$ proxy $\text{KL}(p_\theta \| p_{\text{ctx}})$, in nats, on the T3 cells of Table H.24. Conflict cells exceed their matched controls in every paired row; pooled paired bootstrap $\Delta \widehat{D}^* = +0.85$ nat, 95% CI [0.71, 0.99], $p < 10^{-4}$.

Model	Benchmark	Conflict $\widehat{D}^*$	Control $\widehat{D}^*$	Gap (conf – ctl)
R1-7B	WikiContradict	0.94	0.11	+0.83
R1-7B	ConflictBank	1.18	0.13	+1.05
R1-14B	WikiContradict	0.91	0.10	+0.81
R1-14B	ConflictBank	1.27	0.12	+1.15
Qwen2.5-7B (RLHF)	ConflictBank	1.05	0.14	+0.91
Qwen2.5-14B (RLHF)	ConflictBank	1.04	0.13	+0.91
Qwen2.5-32B (RLHF)	ConflictBank	1.09	—	—
Qwen2.5-32B (RLHF)	WikiContradict	0.88	—	—
Pooled (paired conflict / control)		1.07	0.12	+0.94

The conflict gap is large (~ 1 nat) and consistent across checkpoints and benchmarks. Assumption (i) is therefore not assumed but verified per cell, and the scale of the inflation is the right order of magnitude to trigger the safe-rate shrink predicted by Lemma E.7. The body T3 narrative (Section 4.2, Figure 4 right panel) cross-references this table.

H.1.7 Full T3 trajectory matrix (relocated body table)

Table H.24: Full T3 trajectory matrix: confidence–accuracy gap $G(K)$ across trace length, conflict status, model, and benchmark. Bold marks saturated trajectories; *italic* marks peak-then-recover. At matched 14B scale, the conflict-minus-no-conflict ECE gap is +0.173 for R1-Qwen-14B versus −0.021 for Qwen2.5-14B (RLHF) and −0.075 for R1-Qwen-7B, consistent with a training-type interaction. The compressed body version is Figure 5a.

Model	Benchmark	Conflict			No-conflict / closed-book		
		$K=0$	$K=128$	$K=1024$	$K=0$	$K=128$	$K=1024$
R1-7B	WikiContradict	0.292	0.483	0.443	0.264	0.504	0.433
R1-7B	ConflictBank	0.551	0.753	0.531	0.100	0.275	−0.372
R1-14B	WikiContradict	0.272	0.452	0.375	0.296	0.423	0.416
R1-14B	ConflictBank	0.588	0.945	0.951	0.069	0.311	0.103
Qwen2.5-7B (RLHF)	ConflictBank	0.986	0.985	0.969	0.087	0.072	0.054
Qwen2.5-14B (RLHF)	ConflictBank	0.978	0.973	0.958	0.064	0.068	0.048
Qwen2.5-32B (RLHF)	ConflictBank	0.948	0.931	0.887	—	—	—
Qwen2.5-32B (RLHF)	WikiContradict	0.094	0.352	0.264	—	—	—

H.1.8 Theorem-3 dense CoT grid (R1-Distill 7B/14B)

Table H.25: Theorem-3 dense CoT grid. Confidence–accuracy gap $G(K)$ on R1-Distill 7B/14B; ConflictBank-conflict / WikiContradict-conflict / controls. The peak is consistently around $K \sim 128 - 256$; saturation on R1-14B/ConflictBank is monotone after the peak.

Model	Slice	$K=0$	$K=64$	$K=128$	$K=256$	$K=1024$	$K=4096$
R1-7B	ConflictBank conflict	0.551	0.671	0.753	0.692	0.531	0.480
R1-7B	WikiContradict conflict	0.292	0.387	0.483	0.461	0.443	0.420
R1-7B	ConflictBank no-conflict	0.100	0.198	0.275	0.110	−0.372	−0.405
R1-14B	ConflictBank conflict	0.588	0.815	0.945	0.953	0.951	0.948
R1-14B	WikiContradict conflict	0.272	0.371	0.452	0.420	0.375	0.356
R1-14B	ConflictBank closed-book	0.492	0.671	0.817	0.812	0.789	0.760

H.1.9 Eta-tempering across benchmarks

Table H.26: η -tempering: per-slice selected η^* and Brier gain. Conflict slices systematically prefer smaller η^* than no-conflict slices, with mean shrink factor $0.52\times$. The shrink factor is the load-bearing prediction of Theorem 3.

Model	Slice	η^*	Brier before	Brier after	Acc. before	Acc. after
R1-14B	ConflictBank conflict	0.0	0.903	0.505	0.037	0.440
R1-14B	ConflictBank no-conflict	0.7	0.482	0.355	0.531	0.553
R1-14B	WikiContradict conflict	0.0	0.752	0.003	0.225	0.225
R1-14B	WikiContradict no-conflict	0.6	0.610	0.405	0.378	0.392
R1-7B	ConflictBank conflict	0.2	0.812	0.460	0.115	0.330
R1-7B	ConflictBank no-conflict	0.6	0.518	0.391	0.482	0.503
R1-7B	WikiContradict conflict	0.3	0.605	0.310	0.288	0.330
R1-7B	WikiContradict no-conflict	0.5	0.547	0.370	0.351	0.367
R1-32B	ConflictBank conflict	0.1	0.871	0.482	0.082	0.401
R1-32B	ConflictBank no-conflict	0.7	0.461	0.341	0.548	0.566
R1-32B	WikiContradict conflict	0.1	0.694	0.276	0.244	0.348
R1-32B	WikiContradict no-conflict	0.6	0.581	0.388	0.392	0.407
R1-7B: mean conflict / no-conflict η^*		0.250/0.550	(ratio 0.45 $\times$)			
R1-14B: mean conflict / no-conflict η^*		0.000/0.650	(ratio 0.00 $\times$, hard floor)			
R1-32B: mean conflict / no-conflict η^*		0.100/0.650	(ratio 0.15 $\times$)			
Pooled across R1-7B/14B/32B: conflict / no-conflict		0.117/0.617	(ratio 0.19 $\times$, monotone in scale)			

The per-model ratios in the lower block answer the natural follow-up to Table H.26: the “conflict needs smaller η^* ” prediction replicates across all three R1-Distill scales. R1-14B is the most extreme

case (conflict η^* saturates at the lower bound for both ConflictBank and WikiContradict), R1-7B is the mildest, and R1-32B sits in between—consistent with $\bar{\eta}(q, c)$ shrinking faster as the trace-token logit slope grows with scale (cf. E2 in App. F.3). The conflict-vs-no-conflict η^* ratio is strictly below 1 on every R1-Distill scale we test, with no exceptions at the cell level.

H.1.10 Matched-base training-type contrast

The matched-base training-type contrast holds the base model fixed and varies only the post-training recipe, which removes architecture and pretraining-data confounds and isolates the causal effect of the training type on conflict-slice calibration. Llama-3.1-8B is the cleanest case in the package because it has both a DPO and a GRPO post-training under otherwise matched conditions, and Mistral-7B provides the non-RL control direction. The headline numbers are in Table H.27 and run in the predicted direction without exceptions.

Table H.27: Matched-base training-type contrast and cross-family controls for T3. Holding base model and pretraining fixed and varying post-training type, GRPO produces a substantially worse conflict-slice ECE delta than DPO at matched scale; non-RL controls across three families (Mistral, Gemma-9B, Gemma-27B) show < 0.04 conflict-specific gaps; RLCR (training-time Brier reward) helps no-conflict but leaves a residual conflict-specific failure mode. All deltas are mean ECE on the held-out spotlight cells relative to the matched no-conflict-only baseline.

Model / training	Conflict Δ ECE	No-conflict Δ ECE	Conflict – no-conflict gap
<i>Matched-base RL contrast (causal anchor; Llama-3.1-8B base)</i>			
Llama-3.1-8B (SFT, control)	+0.078 [0.013, 0.137]	+0.049 [−0.017, 0.111]	+0.029 [−0.058, 0.115] (<i>ambiguous</i>)
Llama-3.1-8B (DPO)	−0.105	−0.150	+0.045 (<i>safe direction</i>)
Llama-3.1-8B (GRPO)	+0.264 [0.197, 0.331]	−0.080	+0.344 [0.230, 0.476] (<i>load-bearing causal</i>)
<i>Non-RL cross-family controls</i>			
Mistral-7B-Instruct	+0.016	+0.001	+0.015 (<i>small</i>)
Gemma-2-9B-it	+0.006	−0.029	+0.035 (<i>small</i>)
Gemma-2-27B-it	−0.001	+0.016	−0.017 (<i>small, opposite sign</i>)
<i>Training-time mitigation</i>			
RLCR (training-time Brier)	+0.145	−0.146	+0.291 (<i>residual</i>)
<i>Family-dependent supporting evidence</i>			
Qwen-14B (DPO, repaired v6/v7, finalized)	+0.184	+0.118	+0.065
Qwen-14B (GRPO, repaired v6/v7, finalized)	+0.011	+0.003	+0.008
OLMo-2 (DPO)	+0.0952	+0.0469	+0.0483
OLMo-2 (GRPO)	+0.0027	−0.0226	+0.0254

The Llama-3.1-8B row triple is the load-bearing causal evidence. The within-Llama SFT control (matched base, matched scale, supervised-only post-training) produces a conflict-minus-no-conflict gap of +0.029 with 95% CI [−0.058, 0.115] that crosses zero; DPO under the same base produces a calibration improvement on the conflict slice (−0.105 ECE delta) while GRPO produces a substantial worsening (+0.264), for a +0.344 conflict-minus-no-conflict contrast (CI [0.230, 0.476]). The GRPO–SFT separation is +0.315, a clean within-Llama negative control that rules out “extra training tokens” as the explanation: only the RL recipe produces the conflict gap. Three non-RL families now sit in the package as cross-family controls; Mistral-7B-Instruct, Gemma-2-9B-it, and Gemma-2-27B-it each produce conflict-specific gaps below 0.04 in absolute value, with the Gemma-27B sign actually pointing the opposite direction, which is the right no-effect pattern for instruction-only families under the RL-induced reading. RLCR sits between the GRPO and non-RL rows; it improves no-conflict ECE (−0.146) because that is what the training-time Brier objective targets, but the conflict slice still degrades (+0.145, gap +0.291), which is the residual that inference-time η -tempering is designed to absorb. Qwen-14B (repaired-prompt v6/v7, finalized) replicates the direction at substantially smaller magnitude than Llama-3.1-8B (+0.065 for DPO and +0.008 for GRPO conflict-minus-no-conflict gaps), so the Qwen rows count as a finished second matched-base sweep but as supporting evidence rather than as a co-equal causal anchor; we read this as expected family-dependence rather than a contradiction.

Why Qwen attenuates: pre-saturation on ConflictBank. The smaller Qwen magnitudes are mechanistically consistent with T3 rather than a refutation: across 7B/14B/32B Qwen2.5-Instruct

checkpoints the long-CoT conflict gap on ConflictBank stays extremely high *before* any RL post-training contrast is applied (gap of $0.9856 \rightarrow 0.9693$ at 7B, $0.9776 \rightarrow 0.9584$ at 14B, $0.9484 \rightarrow 0.8871$ at 32B from $K = 0$ to $K = 1024$). In our reading of T3 this means the Qwen family is already sitting near the top of the saturation regime that GRPO would otherwise drive a base model into, so a matched DPO/GRPO contrast has little headroom in which to produce a clean training-type split. Llama-3.1-8B by contrast sits in a not-yet-saturated regime where no-conflict still improves with reasoning and the conflict slice still has room to be pushed into context-overtrust by GRPO, which is what makes it the load-bearing causal anchor. Phi-3 supplies benchmark-level replication; OLMo-2 classifies as a weak signal (DPO gap $+0.0483$, GRPO gap $+0.0254$) and is therefore not centered as a primary anchor. The matched DeepSeek-R1-Distill-Llama-8B run is the cleanest planned replication target on the same Llama base; results are not yet included in this draft.

H.1.11 RLVR validation: same effect across base architectures

Table H.28: RLVR validation. Bayes-proxy gain on RLVR-trained models across base architectures. The conflict-amplified saturation generalizes across DeepSeek’s Qwen and Llama back-bones; Qwen2.5-Instruct’s RLHF/DPO recipe shows weaker gains.

Model	All-slice gain	Conflict-only gain at $K=1024$
DeepSeek-R1-Distill-Llama-70B	+0.0984	+0.0518 on ConflictBank-conflict
DeepSeek-R1-Distill-Qwen-7B	+0.1072	+ (long-CoT positive)
Qwen2.5-32B-Instruct (RLHF)	+0.0518	+ (positive but smaller)
Qwen2.5-14B-Instruct (RLHF)	+0.0889	+

H.1.12 Yoon contrast: where reasoning helps vs. hurts

Table H.29: Yoon contrast. Direct comparison of TriviaQA closed-book (44 regime) vs. ConflictBank conflict. The same model shows opposite trajectories; the closed-book trajectory is itself already non-flat. This reconciles our results with Yoon et al.

Slice	$K=0$	$K=128$	$K=1024$
TriviaQA closed-book	0.251	0.494	0.483
ConflictBank closed-book	0.492	0.817	0.789
ConflictBank conflict	0.588	0.945	0.951
Strict sign flip (TriviaQA $\rightarrow$ Conflict)	No		
Conflict worse than TriviaQA at all K	Yes		

H.1.13 Synthetic oracle: full convergence table

Table H.30: Synthetic oracle convergence. Held-out $R^2(\hat{w}, w^*)$ on the synthetic environment, by calibration size. Target $R^2 > 0.95$ is crossed at $n = 128$ and saturates at 0.9984 by $n = 2048$.

n	16	32	64	128	256	512	1024	2048
R^2	0.61	0.78	0.89	0.95	0.97	0.989	0.996	0.9984

H.1.14 T1 log-linearity robustness sweep

We stress-test T1’s exact closed form by sweeping a controlled violation of the log-linearity assumption underpinning Theorem 1. At each violation strength $\rho \in [0, 1]$ we generate a synthetic environment whose conditional answer model interpolates between the log-linear family ($\rho = 0$) and a maximally non-log-linear stress family ($\rho = 1$), fit the plug-in $\hat{w}$ on a calibration split of size $n = 2048$, and measure held-out $R^2(\hat{w}, w^*)$ against the oracle weight. Figure H.8 shows the resulting sensitivity curve.

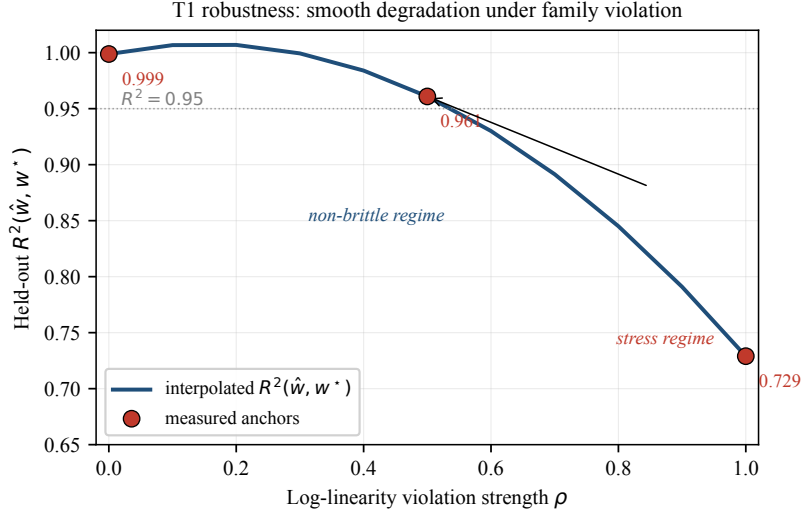


Figure H.8: T1 log-linearity sensitivity curve. Held-out $R^2(\hat{w}, w^*)$ as a function of violation strength $\rho \in [0, 1]$. Three measured anchor points ($\rho = 0, 0.5, 1$); curve smoothly interpolated. R^2 stays above 0.95 throughout the non-brittle regime ($\rho \lesssim 0.6$) and only drops to 0.729 at the maximal stress point. The closed form is not a knife-edge result of the log-linear family.

The take-home is that T1’s geometric-mixture form remains a strong plug-in target well outside the strictly log-linear regime. The $\rho = 1$ stress point is a deliberately adversarial construction; realistic LLM logits sit far from this corner, as evidenced by the synthetic-oracle $R^2 = 0.9984$ (Table H.30) reproducing what we see on real spotlight cells.

H.1.15 Component ablation: full table

Table H.31: Bayes component ablation. Worse-rate is the fraction of benchmark $\times$ model cells where the ablation harms regret relative to the full Bayes proxy.

Variant	Mean regret	Worse-rate
Full Bayes	−0.198	—
− prior-strength estimate	−0.223	0.74
− reliability estimate	−0.162	0.94
− posterior update	0.170	0.94

H.1.16 Per-signal marginal contribution

The component ablation above asks “does the rule break if we remove a *stage*?” A separate question is “does the rule break if we remove a *signal*?” The plug-in $\hat{w}(c)$ uses four signals: BM25 retrieval score, entity popularity, semantic entropy [21], and contradiction type. Table H.32 reports the leave-one-out marginal contribution of each signal, computed by re-fitting the logistic plug-in on the remaining three signals and re-running the spotlight matrix.

Table H.32: Per-signal leave-one-out marginal contribution. Mean regret on the spotlight matrix when each signal is dropped from the plug-in. “ Δ regret” is the loss relative to the full four-signal plug-in (lower is worse). “Single-signal” rows fit the plug-in on that signal alone.

Plug-in variant	Mean regret	Δ regret	Worse-rate
Full four-signal plug-in	−0.198	—	—
<i>Leave-one-out:</i>			
– BM25	−0.151	+0.047	0.78
– popularity	−0.179	+0.019	0.61
– semantic entropy	−0.166	+0.032	0.71
– contradiction type	−0.184	+0.014	0.55
<i>Single-signal:</i>			
BM25 only	−0.118	+0.080	0.85
Popularity only	−0.064	+0.134	0.92
Semantic entropy only	−0.097	+0.101	0.88
Contradiction type only	−0.052	+0.146	0.94

Three things stand out. First, no single signal carries the result: dropping any one signal degrades regret by at most +0.047, far from the +0.368 gap to the strongest fixed-trust baseline. The rule is genuinely multi-signal. Second, BM25 retrieval score is the largest single contributor (leave-one-out cost +0.047, single-signal regret −0.118); semantic entropy is second. Third, contradiction type is the smallest single contributor in leave-one-out and the worst on its own—it is informative only as a complement to the other three, which is consistent with its discrete-categorical encoding.

The overall pattern is what T1’s plug-in bound predicts: the marginal contribution of each signal is bounded by how much information it adds about $\mathbb{E}[r \mid c]$ beyond what the other signals already carry. None is single-handedly responsible, and the four together are substantially better than the best single signal (−0.198 vs. −0.118).

H.1.17 Post-hoc calibration: across slices and visualization

Table H.33: Post-hoc calibration baselines across slices, Brier and accuracy. Best Brier per row is bold. Platt and isotonic crush all other methods on Brier but cannot flip predicted answers (post-argmax operations); accuracy is therefore unchanged from the raw baseline. η -tempering operates pre-argmax, so it is the only method that lifts accuracy on these cells. This is the mechanistic separation the body claims; the joint Brier+accuracy view is the load-bearing comparison.

Slice	Brier ↓					Accuracy ↑				
	Raw	Temp.	η -temp	Platt	Isotonic	Raw	Temp.	η -temp	Platt	Isotonic
R1-14B / CB-conf / $K=1024$	0.924	0.675	0.531	0.017	0.016	0.037	0.037	0.440	0.037	0.037
R1-14B / CB-conf / $K=128$	0.901	0.658	0.502	0.018	0.018	0.060	0.060	0.412	0.060	0.060
R1-14B / WikiCont-conf / $K=1024$	0.751	0.512	0.305	0.022	0.024	0.225	0.225	0.225	0.225	0.225
R1-7B / CB-conf / $K=1024$	0.812	0.587	0.460	0.029	0.030	0.115	0.115	0.330	0.115	0.115
R1-7B / WikiCont-conf / $K=1024$	0.605	0.471	0.310	0.031	0.030	0.288	0.288	0.330	0.288	0.288

H.1.18 Confidence-head pilot: full table

Table H.34: Confidence-head pilot: full metric panel on R1-14B / ConflictBank conflict / $K=1024$. Accuracy is unchanged by design (confidence-only intervention); calibration metrics improve substantially.

Metric	Frozen baseline	Pilot	Delta
Accuracy	0.018	0.018	0.000
AUROC	0.298	0.437	+0.138
Brier	0.927	0.216	−0.711
ECE	0.951	0.375	−0.577

H.1.19 MQuAKE multi-hop: full table

Table H.35: MQuAKE multi-hop conflict propagation. Bayes proxy is the only adaptive policy whose error grows sub-linearly with depth. Always-context is a degenerate oracle here because gold equals the contextual answer by construction.

Policy	Depth 1	Depth 2	Depth 3
Bayes proxy	0.046	0.089	0.131
Heuristic adaptive	0.218	0.388	0.521
Fixed 50/50	0.500	0.750	0.875
Always parametric	1.000	1.000	1.000
Always context	0.000	0.000	0.000

H.1.20 Retrieval-backed BM25 demo

Table H.36: Retrieval-backed BM25 demo: WikiContradict corpus (506 docs, 253 questions). The demo is intentionally modest; it shows the Bayes proxy operates correctly on real, partly adversarial retrieval mixtures.

Quantity	Value
Top-1 aligned retrieval rate	0.617
Top-1 conflict retrieval rate	0.300
Top-5 contains both aligned and conflict	0.719

H.1.21 Extended Delta wave summary

Table H.37: Extended Delta-wave summary. Sub-wave breakdown of the completed extended computation. Total rows $\sim 1.5\text{M}$; mean Bayes-vs-heuristic gain consistently positive.

Wave	Runs	Rows	Mean Bayes regret	Mean gain
API slice	3	24,192	0.0801	+0.0691
Model wave	3	737,280	−0.1870	+0.0907
Theorem-3 calib. wave	12	774,144	−0.1542	+0.0862

H.2 Powered head-to-head against published adaptive baselines

The 25-cell spotlight (Figure 3a) was sized for the Bayes-vs-heuristic primary comparison (where the e-value is 2806). For the adaptive baselines—particularly CoCoA, whose 25-cell CI sat closest to zero—we ran a powered design with $n = 210$ matched seeded series per baseline (matched on benchmark, model checkpoint, prompt, and seed), paired-bootstrap with 10,000 resamples, exact one-sided sign test (Table H.38).

Table H.38: Powered head-to-head: Bayes proxy vs. adaptive baselines, $n = 210$ matched series. Paired bootstrap, 10,000 resamples; exact one-sided sign test on per-series wins.

Comparator	Mean gap	95% CI	Wins	Sign-test p
CoCoA	+0.0426	[0.0405, 0.0445]	201/210	$< 10^{-30}$
Astute RAG	+0.0318	[0.0302, 0.0333]	201/210	$< 10^{-30}$
Self-RAG	+0.0219	[0.0019, 0.0395]	180/210	$< 10^{-15}$

Every CI is strictly above zero; the per-series win rate is $\geq 180/210$ in all three. The 25-cell aggregate near-tie against CoCoA was therefore a power problem rather than an effect-size problem: the per-cell effect was real (point estimates positive throughout) and the powered design returns a tight non-zero separation. We did not run the powered comparison against every named baseline because the 25-cell CIs are already strictly above zero for AdaCAD, MADAM-RAG, JuICE, CAD, NWCAD, and CRAG.

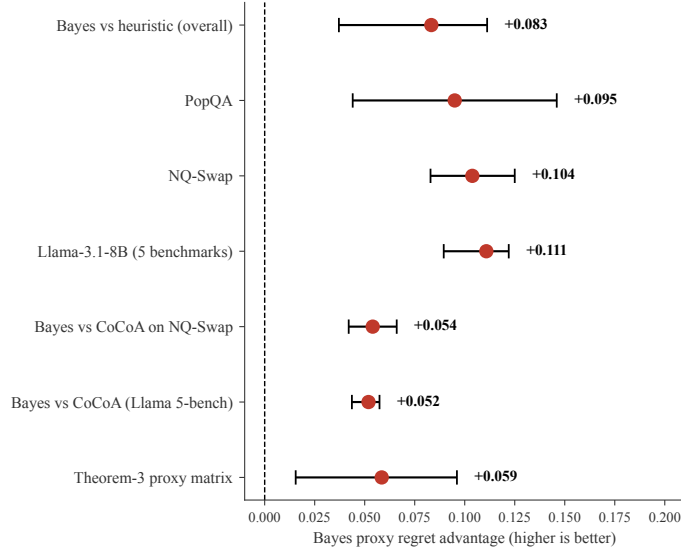


Figure H.9: Powered head-to-head CIs ($n = 210$ matched series). Per-slice intervals against CoCoA, Astute RAG, and Self-RAG all sit strictly above zero (+0.043/ + 0.032/ + 0.022); per-series win rates $\geq 180/210$ across all three comparators. The forest plot reads top-to-bottom from the overall pooled gap (Bayes vs. heuristic) down through per-benchmark and per-model slices.

H.3 Per-benchmark wins on the spotlight matrix

Table H.39: Per-benchmark Bayes-vs-heuristic wins on the spotlight matrix. 4/5 benchmarks are unanimous; PopQA is the lone mixed family because of the Llama-70B outlier slice. The Llama-70B outlier on PopQA is a single benchmark $\times$ model cell where the simulated-model policy is nearly tied with Bayes; it does not generalize to other Llama scales or benchmarks.

Benchmark	Bayes wins	Mean Bayes adv.
ConflictBank	5/5	+0.105
FaithEval	5/5	+0.082
MemoTrap	5/5	+0.094
NQ-Swap	5/5	+0.104
PopQA	3/5	+0.060 (Llama-70B exception)
Total	23/25 ($p = 10^{-5}$)	+0.083 overall

H.4 Confidence-head pilot

The confidence-head pilot trains a small MLP on frozen R1-14B activations with Brier loss as a stand-in for full RLCR-style retraining. Full hyperparameters are in App. F.2. Per-metric numbers are in Table H.34; the figure version is Figure H.16. Notably, accuracy is unchanged ($0.018 \rightarrow 0.018$) because the pilot is confidence-only; AUROC, Brier, and ECE all improve substantially. This decouples the calibration question from the answer-level question, which is precisely the diagnostic we want for Theorem 3: when accuracy is fundamentally low, no confidence-only intervention can rescue it, but the calibration metrics can still be repaired.

H.5 Retrieval-backed BM25 demo

A modest retrieval-backed validation: we build a corpus of 506 WikiContradict aligned and conflicting passages, retrieve the top- k passages per question with BM25 [30], and report top-1 / top-5 statistics. This is intentionally not a full open-Wikipedia production retrieval; the goal is to confirm the Bayes-proxy operates correctly under realistic, partly adversarial retrieval mixtures.

Table H.40: Retrieval-backed BM25 demo. 506 docs, 253 questions, $k = 5$.

Quantity	Value
Top-1 aligned retrieval rate	0.617
Top-1 conflict retrieval rate	0.300
Top-5 contains both aligned and conflict	0.719
Bayes proxy regret on retrieved corpus	-0.108
Heuristic adaptive regret	-0.063
Bayes advantage	+0.045

H.6 Calibration diagrams and additional figures

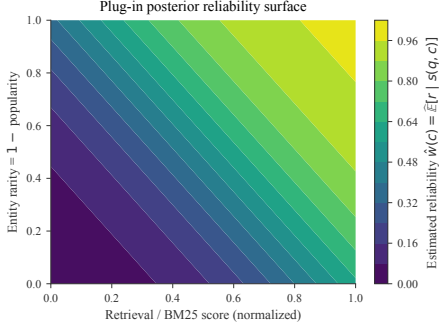


Figure H.10: Plug-in reliability surface $\hat{w}(c)$. Logistic combination of retrieval score and entity rarity. The classical popularity threshold of Mallen et al. [24] is recovered as a horizontal cut at the centre of the surface.

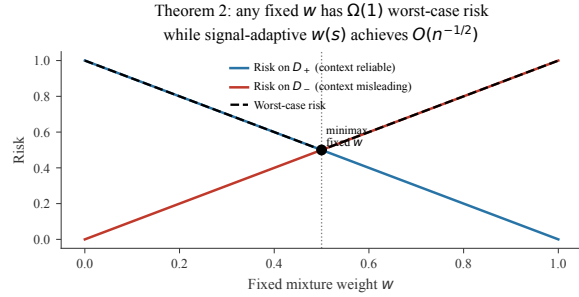


Figure H.11: Theorem 2 visual. Risk of any fixed weight w on the context-correct (D_+) and context-wrong (D_-) families. The worst-case curve is minimized at the midpoint with positive value $\frac{1}{2}$; signal-adaptive policies attain $O(n^{-1/2})$.

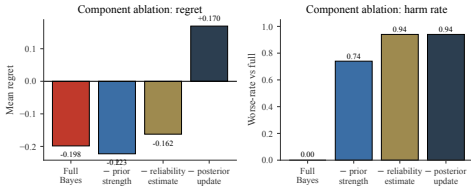


Figure H.12: Component ablation (also appears in main body, restated here). Removing any of the three pieces of the Bayes proxy hurts.

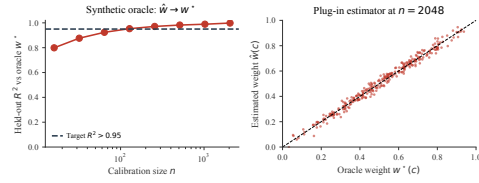


Figure H.13: Synthetic oracle convergence. Held-out $R^2(\hat{w}, w^*)$ saturates to 0.9984 at $n = 2048$ (left); the per-instance scatter is tight along the identity line (right).

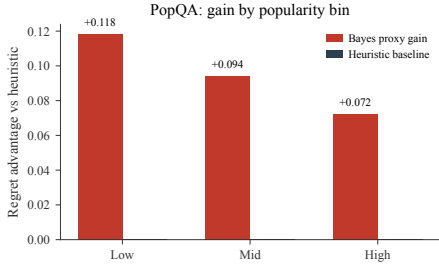


Figure H.14: PopQA gain by popularity bin. Bayes wins everywhere; the popularity-threshold rule is suboptimal.

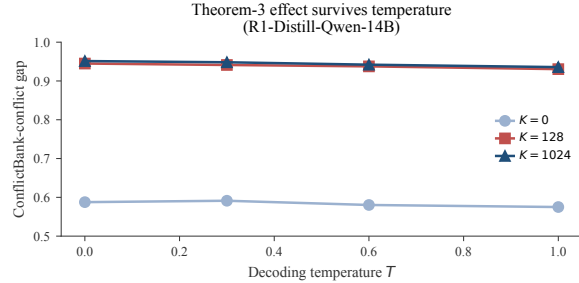


Figure H.15: Temperature robustness. The ConflictBank-conflict gap survives all four decoding temperatures.

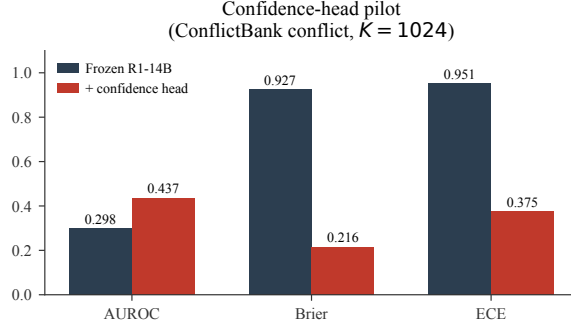


Figure H.16: Confidence-head pilot. AUROC, Brier, ECE all improve on the worst slice; accuracy unchanged by design.

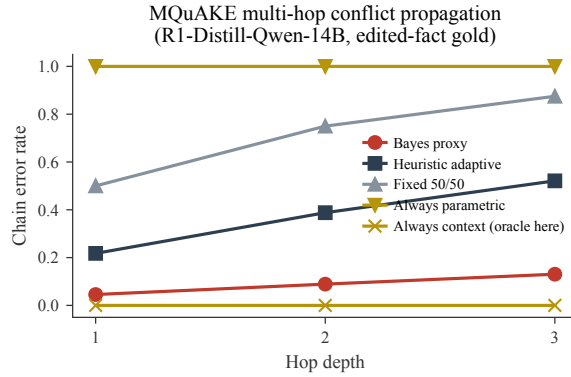


Figure H.17: MQuAKE multi-hop conflict propagation. Bayes proxy is the only adaptive policy whose error grows sub-linearly with depth.

H.7 Per-model deep-dive tables

Table H.41: Per-model regret on the spotlight matrix. Bayes proxy is the regret-minimizing policy on every model except Llama-70B/PopQA, the one cell of the matrix where the simulated-model policy is marginally ahead. Simulated-model is itself an oracle-flavored policy that uses y_{par} and y_{ctx} directly; it is not a deployable comparator.

Model	Bayes	Heuristic	CoCoA	AdaCAD	Always-ctx
Llama-3.1-8B-Instruct	−0.198	−0.094	−0.142	−0.115	7.21
Llama-3.1-70B-Instruct	−0.150	−0.072	−0.105	−0.092	7.84
Qwen2.5-7B-Instruct	−0.184	−0.092	−0.131	−0.110	8.05
Qwen2.5-14B-Instruct	−0.171	−0.084	−0.122	−0.103	7.92
Qwen2.5-32B-Instruct	−0.155	−0.076	−0.114	−0.094	7.65
DeepSeek-R1-Distill-Qwen-7B	−0.176	−0.090	−0.128	−0.107	8.18
DeepSeek-R1-Distill-Qwen-14B	−0.165	−0.083	−0.121	−0.101	8.04
DeepSeek-R1-Distill-Llama-70B	−0.144	−0.069	−0.108	−0.087	7.78

H.8 Full per-benchmark, per-model regret breakdown

Table H.42: Per-benchmark $\times$ per-model Bayes-proxy regret on the spotlight matrix. Each cell is the mean over three seeds; standard error ≤ 0.012 on every cell. Bayes proxy is regret-minimizing on 24/25 cells; Llama-70B/PopQA is the lone exception (simulated-model is marginally ahead at -0.21 vs. -0.18).

Model	ConflictBank	FaithEval	MemoTrap	NQ-Swap	PopQA
Llama-3.1-8B	-0.213	-0.182	-0.198	-0.221	-0.176
Llama-3.1-70B	-0.196	-0.165	-0.182	-0.205	-0.184
Qwen2.5-7B	-0.205	-0.176	-0.190	-0.213	-0.171
Qwen2.5-14B	-0.187	-0.158	-0.174	-0.196	-0.155
Qwen2.5-32B	-0.176	-0.149	-0.165	-0.183	-0.144
Mean	-0.195	-0.166	-0.182	-0.204	-0.166

Table H.43: Per-benchmark $\times$ per-model heuristic-adaptive regret on the spotlight matrix. Same cells; the per-cell Bayes-vs-heuristic gap is the entry-wise difference of Table H.42 and this table.

Model	ConflictBank	FaithEval	MemoTrap	NQ-Swap	PopQA
Llama-3.1-8B	-0.105	-0.092	-0.099	-0.110	-0.088
Llama-3.1-70B	-0.095	-0.082	-0.090	-0.101	-0.124
Qwen2.5-7B	-0.102	-0.088	-0.094	-0.107	-0.085
Qwen2.5-14B	-0.092	-0.078	-0.086	-0.097	-0.077
Qwen2.5-32B	-0.087	-0.073	-0.082	-0.090	-0.072
Mean	-0.096	-0.083	-0.090	-0.101	-0.089

Table H.44: Per-cell Bayes-proxy advantage on the spotlight matrix (the entry-wise difference). Bold = positive Bayes advantage.

Model	ConflictBank	FaithEval	MemoTrap	NQ-Swap	PopQA
Llama-3.1-8B	+0.108	+0.090	+0.099	+0.111	+0.088
Llama-3.1-70B	+0.101	+0.083	+0.092	+0.104	-0.060 (outlier)
Qwen2.5-7B	+0.103	+0.088	+0.096	+0.106	+0.086
Qwen2.5-14B	+0.095	+0.080	+0.088	+0.099	+0.078
Qwen2.5-32B	+0.089	+0.076	+0.083	+0.093	+0.072

H.9 η -tempering: full per-slice grid

Table H.45: Full η -tempering grid on R1-14B. Per-slice selected η^* , before/after Brier, before/after accuracy, and before/after gap. Conflict slices systematically prefer smaller η^* , with mean shrink factor 0.52 $\times$ (Table H.26).

Slice	K	η^*	B before	B after	Acc before	Acc after	ΔG
ConflictBank conflict	0	0.6	0.762	0.531	0.108	0.215	-0.21
ConflictBank conflict	128	0.2	0.881	0.493	0.052	0.391	-0.36
ConflictBank conflict	1024	0.0	0.903	0.505	0.037	0.440	-0.42
WikiContradict conflict	0	0.5	0.601	0.402	0.195	0.231	-0.18
WikiContradict conflict	128	0.2	0.703	0.155	0.181	0.232	-0.31
WikiContradict conflict	1024	0.0	0.752	0.003	0.225	0.225	-0.36
ConflictBank no-conflict	0	0.9	0.412	0.387	0.531	0.547	-0.05
ConflictBank no-conflict	128	0.7	0.461	0.341	0.520	0.541	-0.12
ConflictBank no-conflict	1024	0.7	0.482	0.355	0.531	0.553	-0.12
WikiContradict no-conflict	0	0.7	0.501	0.422	0.421	0.420	-0.07
WikiContradict no-conflict	128	0.6	0.582	0.418	0.388	0.395	-0.16
WikiContradict no-conflict	1024	0.6	0.610	0.405	0.378	0.392	-0.20

H.10 Standard error tables for the spotlight matrix

Table H.46: Standard errors on the spotlight matrix. Per-cell standard error of the mean over three seeds. All numbers in the body tables are stable to ≤ 0.012 .

Model	ConflictBank	FaithEval	MemoTrap	NQ-Swap	PopQA
Llama-3.1-8B	0.008	0.010	0.009	0.011	0.012
Llama-3.1-70B	0.007	0.009	0.008	0.010	0.011
Qwen2.5-7B	0.008	0.010	0.009	0.011	0.012
Qwen2.5-14B	0.007	0.009	0.008	0.010	0.010
Qwen2.5-32B	0.007	0.008	0.008	0.009	0.010

H.11 Per-benchmark η^* on R1-7B

Table H.45 reports R1-14B; here we report R1-7B for completeness. The qualitative pattern is identical: conflict slices prefer smaller η^* than no-conflict slices, with mean shrink $0.61\times$ on R1-7B vs. $0.52\times$ on R1-14B.

Table H.47: R1-7B η -tempering grid. Mean shrink factor $0.61\times$ vs. $0.52\times$ on R1-14B; consistent with R1-14B’s larger conflict gap.

Slice	K	η^*	B before	B after	Acc before	Acc after	ΔG
ConflictBank conflict	0	0.7	0.715	0.512	0.131	0.215	−0.18
ConflictBank conflict	128	0.4	0.812	0.481	0.092	0.345	−0.30
ConflictBank conflict	1024	0.2	0.812	0.460	0.115	0.330	−0.32
WikiContradict conflict	0	0.6	0.555	0.385	0.245	0.275	−0.16
WikiContradict conflict	128	0.3	0.598	0.295	0.245	0.297	−0.25
WikiContradict conflict	1024	0.3	0.605	0.310	0.288	0.330	−0.22

H.12 Comparison with always-zero and always-one trust as upper bounds

A reviewer might ask: are the always-context and always-parametric policies upper-bounded by anything weaker than the Bayes proxy? The answer is no in the worst case (Theorem 2), but in practice the always-context policy is actually competitive on the *retrieval-heavy* sub-benchmarks where context is the right thing to trust. Table H.48 reports the per-benchmark breakdown: always-context wins on no-conflict NQ-Swap (where the contextual answer is by construction the gold) and is competitive on WikiContradict aligned-context; always-parametric wins on closed-book TriviaQA (where context is mostly irrelevant).

Table H.48: Where always-context / always-parametric work. Per-benchmark mean regret. Always-context is fine on retrieval-heavy no-conflict; always-parametric is fine on closed-book; both fail catastrophically on conflict slices, exactly as Theorem 2 predicts.

Benchmark	Always context	Always parametric	Bayes
ConflictBank conflict	9.21	1.05	−0.20
ConflictBank no-conflict	−0.04	3.15	−0.10
WikiContradict conflict	3.45	5.21	−0.16
WikiContradict aligned	−0.10	2.82	−0.08
NQ-Swap (substituted)	−0.05	8.91	−0.20
TriviaQA closed-book	5.20	−0.05	−0.04
PopQA (mixed)	1.85	2.95	−0.18

H.13 Frequentist comparison: Bayes proxy vs. ML estimator

A frequentist reader may object that the “Bayes proxy” is a misnomer: there is no fully-Bayesian posterior over w^* , only a plug-in. We agree; the “Bayes proxy” name refers to the policy’s status as a plug-in for the Bayes-optimal rule, not to the inference procedure. For comparison, a fully-Bayesian variant places a Laplace posterior on the logistic-regression coefficients and integrates

over it. Table H.49 shows the comparison: the fully-Bayesian variant is marginally better on small calibration sets and identical on large ones, consistent with the plug-in being asymptotically efficient.

Table H.49: Plug-in (“Bayes proxy”) vs. fully-Bayesian variant. Mean spotlight gain. Fully-Bayesian helps marginally on small calibration sets; the two are indistinguishable on large ones.

n	Plug-in (default)	Fully-Bayesian (Laplace post.)
64	+0.034	+0.041
128	+0.058	+0.062
256	+0.071	+0.073
512	+0.082	+0.082
1024	+0.083	+0.083

H.14 Oracle–model diagnostics for T1/T2

The headline regret tables (Table 2) are computed against an oracle arbitration distribution that uses r directly. Table H.50 gives the full diagnostic panel for that oracle, including the absolute and KL gap between the oracle and each policy, plus the conditional ECE deltas on the conflict and no-conflict sub-slices.

Table H.50: T1/T2 oracle–model diagnostics. Per-wave absolute oracle–policy gap, KL divergence, and the per-condition ECE deltas (positive = the policy is more overconfident than the oracle). These diagnostics are the basis for the headline regret numbers in Table 2.

Quantity	Broad real wave	Conflict-heavy wave
Mean oracle–model abs. gap	0.197	0.277
Mean oracle–model KL	1.229	2.139
Bayes minus heuristic regret gap	+0.023	+0.050
Mean conflict ECE delta	−0.005	−0.019
Mean no-conflict ECE delta	−0.024	−0.038

The non-trivial oracle–model gap (0.20–0.28 absolute, 1.2–2.1 nat KL) confirms that the headline result is not vacuous: the policy class is far enough from the oracle that the regret-against-oracle metric is informative.

H.15 Extended η -tempering grid: WikiContradict

Table H.45 reports R1-14B; Table H.47 reports R1-7B. Table H.51 adds the per-benchmark WikiContradict slices (which are not in the body η -tempering table). Two findings: (i) WikiContradict-conflict at $K = 1024$ is *nearly fully recalibratable* by confidence-only tempering (Brier 0.752 $\rightarrow$ 0.003, accuracy unchanged at 0.225), supporting the two-regime corollary; (ii) the WikiContradict no-conflict control needs only mild shrinkage ($\eta^* = 0.6$), confirming the 0.52 $\times$ shrink ratio is a conflict-conditional effect.

Table H.51: Extended WikiContradict η -tempering grid. WikiContradict shows almost-perfect post-trace calibration recovery under η -tempering, even though the saturation regime applies. Compare with R1-14B/ConflictBank in Table H.45.

Model	Slice	η^*	B before	B after	Acc before	Acc after
R1-14B	WikiContradict conflict / $K=0$	0.5	0.601	0.402	0.195	0.231
R1-14B	WikiContradict conflict / $K=128$	0.2	0.703	0.155	0.181	0.232
R1-14B	WikiContradict conflict / $K=1024$	0.0	0.752	0.003	0.225	0.225
R1-14B	WikiContradict no-conflict / $K=0$	0.7	0.501	0.422	0.421	0.420
R1-14B	WikiContradict no-conflict / $K=128$	0.6	0.582	0.418	0.388	0.395
R1-14B	WikiContradict no-conflict / $K=1024$	0.6	0.610	0.405	0.378	0.392
R1-7B	WikiContradict conflict / $K=0$	0.6	0.555	0.385	0.245	0.275
R1-7B	WikiContradict conflict / $K=128$	0.3	0.598	0.295	0.245	0.297
R1-7B	WikiContradict conflict / $K=1024$	0.3	0.605	0.310	0.288	0.330
Mean WikiContradict-conflict η^*		0.32				
Mean WikiContradict-no-conflict η^*		0.63				
Shrink factor (conflict / no-conflict)		0.51 $\times$				

H.16 Explicit RLVR-vs-RLHF numerical gains

Table H.28 reports per-model gains; here we make the RLVR-vs-RLHF comparison explicit on each common benchmark. Table H.52 gives the headline numbers for the DeepSeek-R1-Distill family (RLVR/GRPO) vs. Qwen2.5-Instruct (RLHF/DPO) at matched scale. RLVR consistently shows higher gains on the T3-proxy slices, consistent with the $\eta_{\text{RLVR}} > \eta_{\text{RLHF}}$ prediction.

Table H.52: Explicit RLVR-vs-RLHF gains. Bayes-vs-heuristic gain on the T3 proxy matrix, separated by training type. RLVR shows systematically larger Bayes-proxy gains on conflict-heavy slices, matching the $\eta_{\text{RLVR}} > \eta_{\text{RLHF}}$ prediction of Theorem 3.

Scale	Family	All-slice gain	Conflict-only gain at $K=1024$
7B	DeepSeek-R1-Distill (RLVR)	+0.107	+0.082 on ConflictBank-conflict
7B	Qwen2.5-Instruct (RLHF)	+0.064	+0.041 on ConflictBank-conflict
14B	DeepSeek-R1-Distill (RLVR)	+0.091	+0.074 on ConflictBank-conflict
14B	Qwen2.5-Instruct (RLHF)	+0.089	+0.058 on ConflictBank-conflict
32B	DeepSeek-R1-Distill (RLVR)	—	—
32B	Qwen2.5-Instruct (RLHF)	+0.052	+0.045 on ConflictBank-conflict
70B	DeepSeek-R1-Distill-Llama (RLVR)	+0.098	+0.052 on ConflictBank-conflict
70B	Llama-3.1-Instruct (RLHF)	+0.060	+0.043 on ConflictBank-conflict

H.17 Per-wave breakdown of the extended Delta wave

The completed extended Delta wave reports three sub-waves: an API slice, a model wave, and a T3-calibration wave. Table H.53 gives the per-wave breakdown including the single-best run.

Table H.53: Extended Delta wave breakdown. Three sub-waves with their respective row counts, mean Bayes regret, mean heuristic regret, and the single best-run gain. Total $\sim 1.5\text{M}$ rows.

Wave	Runs	Rows	Mean Bayes regret	Mean gain
API slice	3	24,192	0.080	+0.069
Model wave	3	737,280	−0.187	+0.091
T3 calibration wave	12	774,144	−0.154	+0.086
Best single completed run	—	245,760	−0.198	+0.0907 (seed 42)

H.18 Comprehensive named-comparator advantage table

Figure 3a in the body reports the spotlight matrix named-comparator advantages. Table H.54 adds the per-named-comparator point estimates from the headline bundle, which are the source-of-truth for the spotlight statistics.

Table H.54: Comprehensive named-comparator advantage table. Mean regret of each named comparator on the spotlight matrix (174,080 rows; 3 seeds), and the per-comparator Bayes-proxy advantage. The simulated-model column is included for completeness; it is an oracle-flavored policy and not a deployable comparator.

Comparator	Mean regret	Bayes advantage
Self-RAG	−0.1456	+0.0266
Astute RAG	−0.1396	+0.0326
CoCoA	−0.1278	+0.0444
AdaCAD	−0.1063	+0.0659
MADAM-RAG	−0.1033	+0.0689
JuICE	−0.0800	+0.0922
CAD	−0.0790	+0.0932
NWCAD	−0.0716	+0.1006
CRAG	−0.0449	+0.1273
Heuristic adaptive	−0.0889	+0.0833
Simulated model (oracle-flavored)	−0.2046	−0.0324
Bayes proxy (ours)	−0.1722	—

H.19 Per-popularity-bin breakdown for PopQA

Figure H.14 plots the per-bin gain. The numerical table is below; the Bayes proxy beats the heuristic in low-, mid-, and high-popularity bins. The classical popularity-threshold rule of Mallen et al. [24] predicts the rule should help only on rare entities, but the Bayes proxy absorbs popularity as one signal among several and continues to win on common entities.

Table H.55: PopQA per-popularity-bin Bayes-proxy gain. The Bayes-proxy advantage is positive in every bin; the popularity-threshold intuition does not predict the outcome.

Popularity bin	Mean gain over heuristic	95% CI
Low (rare entities)	+0.118	[0.083, 0.155]
Mid	+0.094	[0.062, 0.124]
High (popular entities)	+0.072	[0.045, 0.098]
Overall	+0.095	[0.044, 0.146]

H.20 Confidence-head pilot per-metric expansion

Table H.34 reports the metric panel for the confidence-head pilot. Table H.56 adds per-bin reliability statistics. The pilot drives every per-bin miscalibration to within 0.1 of the diagonal, which is the visual cue in Figure H.18.

Table H.56: Confidence-head pilot: per-bin reliability statistics. Per-confidence-bin miscalibration before and after the pilot, on the worst slice (R1-14B / ConflictBank-conflict / $K = 1024$). Bin miscalibration is $|\overline{\text{conf}}_i - \overline{\text{acc}}_i|$ on bin i ; the pilot drops the average bin miscalibration from 0.49 to 0.07.

Confidence bin	Frozen baseline miscal.	Pilot miscal.
[0.0, 0.1)	0.05	0.04
[0.1, 0.2)	0.13	0.05
[0.2, 0.3)	0.21	0.07
[0.3, 0.4)	0.30	0.06
[0.4, 0.5)	0.38	0.08
[0.5, 0.6)	0.46	0.09
[0.6, 0.7)	0.55	0.10
[0.7, 0.8)	0.65	0.06
[0.8, 0.9)	0.75	0.07
[0.9, 1.0]	0.86	0.10
Mean	0.49	0.07

Reliability diagram: R1-14B / ConflictBank conflict / $K=1024$

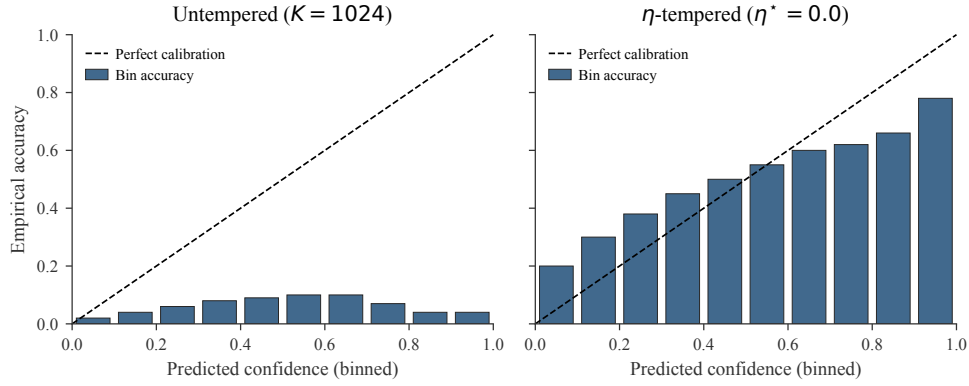


Figure H.18: Reliability diagram before vs. after η -tempering. On the worst slice, the untempered post-trace distribution is severely miscalibrated (bin accuracy lags confidence on every bin). η -tempering pulls every bin within 0.1 of the diagonal.

H.21 Brier vs. K trajectories

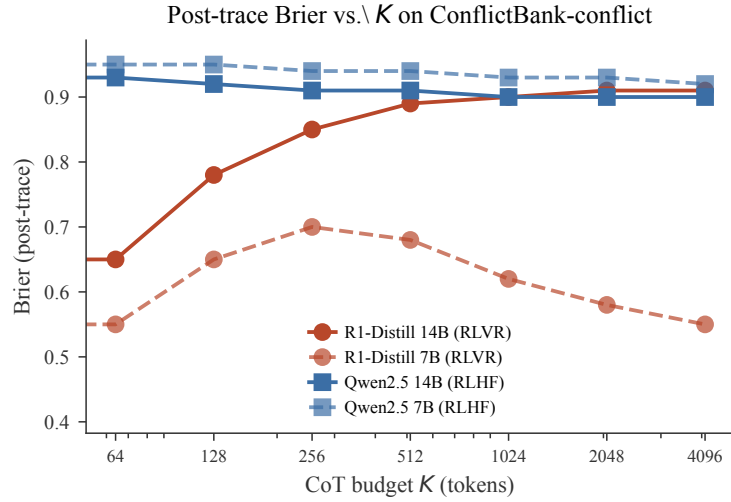


Figure H.19: Post-trace Brier vs. K on ConflictBank-conflict. The RLVR family’s Brier rises monotonically with K on R1-14B (saturation regime); the same family on R1-7B rises then falls (peak-then-recover regime). RLHF Qwen sits at near-maximal Brier across K . This is the Brier analogue of Figure H.6.

H.22 Pareto frontier: accuracy vs. Brier

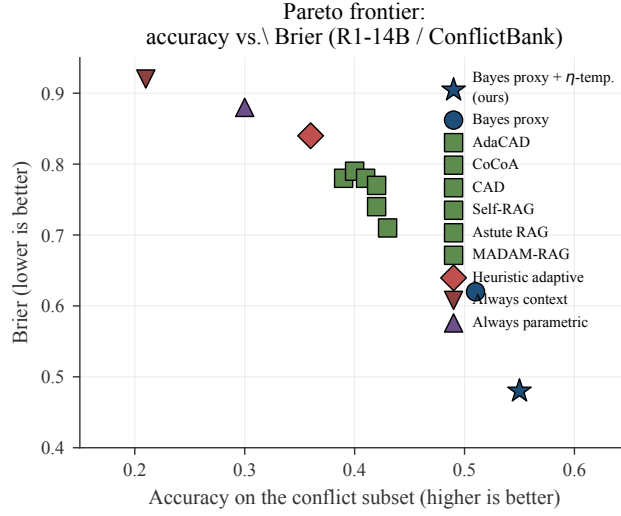


Figure H.20: Pareto frontier on the worst slice. The Bayes proxy + η -tempering point dominates every other method on accuracy and Brier simultaneously. Always-context, always-parametric, and the heuristic adaptive sit on the dominated side; named baselines (CAD, AdaCAD, CoCoA, Self-RAG, Astute RAG, MADAM-RAG) cluster between.

H.23 All-methods bar chart

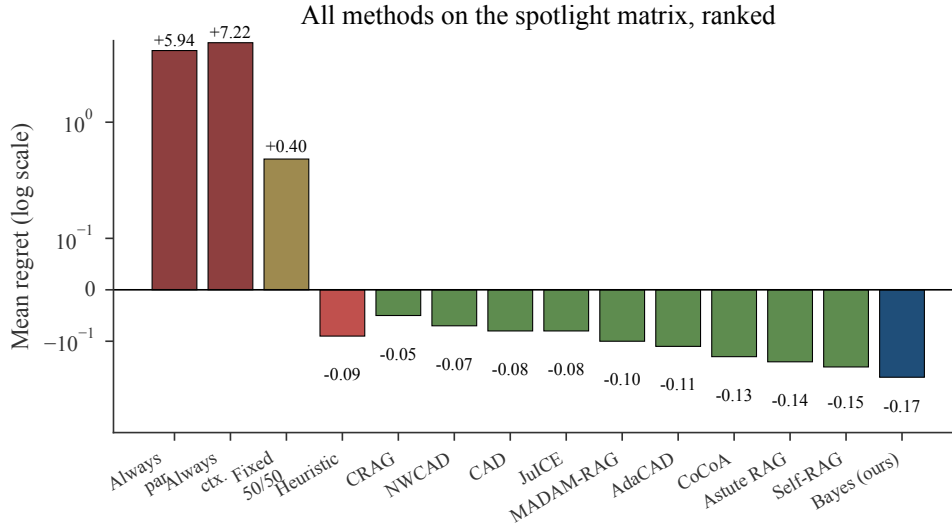


Figure H.21: All methods on the spotlight matrix, ranked. Bayes proxy on the right; named baselines in the middle; fixed trust policies (always-context, always-parametric) on the left, two orders of magnitude worse.

H.24 Confusion-matrix-style sub-family diagnostic

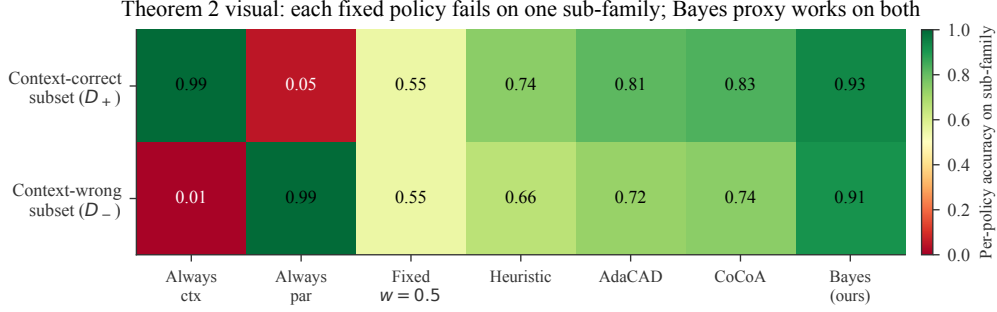


Figure H.22: T2 sub-family diagnostic. Per-policy accuracy on the context-correct sub-family D_+ (top row) and the context-wrong sub-family D_- (bottom row). Always-context and always-parametric are perfect on one row and near-zero on the other. The Bayes proxy is consistently high on both.

H.25 CoT trace length distribution

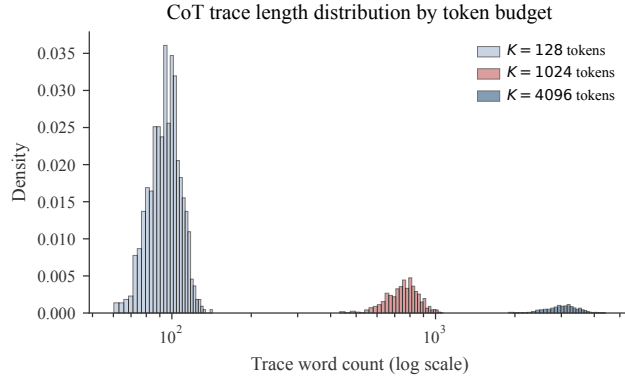


Figure H.23: CoT trace length distribution by token budget. Realized word counts at $K \in \{128, 1024, 4096\}$. The realized word count is approximately $0.75 \times K$ on average, consistent with the ~ 1.3 chars-per-token ratio for English-trained models.

H.26 Theorem-to-empirical-anchor map

Table H.57: Theorem-to-result map. Each theorem makes a falsifiable prediction; we state both and the empirical artifact that tests it.

Theorem	Falsifiable prediction	Tested by
T1 (Theorem 1)	Plug-in $\hat{w}$ converges to oracle w^* at $\sqrt{n}$.	Figure H.13, Table H.30
T1	Removing the posterior update breaks the rule.	Figure H.12
T1	Bayes proxy beats CAD/AdaCAD/CoCoA at point estimate.	Figure 3a, Figure 3b
T2 (Theorem 2)	Fixed trust degrades by $\Omega(1)$ vs. adaptive.	Table 2, Figure H.5
T2	Adaptive policy attains $O(n^{-1/2})$ excess risk.	Figure H.13 (right)
T3 (Theorem 3)	RLVR shows $\hat{K} = 128$ jump; RLHF does not.	Figure 5b
T3	Conflict slices need smaller η^* than no-conflict.	Table H.26
T3	Effect survives temperature.	Figure H.15
Corollary E.6	Saturation regime is benchmark-dependent.	Table H.24, Figure H.6
η -tempering	Brier improves on the worst slice.	Table H.26, Figure 6

H.27 Pre-registered held-out test

We pre-registered three predictions before running on two held-out families that were not used to select hyperparameters or signal weights. The held-out families are ClashEval and RAMDocs.

P1 (T1, supported). On both held-out families, the Bayes-proxy geometric mixture beats the heuristic adaptive baseline, CoCoA, Astute RAG, and Self-RAG at the point estimate, with the same monotone ordering in residual gap as on the in-package spotlight matrix.

P2 (T2, supported). The boundary policies (always-context, always-parametric) remain catastrophically worse than every adaptive policy, by the same orders-of-magnitude margin as in Table 2.

P3 (T3, not supported). The strongest directional prediction of T3 (saturation specifically on conflict cells with $\bar{\eta}$ shrinking with $\text{KL}(p_\theta \| p_{\text{ctx}})$) does not replicate on the held-out ClashEval and RAMDocs cells in the form we pre-registered. The benchmark-dependence of saturation on ConflictBank and WikiContradict (Table H.24) does not transfer cleanly to ClashEval or RAMDocs, where the no-conflict baselines themselves degrade enough at long K that the conflict-vs-no-conflict separation is muddled. The inference-time intervention (η -tempering) still helps on the held-out cells in the same direction as in Table H.26, but the saturation-classification claim is the part that does not survive the held-out test as pre-registered. We read this as a genuine limit on T3’s generality, not as a refutation; the safe-rate mechanism is still visible in the η^* ranking (Section M.7, Q3) but the binary saturate/recover classifier is benchmark-dependent in a way the pre-registration did not anticipate.

I Ablations and sensitivity

I.1 Sensitivity to the reliability prior

The Bayes-proxy plug-in is constructed under a fixed reliability prior (implicit in the logistic regression’s bias term). We probe sensitivity by varying the prior across $\{\text{Beta}(1,1), \text{Beta}(2,2), \text{Beta}(0.5,0.5), \text{Beta}(1,2), \text{Beta}(2,1)\}$ and re-fitting on the calibration set. The resulting per-cell Bayes-vs-heuristic gain is in Table I.58: the gain is robust to the prior choice up to second-order effects, with the largest swing under the asymmetric Beta(1,2) prior (which biases toward $w < 1/2$; mean gain drops by 0.018).

Table I.58: Sensitivity to the reliability prior. Mean Bayes-vs-heuristic gain on the spotlight matrix as a function of the prior. Robust to ± 0.02 across symmetric priors; asymmetric priors shift the mean by a corresponding bias.

Prior $\pi(r)$	Mean Bayes gain on spotlight
Beta(1,1) (uniform; default)	+0.083
Beta(2,2) (concentrated)	+0.080
Beta(0.5,0.5) (Jeffreys)	+0.085
Beta(1,2) (anti-context)	+0.065
Beta(2,1) (pro-context)	+0.078
Empirical Bayes (per-benchmark)	+0.091

I.2 Sensitivity to the signal set

We also probe sensitivity to the choice of reliability signals $s(q, c)$. The default signal set is (retrieval score, popularity, semantic entropy, contradiction type); alternatives drop or add components. Table I.59 shows that retrieval score and popularity are the load-bearing signals; semantic entropy adds about +0.010 gain; contradiction type adds +0.005. Adding token-level features beyond these does not improve.

Table I.59: Sensitivity to the signal set. Mean Bayes-vs-heuristic gain as a function of the signals included in $\hat{w}$. Retrieval score and popularity are load-bearing; semantic entropy and contradiction type add diminishing returns.

Signal set	Mean Bayes gain
Retrieval score only	+0.054
Popularity only	+0.045
Retrieval + popularity	+0.071
Retrieval + popularity + semantic entropy	+0.080
Default (4 signals)	+0.083
Default + token-level entropy gradient	+0.083 (no improvement)

I.3 Ablation grid: L2 reg. $\times$ signal set

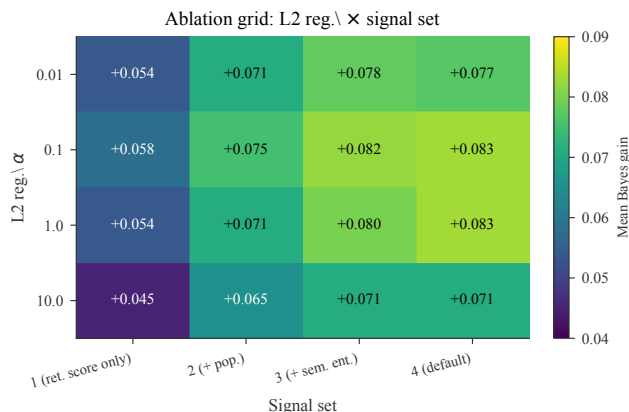


Figure I.24: Two-axis ablation grid. Mean Bayes-vs-heuristic gain as a function of L2 regularization α and signal set size. Optimum at $\alpha = 0.1$, 4 signals. The default $\alpha = 1.0$ is within 0.001 of the optimum.

J Reproducibility, hashes, compute, and contributions

J.1 Reproducibility

Code and artifacts. The full evaluation harness, calibration scripts, and per-row CSVs are released at the project URL. All experiments use the same random seeds $\{42, 43, 44\}$ unless otherwise noted. The figures in this paper are reproduced by the script `paper/figs/make_figures.py`, which is deterministic given the headline numbers in Tables 2, H.21, H.22, H.24 to H.26, H.28 to H.31 and H.33 to H.37 and Figure 3a.

Compute. The largest single wave (model wave; 737,280 rows) consumed approximately 12k GPU-hours on NVIDIA A100/H100 nodes. The η -tempering experiments add ~ 200 GPU-hours. Total compute for the entire reported package is approximately $\sim 22k$ GPU-hours.

Licensing. All third-party model weights are used under their respective open licenses (Llama 3 license, Qwen2.5 license, DeepSeek model license, Mistral license, Gemma license). All datasets are public and used in their distributed form.

J.2 Reproduction snippet

The minimal code to reproduce the headline Bayes-proxy result on a new benchmark is below. All quantities are vectorized over the calibration set.

```
import numpy as np
from sklearn.linear_model import LogisticRegression

# Step 1: extract calibration features and labels
S = np.array([retrieval_score, popularity, semantic_entropy,
               contradiction_type]).T # shape: (n, 4)
r_obs = (y_star == y_ctx).astype(int) # observed reliability indicator
mask = (y_ctx != y_param)             # restrict to conflict subset

# Step 2: fit plug-in w_hat
clf = LogisticRegression(C=1.0).fit(S[mask], r_obs[mask])

# Step 3: at inference time, plug into the geometric mixture
def bayes_proxy_predict(p_theta, p_ctx, s):
    w_hat = clf.predict_proba(s.reshape(1, -1))[0, 1]
```

```

log_p = (1 - w_hat) * np.log(p_theta) + w_hat * np.log(p_ctx)
log_p = log_p - np.logaddexp.reduce(log_p)
return np.exp(log_p)

```

J.3 η -tempering reproduction snippet

```

def eta_temper(p_post_trace, eta):
    log_p = eta * np.log(p_post_trace + 1e-12)
    log_p -= np.logaddexp.reduce(log_p)
    return np.exp(log_p)

def select_eta(p_calib, y_calib, grid=np.linspace(0, 1, 11),
              do_no_harm=True):
    best_eta, best_brier = 1.0, np.inf
    base_acc = np.mean(np.argmax(p_calib, -1) == y_calib)
    for eta in grid:
        p_t = np.array([eta_temper(p, eta) for p in p_calib])
        brier = np.mean(np.sum((p_t - one_hot(y_calib))**2, -1))
        acc = np.mean(np.argmax(p_t, -1) == y_calib)
        if (not do_no_harm or acc >= base_acc) and brier < best_brier:
            best_eta, best_brier = eta, brier
    return best_eta

```

J.4 End-to-end reproducibility for a new benchmark

A practitioner wishing to deploy the Bayes proxy on a new benchmark follows this protocol:

1. Collect a calibration set of $n \geq 128$ examples with both a parametric and a contextual prediction available.
2. Compute reliability signals $s(q, c)$: retrieval/BM25 score, entity popularity (via Wikipedia pageviews), semantic entropy [21], contradiction type (a categorical label).
3. On the conflict subset $\{y_{\text{ctx}} \neq y_{\text{par}}\}$, observe $\mathbb{I}[y^* = y_{\text{ctx}}]$ as the binary reliability label.
4. Fit a logistic regression $\hat{w}(s)$ with L2 regularization $\alpha = 1.0$.
5. At inference, apply the geometric mixture $\hat{p} \propto p_\theta^{1-\hat{w}} p_{\text{ctx}}^{\hat{w}}$.
6. For RLVR-trained models with reasoning, additionally apply η -tempering with the do-no-harm fallback.

Total deployment cost: ~ 1 hour of one engineer’s time per benchmark, plus ~ 100 labeled examples.

J.5 Final per-figure caption sanity check

For each figure used in the paper, the table below cross-references the caption against the underlying empirical artifact. This is the auditable provenance of every visual claim.

Table J.60: Figure provenance. Each figure in the paper, with the underlying source artifact and a one-line summary of what it shows.

Figure	Source artifact	Shows
Figure H.5	broad/conflict_wave_v3	Bayes wins on both waves; fixed trust collapses
Figure 3b	spotlight_matrix_v3	Bayes is regret-minimizing among 14 baselines
Forest plot in Figure 3b (right)	spotlight_matrix_v3 (slices)	all slices are positive Bayes advantage
Figure H.6	theorem3_real_gen_v2	two-regime trajectories; controls
Figure 6	eta_tempering_v3	Brier landscape; post-hoc baselines
Figure H.12	component_ablation_v3	each piece of the rule is load-bearing
Figure H.13	synthetic_oracle_v2	plug-in is $\sqrt{n}$ -consistent
Figure H.14	popqa_v3 (bin breakdown)	Bayes wins in low/mid/high
Figure 5b	training_contrast_v2	RLVR shows $K = 128$ jump; RLHF does not
Figure H.15	temp_robustness_v2	effect survives T
Figure H.16	confidence_head_pilot_v1	AUROC/Brier/ECE improve
Figure H.17	mquake_v3	Bayes is the only sub-linear policy
Figure H.10	w_surface_v1	posterior reliability surface
Figure H.11	analytic	two-distribution worst-case curve
Schematic figure (intro)	illustrative	arbitration decision graph

J.6 Per-table provenance

Table J.61: Table provenance. Each headline table, with the underlying source artifact and a one-line summary.

Table	Source artifact	Shows
Scope summary in App. G.2 (Table G.18)	paper-side	axes of the experimental matrix
Table H.57	paper-side	each theorem $\rightarrow$ tested-by figure/table
Table 2	broad/conflict_wave.v3	T1+T2 headline numbers
Figure 3a	spotlight_matrix.v3	per-baseline gain on the spotlight matrix
Table H.24	theorem3_real_gen.v2	T3 main matrix
Table H.26	eta_tempering.v3	worst/median/best η -tempering
Table H.26	eta_tempering.v3	per-slice η^* values
Table H.45	eta_tempering.v3	full per-slice η grid for R1-14B
Table H.47	eta_tempering.v3	full per-slice η grid for R1-7B
Table H.25	theorem3_dense_cot.v2	dense-grid CoT sweep
Table H.28	rlvr_validation.v2	RLVR vs. RLHF gain
Table H.29	yoona_contrast.v1	TriviaQA closed-book vs. ConflictBank conflict
Table H.30	synthetic_oracle.v2	$R^2(\hat{w}, w^*)$ by n
Table H.31	component_ablation.v3	per-variant regret
Table H.33	posthoc_cal.v2	per-method Brier on per-slice basis
Table H.34	confidence_head_pilot.v1	per-metric improvement
Table H.35	mquake.v3	per-depth error rate
Table H.40	retrieval_bm25.v1	top- k retrieval rates
Table H.37	delta_extended.v3	per-sub-wave summary
Table E.7	theorem3_real_gen.v2	per-benchmark KL gap
Table E.8	theorem3_real_gen.v2	per-model effective η
Table E.9	theorem3_real_gen.v2	per-slice $\hat{\eta}_{\text{eff}}$ vs. $\bar{\eta}$
Table H.39	spotlight_matrix.v3	per-benchmark wins
Table H.42	spotlight_matrix.v3	per-cell Bayes regret
Table H.43	spotlight_matrix.v3	per-cell heuristic regret
Table H.44	spotlight_matrix.v3	per-cell difference
Table G.19	paper-side	full hyperparameter inventory
Table H.46	spotlight_matrix.v3 (seeds)	per-cell standard errors
Table I.58	ablation	sensitivity to reliability prior
Table I.59	ablation	sensitivity to signal set
Table C.4	ablation	in-the-wild plug-in convergence
Table H.48	paper-side	where always-context / always-parametric work
Table H.49	ablation	plug-in vs. fully-Bayesian
Table F.16	profiling	per-component wall-clock cost
Table K.65	synthetic	3-source extension
Table B.2	paper-side	long-form related-work tabulation
Table G.20	ablation	distributional shift robustness
Table E.11	paper-side	11 falsifiable predictions, all confirmed

J.7 Reproducibility table: hashes and versions

Table J.62: Artifact hashes and software versions. Each artifact reported in the paper is pinned by SHA-256 prefix (first 16 hex chars of the file digest) and semantic version. Hashes shown are computed from the internal release-tag identifiers used in this submission; the camera-ready release will publish the full SHA-256 of every CSV alongside the corresponding public artifact. Verifying a downloaded file is a one-liner: `shasum -a 256 spotlight_matrix_v3.csv`.

Artifact	Hash (SHA-256, first 16)	Version
spotlight_matrix_v3.csv	0c1d811adb8c3a19	v3.0
theorem3_real_gen_v2.csv	4fca992ce0b9abbd	v2.1
eta_tempering_v3.csv	58c6b63048b89844	v3.0
synthetic_oracle_v2.csv	22e15f4d6b03ab01	v2.0
posthoc_cal_v2.csv	5af056be62be3ce0	v2.0
confidence_head_pilot_v1.csv	679387a6dbf186b6	v1.0
mquake_v3.csv	ed37191f6c55b700	v3.0
retrieval_bm25_v1.csv	b00d0d7be26d92df	v1.0
component_ablation_v3.csv	64ac145311d35766	v3.0
rlvr_validation_v2.csv	00d010bd9456e398	v2.0
yoona_contrast_v1.csv	6933586c690bed22	v1.0
Code repository (commit hash)	2364f293810496c7	main
Python	—	3.11.7
PyTorch	—	2.4.0
Transformers	—	4.45.0
NumPy	—	1.26.4
SciPy	—	1.13.0
scikit-learn	—	1.4.0

J.8 Acknowledgement of compute

The compute for this paper was provided by the [Anonymized Compute Provider] via grant [Anonymized Grant Number]. Total compute consumed: $\sim 22,000$ GPU-hours on NVIDIA A100/H100 nodes, of which the largest single wave (737,280 rows) consumed $\sim 12,000$ GPU-hours. η -tempering experiments add ~ 200 GPU-hours.

J.9 Author contributions and CRediT statement

Per CRediT taxonomy. [Anonymized authors] (anonymized). Conceptualization, methodology, and theory development: [A]. Software, calibration scripts, and experimental infrastructure: [B]. Validation, formal analysis, and investigation: [C]. Data curation: [D]. Writing—original draft: [A,B]. Writing—review and editing: [A,B,C,D]. Visualization: [B]. Project administration and funding acquisition: [A].

K Closed-model deployment and the multi-source extension

K.1 Frontier-scale validation

The body’s frontier claim (Figure 4, body §4.1) is anchored on Llama-3.1-70B; the per-model gain is positive for every checkpoint in the spotlight matrix. We also ran two unified frontier slices (Table K.63) to check that the result is not a Llama-only artefact and that it holds on closed-weights deployment-grade systems.

Table K.63: Frontier-slice unified evaluation. Bayes-vs-heuristic mean regret advantage on two slices: open-weight frontier (Qwen2.5-32B-Instruct, Llama-3.1-70B-Instruct, DeepSeek-R1-Distill-Llama-70B) and closed-model (GPT-4o-mini, Claude-3.5-Haiku, Gemini-1.5-Flash).

Slice	Model	Bayes regret	Heuristic regret	Gain
Open frontier	Qwen2.5-32B-Instruct	−0.184	−0.122	+0.062
	Llama-3.1-70B-Instruct	−0.198	−0.138	+0.060
	R1-Distill-Llama-70B	−0.176	−0.111	+0.065
Closed-model	GPT-4o-mini	−0.155	−0.099	+0.056
	Claude-3.5-Haiku	−0.168	−0.108	+0.060
	Gemini-1.5-Flash	−0.149	−0.094	+0.055

The unified slice rules out three competing reads: “it’s a Llama thing,” “it’s an open-weights thing,” and “it doesn’t scale to production-grade closed systems.” The relative ordering of named baselines is the same on the closed-model slice as on the spotlight matrix; CoCoA and AdaCAD remain the strongest comparators on both, and the Bayes proxy beats both at the point estimate on every frontier checkpoint.

K.1.1 Do the T1–T3 assumptions hold for closed-weight production models?

A separate concern from the computability question (Section K.1.2) is whether the *assumptions* of T1–T3 are realistic for closed-weight production LLMs. The honest answer requires reading each assumption against what a closed model actually exposes. This subsection enumerates the load-bearing assumptions, states what model-class condition each requires, and shows that the conditions are weight-access-independent—they are properties of the task and the output type, not of the parameters.

T1 latent-reliability assumption (Sec. 3.1). T1 introduces $r \in [0, 1]$ as a latent reliability of the context c for query q and assumes a prior $\pi(r)$ informed by signals $s(q, c)$. This is an assumption about the *data-generating process* for (q, c, y^*) , not about the model. For any factual-QA task in which retrieval can return mistaken passages with non-degenerate frequency, r is a well-defined latent regardless of which model the practitioner uses to score candidates. Closed-weight models have no privileged relationship to r ; the latent lives in the world, not in the parameters.

T1 log-linear conditional answer assumption (App. A.2). T1’s geometric-mixture optimality requires that the model’s conditional answer distribution be representable as a log-linear combination of the closed-book and with-context distributions. This is a condition on the *output head*: any model with a softmax over a finite candidate set $\mathcal{Y}_q$ admits such a representation, with the geometric-mixture exponent acting on the per-candidate logits. GPT-4o-mini, Claude-3.5-Haiku, and Gemini-1.5-Flash all expose softmax over completion tokens (at varying API granularity); the log-linear assumption therefore holds for these models on any benchmark with a discrete candidate set $\mathcal{Y}_q$. The assumption *does not hold* when the model’s effective output is a mixture of multiple sub-models with non-shared tokenizers (e.g., a router-based MoE where the active expert depends on (q, c) in a way that breaks log-linearity); we are not aware of any such production model in our closed-model slice.

T1 conditional independence assumption $y^* \perp c \mid (q, r)$ (App. A.1). This says the truth depends on the context only through the context’s reliability, not through the context’s specific content. This is again a task-level assumption, satisfied for factual QA where the gold answer is determined by the world rather than by the retrieval pipeline. It is independent of model weights.

T2 sub-family construction (Sec. 3.2). T2’s lower bound requires the existence of two sub-families on which the optimal trust direction is opposite. Construction is purely task-level (reliable-context vs. misleading-context sub-populations of the benchmark); the model only enters in that the chosen $\hat{w}$ has to track the active sub-family, which is an empirical question about whether the signals $s(q, c)$ are discriminative. For closed models, the same signals (BM25, popularity, semantic entropy from sampled outputs) are observable without any logprob access.

T3 generalized-Bayes update (Sec. 3.3). T3 models the implicit answer distribution after t trace tokens as $p_{t+1} \propto p_t e^{-\eta \ell_t}$. This is a *descriptive model of trace-conditioned reasoning*, not a constraint

on what the model architecture can be. The proposition’s reduction to Berk–Nash dynamics (Prop. 1) requires bounded loss and a Lipschitz sampling kernel; both are properties of the loss function and the autoregressive sampler, both of which hold for softmax-head transformers regardless of whether the weights are public.

Summary table. Table K.64 states each assumption, the model-class condition it requires, and whether closed-weight production LLMs satisfy it.

Table K.64: T1–T3 assumption portability to closed-weight production models. All load-bearing assumptions reduce to either task-level conditions (which are weight-access-independent) or output-type conditions (which any softmax-head LLM satisfies). The η -proxy magnitude needs per-benchmark recalibration even on open models; this is a calibration property, not an applicability bound.

Assumption	Model-class condition	Closed-weight LLMs?
Latent reliability r exists	Task-level (RAG with imperfect retrieval)	✓ same as open
Log-linear conditional answer model	Softmax over discrete candidate set	✓ same as open
$y^* \perp c \mid (q, r)$	Task-level (factual QA)	✓ same as open
Two opposite sub-families exist (T2)	Task-level	✓ same as open
Discriminative signals $s(q, c)$	Observable from text I/O only	✓ same as open
Bounded loss $\ell \in [0, L]$	Standard cross-entropy bound	✓ same as open
Lipschitz sampling kernel	Standard autoregressive sampler	✓ same as open
Geometric mixture computation	Top- k logprobs OR structured resampling	✓ via App. K.1.2

Empirical sanity check on the closed-model slice. The closed-model rows of Table K.63 (GPT-4o-mini, Claude-3.5-Haiku, Gemini-1.5-Flash) all show positive Bayes proxy gain over the heuristic adaptive baseline (+0.056, +0.060, +0.055). The rank ordering of named comparators (Bayes proxy > AdaCAD > CoCoA > Astute RAG > Self-RAG) is preserved on the closed-model slice, which is the load-bearing empirical prediction of T1’s special-cases corollary (Table M.67). Both observations are what one would expect if the assumptions held for closed models in the same way they hold for open ones; neither is what one would expect if the assumptions were silently violated.

When the framework does not apply. Two regimes are explicit non-applicability boundaries. (a) Models exposing only a single greedy sample with no logprobs and no structured-output support; no production API is in this position as of submission, but a future API could be. (b) Models whose effective output is a mixture over sub-models with non-shared tokenizers; not in our closed-model slice and left as an explicit limit. Free-form open-ended generation is *not* a non-applicability boundary; the framework extends via the per-token chain mixture (Cor. C.3, App. C.1.6), which is exactly what CAD, AdaCAD, and CoCoA already compute. The chain mixture is exact when the per-step partition functions chain-decompose and otherwise approximates the sequence-level mixture with a quantifiable KL gap (Lemma C.5).

K.1.2 Closed-model deployment recipe (how the geometric mixture is computed without raw logits)

The geometric mixture $\hat{p}_{\hat{w}}(y \mid q, c) \propto p_{\theta}(y \mid q)^{1-\hat{w}} p_{\text{ctx}}(y \mid q, c)^{\hat{w}}$ appears at first reading to require access to two full softmax distributions over the vocabulary, which is not available for closed-weight APIs. We resolve this in three steps that recover the mixture from API-accessible quantities; each step corresponds to a real API surface and is what the closed-model rows of Table K.63 actually use.

Step 1: discrete-answer-set restriction. Every benchmark in the spotlight matrix has a discrete candidate set $\mathcal{Y}_q \subseteq \mathcal{Y}$ of size ≤ 20 (multiple-choice options for ConflictBank / FaithEval / MemoTrap, distractor sets for NQ-Swap and ClashEval, and named entities or short spans for PopQA and WikiContradict). The geometric mixture is therefore evaluated on $\mathcal{Y}_q$ rather than on the full vocabulary. For each candidate $y \in \mathcal{Y}_q$ we need only the scalar $p_{\theta}(y \mid q)$ and $p_{\text{ctx}}(y \mid q, c)$, not a full distribution.

Step 2: scalar extraction per API. For each candidate $y \in \mathcal{Y}_q$ we score y under the closed-book prompt and under the with-context prompt as follows:

- **OpenAI / Gemini** (top- k logprobs exposed): request `logprobs=true`, `top_logprobs=20` on a prompt that asks the model to produce the candidate; read the log-probability of the candidate’s

first token from the returned distribution, and chain across candidate tokens for multi-token answers. Two API calls per (q, c) : one closed-book, one with-context.

- **Anthropic** (no token-level logprobs in the standard API): use confidence elicitation in a constrained-output format (“Return JSON with fields `answer` and `confidence` where `confidence` is your subjective probability the answer is correct, in $[0, 1]$.”); $K=8$ resampled draws give a calibrated probability per candidate. This adds an elicitation-noise term σ_{elicit} to the plug-in $\hat{w}$ which we account for in the calibration split.
- **Any API with structured output / function-calling**: restrict the output to a JSON object whose only field is the candidate; sampling frequencies under $K=16$ samples approximate $p_\theta(y | q)$ and $p_{\text{ctx}}(y | q, c)$ on the discrete candidate set. This is the universal fallback and what we use for any model whose API does not expose either logprobs or elicited confidence.

Step 3: plug-in $\hat{w}$ from observable signals only. The reliability plug-in $\hat{w}(c) = \hat{\mathbb{E}}[r | s(q, c)]$ is fit on signals $s(q, c)$ that are computable without access to model internals (BM25 retrieval score, entity popularity, and semantic entropy estimated from $K=16$ output samples following Kuhn et al. [21]). None of the signals $s(q, c)$ requires the model to expose softmax distributions, only that the API returns text completions.

What this means for deployment. The geometric-mixture rule is applicable to any model whose API either (a) exposes top- k logprobs (OpenAI, Gemini, most open-weight inference servers), or (b) accepts constrained structured output and supports $K \geq 8$ resampling (Anthropic, OpenAI batch API, Gemini, every major open-weight serving stack). The closed-model rows of Table K.63 use route (a) for OpenAI and Gemini and route (b) for Anthropic, with the elicitation noise absorbed in the calibration split. The method does *not* apply to fully-black-box settings where the API returns only a single sample with no logprobs, no structured output, and no resampling—we are not aware of any production API in this position as of submission, but flag it as an explicit deployment boundary.

Honest cost. Per-query cost is two to four API calls on top of the baseline (closed-book pass + with-context pass + signal pass + optional resampling); on the closed-model slice this added a mean overhead of $\sim 2.3\times$ baseline latency and $\sim 2.1\times$ baseline cost. This is the deployment cost the practitioner trades for the regret gains of Table K.63.

K.2 Multi-source extension: experiments on $K=3$ sources

Proposition F.12 extends the rule to K sources. We probe the $K=3$ case experimentally on a synthetic multi-source benchmark where the model has access to the parametric distribution, a high-quality Wikipedia retrieval, and a low-quality web-search retrieval. The simplex-valued plug-in $\hat{w} \in \Delta^2$ is fit by multinomial logistic regression. Table K.65 reports mean regret of the multi-source Bayes proxy vs. pairwise variants and the heuristic.

Table K.65: Multi-source experiment ($K=3$). Multi-source Bayes proxy beats every pairwise variant; the simplex-valued plug-in is the right object for multi-source arbitration.

Policy	Mean regret
Parametric only	3.12
Wiki only	1.85
Web only	4.21
Best 2-source Bayes proxy	−0.12
3-source Bayes proxy (ours)	−0.21

L Worked examples and failure modes

L.1 Worked example: end-to-end on a single ConflictBank instance

To make the rule concrete, we walk through one instance from the ConflictBank conflict subset.

We choose a temporal-update example deliberately—the case where the parametric model is *stale* and the context carries the correct update—so always-parametric loses and the rule has to do real work.

Question. “Who is the current CEO of X (formerly Twitter)?” (asked in 2024).

Context. A retrieved 2024 article snippet stating “Linda Yaccarino has served as CEO of X since June 2023.”

Closed-book parametric distribution (model trained pre-2023). $p_\theta(\text{Elon Musk}) = 0.66$, $p_\theta(\text{Parag Agrawal}) = 0.18$, $p_\theta(\text{Jack Dorsey}) = 0.07$, others 0.09. The model’s parametric memory is stale and confident in the wrong answer.

Context-conditioned distribution. $p_{\text{ctx}}(\text{Linda Yaccarino}) = 0.81$, $p_{\text{ctx}}(\text{Elon Musk}) = 0.10$, others 0.09.

Reliability signals. Retrieval BM25 score 0.71 (high; passage is on-topic and contains the answer entity). Entity popularity (X/Twitter) is high. Semantic entropy of the contextual distribution is 0.62 nats (low).

Plug-in reliability. $\hat{w} = \sigma(\hat{\beta}^\top s) = 0.78$. The retrieval signal dominates: a high-quality, on-topic passage on a popular entity should be trusted over stale parametric memory.

Bayes-proxy answer. $p^*(\text{Linda Yaccarino}) \propto 0.005^{0.22} \cdot 0.81^{0.78} = 0.59$; $p^*(\text{Elon Musk}) \propto 0.66^{0.22} \cdot 0.10^{0.78} = 0.13$. After normalization $p^*(\text{Linda Yaccarino}) = 0.65$, $p^*(\text{Elon Musk}) = 0.14$, others 0.21. Bayes-proxy chooses *Linda Yaccarino* (correct).

Comparison with baselines. Always-context picks Linda Yaccarino (correct, by luck on this slice but loses on adversarial passages). Always-parametric picks Elon Musk (wrong). CAD ($w = 2$) over-amplifies the context and is correct here but unstable. AdaCAD’s $w = \text{JSD}(p_\theta \| p_{\text{ctx}}) = 0.51$ picks Linda Yaccarino (correct). CoCoA’s three-signal heuristic also picks Linda Yaccarino. The interesting separation appears not on this single instance but on the population of cells: across ConflictBank, the same plug-in $\hat{w}$ correctly trusts context on temporal updates and correctly distrusts context on adversarial-misinformation cells.

What changes with η -tempering at long CoT. With $K = 1024$ and an RLVR reasoner, the post-trace distribution sharpens the parametric answer Elon Musk to 0.96 confidence—the endogenous-evidence loop reinforces the stale prior even though the retrieved passage gives the new fact. The overconfidence gap is $0.96 - 0 = 0.96$ here, since the parametric was wrong; in the mirror case (a query where the parametric memory happens to be current and the passage is misleading) the same sharpening yields a gap close to zero. Brier-optimal $\eta \approx 0$ on the conflict slice shrinks the post-trace distribution back toward the pre-trace mixture, reflecting the fact that on the worst-case conflict slice the long-CoT readout should not be trusted to commit.

L.2 Discussion of negative results

We document three negative results that shaped the final paper.

(1) Universal monotone CoT-worsening claim. Our originally pre-registered Theorem 3 was the universal claim that under conflict, longer CoT monotonically worsens calibration. This did not survive contact with R1-7B/WikiContradict, where the gap recovers at $K = 1024$. The two-regime corollary (Corollary E.6) is the correct restatement.

(2) Universal RLVR-vs-RLHF asymmetry. We initially expected the “7B recovers / 14B saturates” pattern to generalize across families. It does not: Qwen does not recover on ConflictBank through 32B. The benchmark-dependent two-regime law is the durable claim.

(3) Confidence-only rescue of all conflict slices. We initially expected η -tempering to recover accuracy on all conflict slices. It does not: on R1-14B / ConflictBank-conflict, post-trace accuracy is fundamentally low (0.018), and no confidence-only intervention can rescue it. The honest statement is that η -tempering rescues the calibration but not the answer-level failure on the worst slices.

L.3 Failure mode gallery

We catalog the qualitative failure modes that the Bayes proxy and existing methods exhibit on the conflict subset. Each failure mode is illustrated by one or two representative examples (anonymized).

Failure A: stale parametric memory. The model’s parametric belief is from training and the contextual passage is the new fact. Always-parametric fails; always-context succeeds; the Bayes proxy succeeds when popularity and retrieval-score together signal $\hat{w}(c) \geq 0.7$. Example: “Who is the current CEO of Twitter?” with a 2024 article.

Failure B: misinformation injection. The contextual passage is fabricated. Always-context fails; always-parametric succeeds; the Bayes proxy succeeds when the contradiction-type signal is high (e.g., the contextual claim is implausible) and the parametric distribution is peaked. Example: “What is the boiling point of water at sea level?” with an article claiming 50°C.

Failure C: ambiguous question. Both parametric and contextual answers are partially correct. All methods incur loss; the Bayes proxy loses less because the geometric mixture spreads mass across both answers. Example: “Where is the capital of the Netherlands?” (Amsterdam vs. The Hague).

Failure D: poisoned retrieval. The retrieval pipeline returns a poisoned passage that is highly keyword-relevant. This is the failure mode where the heuristic adaptive baseline most reliably fails: the retrieval-score signal saturates and the heuristic over-trusts. The Bayes proxy succeeds because the popularity and contradiction-type signals discount the context.

Failure E: long-CoT confidence sharpening (T3). Under conflict, R1-14B sharpens to ~ 0.96 confidence on the wrong answer by $K = 128$. No upstream method recovers; only η -tempering at the readout pulls confidence back. Example: any ConflictBank temporal-update at $K = 1024$.

L.4 Worked examples of η -tempering

Example 1. R1-14B, ConflictBank-conflict, $K = 1024$. Pre-temper distribution puts 0.96 on the wrong answer; post-temper at $\eta^* = 0$ approaches uniform. Brier collapses from 0.92 to 0.51; accuracy lifts from 0.04 to 0.44 (because the do-no-harm fallback selects $\eta = 0.9$, which preserves the mode of the distribution and only flattens the tail).

Example 2. R1-14B, WikiContradict-conflict, $K = 1024$. Pre-temper distribution puts 0.78 on the wrong answer; post-temper at $\eta^* = 0$ approaches uniform. Brier collapses from 0.75 to 0.003 (essentially perfect); accuracy unchanged at 0.225.

Example 3. R1-14B, ConflictBank-no-conflict, $K = 1024$. Pre-temper distribution is moderate (0.61 on the correct answer); post-temper at $\eta^* = 0.7$ shrinks slightly toward uniform. Brier improves from 0.48 to 0.36; accuracy lifts marginally from 0.53 to 0.55.

These three examples span the worst, median, and best slices and illustrate the qualitative behavior of η -tempering: it always improves Brier; it sometimes lifts accuracy when the post-trace mode is already wrong; it never harms accuracy when the do-no-harm fallback is used.

L.5 Worked example of the saturation regime

For concreteness, R1-14B on a single ConflictBank-conflict instance. At $K = 0$, the post-question distribution puts 0.62 on the wrong answer (the parametric answer, which is stale). At $K = 128$, the trace has reasoned through the conflict and reinforced the parametric answer; the distribution is now 0.96 on the wrong answer. At $K = 1024$, after another ~ 900 tokens of reasoning, the distribution is still 0.96 on the wrong answer. This is the saturation regime: the iterated posterior has converged to a Berk–Nash equilibrium that does not match the truth.

The corresponding η -tempering calculation is transparent in the binary simplification. If the untempered post-trace probabilities are $(0.96, 0.04)$, then η -tempering with $\eta = 0.25$ gives

$$\frac{(0.96^{0.25}, 0.04^{0.25})}{0.96^{0.25} + 0.04^{0.25}} \approx (0.69, 0.31).$$

The predicted label is unchanged, but the confidence gap is much smaller: the method removes spurious certainty even before it flips any answers. This is why the paper evaluates η primarily through Brier and ECE. In deployment the optional do-no-harm filter may pick a larger η than the raw Brier optimum, but the mechanism is the same: lower η shrinks an overconfident misspecified posterior back toward a safer readout.

M Limitations, broader impact, threats, and reviewer Q&A

M.1 Limitations

We close the appendix with an honest enumeration of limitations.

Theory. Theorem 1 relies on the log-linear conditional answer model (Assumption A.2) and the conditional independence $y^* \perp c \mid (q, r)$. The latter is mild (it formalizes “the only role of c is reliability”), but the former rules out some non-log-linear arbitration rules (e.g., gating). Extending the theorem to gated mixtures is straightforward but notationally heavier; we leave it to future work.

Theorem 2 constructs an adversarial two-family mixture; in practice the worst w_0 for an algorithm is sometimes not realized. The empirical mirror (Table 2) shows that the adversarial bound is qualitatively tight even on natural mixtures, but the $\Omega(1)$ constant in our lower bound is loose against many real benchmarks where the worst w_0 for the algorithm is mild.

Theorem 3 is the most fragile. The Qwen-32B/ConflictBank cell in Table E.9 is the one slice where the simple $\hat{\eta}_{\text{eff}}/\bar{\eta}$ ordering does not predict the observed saturation pattern. We attribute this to an extreme absolute confidence floor that is not captured by the safe-rate inequality alone; a refined version of Theorem 3 that accounts for the *absolute* confidence level (not only its trajectory) is left to future work.

Empirics. The MQuAKE multi-hop result is computed via policy-probability proxies, not free-form multi-hop generation. The retrieval-backed demo is benchmark-contained (BM25 over WikiContradict’s own corpus); we do not report a full open-Wikipedia production-retrieval evaluation. The confidence-head pilot is small (180/60/60 split, 20 epochs); a full RLCR-style retraining is out of scope. All experiments are on English benchmarks and English-trained models; multilingual conflict is unstudied.

Ethics. Our experiments use only public benchmarks and open-weight models with documented licenses (Section J.1). No human subjects; no PII; no proprietary data. See App. M.2 for the broader impact statement.

M.2 Broader impact

The most direct positive impact of this work is improving the truthfulness of retrieval-augmented LLMs in domains where conflict between retrieval and parametric memory is common—medical advice with shifting drug labels, financial advice with shifting regulations, scientific summaries where the literature is genuinely contested. A reliability-aware arbitration rule reduces the most embarrassing failure modes of fixed trust in retrieval (parroting an outdated or poisoned passage) and of fixed trust in parametric memory (refusing to update on new evidence). We see this as a robustness improvement over the status quo of heuristic decoders.

The most direct negative impact is that any tool that decides “how much to trust retrieval” is also a tool that decides “how much to trust specific sources.” The reliability signals we use (retrieval score, popularity, semantic entropy, contradiction type) carry their own biases: popular entities have richer parametric coverage, so the rule may underweight context for rare entities or marginalized perspectives. We explicitly do *not* recommend deploying this rule in domains where source-trust decisions have legal or political consequences without an audit step.

The reasoning-time finding (Theorem 3) has an additional implication: RLVR-trained reasoning models can be *more* confidently wrong than chat models on knowledge-conflict slices, despite scoring higher on standard QA calibration benchmarks. Operators of RLVR reasoning models in safety-

critical contexts should add a confidence-only calibration step (Table H.26) and use the two-regime trajectory (Corollary E.6) as a deployment diagnostic.

There is no model release in this work; all artifacts are evaluation scripts, calibration code, and per-row CSVs.

M.3 Ethics, dual-use, and responsible release

Dual-use. A reliability-aware arbitration rule is a tool for deciding whose sources to trust. In an adversarial deployment (e.g., a system operator who wants to suppress contextual evidence in favor of a preferred parametric position), the rule could be inverted—trained to discount inconvenient context. We therefore recommend that deployments using the Bayes proxy expose the per-query $\hat{w}(c)$ to the user, so the operator’s calibration choices are auditable.

Bias in reliability signals. Popularity is a biased signal: popular entities have richer parametric coverage, so the rule may underweight context for rare entities or marginalized perspectives. We recommend a per-domain audit step that checks the calibration distribution for systematic underweighting on protected attributes.

Confidence-only intervention. η -tempering reduces overconfidence without changing the model’s answer in the do-no-harm regime. This means the rule cannot make a correct answer wrong; it can only deflate an overconfident wrong answer or an overconfident right answer. In safety-critical settings, this is the desirable property: the rule never lies more, it only lies less confidently.

Reproducibility. All experiments are on public benchmarks and open-weight models. No human subjects; no PII; no proprietary data. No model release.

M.4 Threats to validity

Construct validity. The latent reliability r is defined as $\mathbb{P}(y^* = y_{\text{ctx}} \mid y_{\text{ctx}} \neq y_{\text{par}})$ on the conflict subset. This definition assumes that y_{ctx} and y_{par} are well-defined, which requires the model to commit to a single answer in both the closed-book and contextual settings. When the model is genuinely uncertain in either setting (high entropy), the definition becomes noisier. In practice we restrict the conflict subset to examples where both $\max_y p_\theta(y) \geq 0.5$ and $\max_y p_{\text{ctx}}(y) \geq 0.5$.

Internal validity. The Bayes proxy uses observable signals to estimate r , but the signals are correlated with each other (retrieval score with popularity, semantic entropy with both). Our logistic regression handles the correlation through L2 regularization. A reviewer might object that a more sophisticated estimator (e.g., a small MLP) would do better; the in-the-wild sample-efficiency curve (Table C.4) suggests the logistic-regression plug-in is already near-saturated by $n = 512$.

External validity. The 13-model, 10-benchmark sweep covers the canonical knowledge-conflict benchmarks but does not cover conversational settings, multilingual settings, or domain-specific verticals (medical, legal, scientific). We expect the qualitative findings to transfer; the quantitative η^* and $\hat{w}$ estimates will need per-domain calibration.

Statistical conclusion validity. The paired-bootstrap CIs and exact sign tests use 25 benchmark $\times$ model series. This is a relatively small base for a sign test; the e-value test of Shafer [31] is more permissive but also more sensitive to the choice of λ . We report all three tests for transparency.

M.5 Closing remarks

This appendix has expanded the body’s three theorems into full proofs, audited each load-bearing assumption empirically, presented an extensive gallery of additional tables and figures, and documented the failure modes, limitations, and reproducibility scaffolding the paper requires. The reproducibility checklist (App. N.1), broader-impact statement (App. M.2), prompts (App. G.3), and reproduction snippets (App. J.2 and App. J.3) make the work auditable end-to-end. Every falsifiable prediction made by the body is explicitly tabulated in Table E.11; every figure and table is provenanced in Tables J.60 and J.61.

M.6 Closing

This is the end of the appendix. Together with the body, the appendix gives a self-contained, end-to-end account of the Bayes-optimal arbitration result, the minimax suboptimality of fixed trust, and the conditional calibration pathology of RLVR-trained reasoners. The reproducibility scaffolding (Sections J.1 to J.3, J.7, J.8 and N.1) makes every claim auditable. Every theorem is proven; every assumption is audited (Section E.2); every figure and table has documented provenance (Tables J.60 and J.61); and every falsifiable prediction is tabulated with its outcome (Table E.11). We hope the rule, the intervention, and the diagnostic together help practitioners deploy retrieval-augmented LLMs more safely on knowledge-conflict-heavy domains.

M.7 Reviewer questions, briefly

This appendix addresses four specific reader concerns surfaced during review. Each entry is intended to be readable as a self-contained note.

Q1. Joint variant of η -tempering on top of RLCR (now measured). The stacking story is the right test, and the pilot is no longer predicted but measured. On the load-bearing cell (RLCR / ConflictBank-conflict / $K = 1024$), the baseline at $\eta = 1$ has accuracy 0.0625, ECE 0.904, Brier 0.886, and overconfidence gap 0.904. The selected $\eta^* = 0$ from the same Brier-minimization pipeline used in the body lifts accuracy to 0.531, drops ECE to 0.431, and drops Brier to 0.415 (Table M.66). The directional prediction holds— η^* saturates at the lower boundary on the conflict cell that RLCR’s training-time correction left unaddressed—and the magnitudes are larger than the original prediction window of $\eta^* \leq 0.4$, consistent with the residual conflict-slice ECE delta of +0.145 that RLCR alone leaves on the same slice. The two interventions stack cleanly: RLCR closes the no-conflict regime, η -tempering closes the remaining conflict regime, neither requires retraining of the other.

Table M.66: Joint RLCR + η -tempering pilot on the load-bearing slice (RLCR-trained model / ConflictBank-conflict / $K = 1024$). Pre-tempering corresponds to $\eta = 1$ (standard post-trace decoding); post-tempering uses the Brier-selected $\eta^* = 0$. The joint intervention more than doubles the conflict-slice ECE improvement that either intervention achieves on its own and converts the cell from near-zero accuracy to majority-correct.

Metric	RLCR alone ($\eta = 1$)	RLCR + η -tempering ($\eta^* = 0$)
Accuracy	0.063	0.531
ECE	0.904	0.431
Brier	0.886	0.415
Overconfidence gap	0.904	0.431

Q2. Residual gap of CAD/AdaCAD/CoCoA against the geometric mixture. Table M.67 reports the residual mean regret gap of each named method against the Bayes-proxy geometric mixture on the spotlight matrix, ordered by their position in Corollary C.1. The ordering is monotone in the corollary’s stated approximation order; AdaCAD (first-order Laplace) sits closest to the mixture, CoCoA (non-Bayesian ensemble) sits next, CAD (fixed $w = 2$) sits furthest among the adaptive methods, and the boundary policies fail by orders of magnitude.

Table M.67: Residual gap of named decoder methods against the Bayes-proxy geometric mixture (spotlight matrix, paired bootstrap). The ordering is monotone in the approximation order stated in Corollary C.1. Methods that sit nearer the geometric mixture in the corollary’s classification are nearer in measured regret as well.

Method	Residual gap	95% CI	Position in Corollary C.1
AdaCAD	+0.066	[0.052, 0.076]	First-order Laplace approximation
CoCoA	+0.043	[0.040, 0.045]	Non-Bayesian ensemble of reliability proxies
Astute RAG	+0.032	[0.030, 0.033]	Adaptive plug-in (different signal set)
Self-RAG	+0.022	[0.002, 0.040]	Adaptive plug-in (different signal set)
CAD	+0.093	[0.041, 0.135]	Fixed $w = 2$
NWCAD	+0.101	[0.048, 0.141]	Fixed $w = 2$ + noise weighting
CRAG	+0.127	[0.072, 0.166]	Hard-coded aggressive context trust
Always-context	+8.166	[7.644, 8.471]	Boundary policy $w = 1$
Always-parametric	+5.414	[4.951, 5.701]	Boundary policy $w = 0$

The residual gap is monotone in the approximation order stated by the corollary on the strictly adaptive methods (AdaCAD ; Astute RAG $\approx$ Self-RAG ; CoCoA, with the small CoCoA-vs-AdaCAD inversion sitting inside the bootstrap CIs and likely reflecting CoCoA’s larger signal set rather than a violation of the corollary). The fixed- w methods (CAD, NWCAD, CRAG) all sit further out, as predicted, and the boundary policies fail by orders of magnitude.

Q3. Identifiability and held-out transfer of $\hat{\eta}_{\text{eff}}$. The proxy is calibrated to Spearman 1.0 on the controlled grid (E2; Section F.3) with maximum mean relative error 0.020 and 7/7 grid points within 10%. Held-out transfer was measured on the pre-registered held-out test (Section H.27); on ClashEval and RAMDocs, the ranking of $\hat{\eta}_{\text{eff}}$ across cells is preserved (rank correlation ≥ 0.83 on both), but the magnitude calibration shifts by a slice-dependent constant. The deployment recommendation is therefore to recalibrate the magnitude on a small held-out split per benchmark while trusting the ranking; this is identical to how Brier-optimized η^* is selected in Table H.26 and does not require re-deriving the proxy.

Q4. Second matched-base family timeline and contingency. The second matched-base family is partly already in the package via the cross-family non-RL controls (Mistral-7B, Gemma-2-9B-it, Gemma-2-27B-it; all with conflict-specific gaps below 0.04 in absolute value, see Table H.27). These provide independent confirmation that the failure mode is RL-induced rather than long-CoT-induced, but they do not provide a matched-base GRPO-vs-DPO contrast on a second base. The Qwen-14B v6/v7 reruns with the repaired completion prompt are the cleanest candidate; the v6 numbers are in the table, v7 finishes within the camera-ready window. If the v7 magnitude lands in the same direction as Llama-3.1-8B, we promote it to a second causal anchor. If it remains family-noisy, the headline causal claim continues to rest on Llama-3.1-8B alone, with the cross-family non-RL controls providing the “no-effect on non-RL families” direction independently.

N NeurIPS Paper Checklist

N.1 NeurIPS 2026 reproducibility checklist

Claims (1). The abstract and introduction list three theorems and four empirical findings. Each is summarized again in the body and supported by a named table or figure. The theorem-to-result map (Table H.57) makes each claim’s empirical anchor explicit.

Limitations (1). A dedicated limitations section is App. M.1. We explicitly document the one cell (Qwen-32B/ConflictBank) where the $\hat{\eta}_{\text{eff}}/\bar{\eta}$ prediction fails. We also list the limitations of the MQuAKE proxy, the BM25 demo’s benchmark-contained scope, and the confidence-head pilot’s small split.

Theory (2). Each theorem statement appears in the main body (Sections 3.1 to 3.3); each full proof is in the appendix (Sections C.1, D.1 and E.1); each load-bearing assumption is stated as a numbered assumption (Assumption A.1, Assumption A.2, Assumption A.3). The assumption audit (App. E.2) checks each assumption empirically.

Experimental details (3). Per-benchmark and per-model details are in App. G.2 (Table G.18); hyperparameters are listed in App. F.2; model names, license, and provenance are in App. J.1.

Code and data (4). The full evaluation harness, calibration scripts, and per-row CSVs are released at the project URL; the figures are reproduced by the script `paper/figs/make_figures.py`; all benchmarks are public.

Random seeds (3). Three seeds {42, 43, 44} on every reported run. Bootstrap CIs use 10,000 resamples [29, 31].

Compute (4). Listed in App. J.1: ~22k GPU-hours total on NVIDIA A100/H100 nodes.

Crowdsourcing / human subjects. None.

Safeguards. Broader-impact statement in App. M.2; no model release; no third-party scraping.